

Native 1-Bit and 1.58-Bit Large Language Models: A Survey and an Independent Re-Evaluation

Habib Ullah Manzoor^{*1} and Basim Alhumaily^{†2}

¹School of Computing, Engineering and Physical Sciences, University of the West of Scotland,
UK

²Department of Electrical Engineering, College of Engineering, Qassim University, Buraydah
52571, Saudi Arabia

Abstract

Large language models whose linear-layer weights are constrained to binary $\{-1, +1\}$ or ternary $\{-1, 0, +1\}$ turn the dominant matrix multiplication into addition. The centre of gravity is the native paradigm, which imposes that constraint during training; adjacent post-training and quantization-aware routes target the same one-to-two-bit range. This paper is a curated, open-access survey of 135 works from 2023 to 2026, organized on a taxonomy and a common effective-bits accounting, together with an independent re-measurement of the flagship open model’s headline claims and, where open matched checkpoints exist, a controlled test of the accuracy claim. In a matched comparison, a native ternary model against its FP16 counterpart at the same parameter count, tokenizer, and training data, the two land within 0.6 points on a seven-task zero-shot suite at the two-billion-parameter scale, with the ternary model’s perplexity about 14% higher: downstream parity, worse language-modeling loss. Post-hoc 4-bit quantization of a full-precision model loses more of the suite than that, and a 1.58-bit model at the one-billion-parameter scale loses much more; this is consistent with, but too thin a basis alone to establish, a scale dependence for the parity result, and whether it holds beyond the measured regime is untested. On efficiency the finding is pointed: on one commodity x86 machine a standard 4-bit quantization of the full-precision baseline matched or beat the flagship 1-bit model on decode speed and on-disk size and was comparable on per-token energy, and held close on the accuracy tasks checked, leaving the 1-bit model only a 24-to-27% smaller resident set; its efficiency case at this scale holds against half precision and an 8-bit baseline but not against 4-bit, and a GPU and a second machine are untested. The reference CPU runtime, moreover, does not run the flagship model correctly as documented, needing a one-line activation fix, a sign the open deployment path is still immature. The activation-precision floor and training stability at frontier scale remain the field’s open problems.

Index Terms: 1-bit and 1.58-bit language models, BitNet, quantization-aware training, efficient inference, hardware co-design.

1 Introduction

1.1 The inference wall

The dominant cost of a deployed large language model (LLM) is no longer training but inference. Every generated token requires a full forward pass, and for autoregressive decoding that pass is bounded not by arithmetic throughput but by the movement of weights from memory into the compute units. Accelerator peak throughput, measured in floating-point operations per second, has

^{*}Corresponding author. Email: habib.manzoor@uws.ac.uk

[†]Email: b.alhumaily@qu.edu.sa

grown far faster than dynamic-memory bandwidth for a decade: as the Spectra analysis notes, GPU operations per second have doubled roughly every 1.26 years against every 2.9 years for memory bandwidth [1]. The arithmetic-to-bandwidth ratio of hardware has therefore drifted steadily away from what dense half-precision inference requires. The practical consequences are by now familiar: multi-GPU servers are needed for models that would otherwise fit on a single card; inference energy budgets draw scrutiny on sustainability grounds; and running a capable model on a phone, a laptop, or an embedded board without a network round trip remains largely infeasible.

Quantization addresses this wall directly by reducing the number of bits per weight, and hence the number of bytes that must cross the memory hierarchy. Eight-bit and four-bit post-training quantization are now routine. Below four bits, however, accuracy degrades sharply for models that are quantized after training [2], [3], and a growing body of scaling-law analysis indicates that this degradation is not a transient engineering problem. For a fixed architecture, the loss added by low-bit quantization grows with the number of training tokens, so that the better trained the base model, the worse post-hoc compression treats it [4, 5].

1.2 The native 1-bit paradigm

A separate line of work avoids that ceiling by making the low-bit constraint part of the model from the first optimization step. BitNet introduced the **BitLinear** layer, a drop-in replacement for a standard linear layer that binarizes weights in the forward pass and trains through the non-differentiable quantizer using a straight-through estimator. The resulting 1-bit Transformers follow a loss-versus-scale curve parallel to that of full-precision models [6, 7]. BitNet b1.58 then moved from binary weights in the set $\{-1, +1\}$ to ternary weights in $\{-1, 0, +1\}$, which carry $\log_2 3 \approx 1.58$ bits of information per parameter, and reported parity with a half-precision Transformer of equal size and training-token budget from roughly the 3B-parameter scale upward, at a fraction of the memory, latency, and energy [8]. In 2025 this became concrete with BitNet b1.58 2B4T, an openly released 2B-parameter model trained natively on 4T tokens that matches similarly sized full-precision open models on standard benchmarks [9].

A native 1-bit model is defined by when the low-bit constraint is imposed, during training rather than after, which is what lets it reach an accuracy a post-hoc quantization of the same architecture does not (Section 8.2). The inference-time arithmetic benefit is a separate property of the resulting representation, not of how the weights got there: whenever every weight lies in $\{-1, 0, +1\}$, whether from native training or post-hoc quantization, the multiplication in a matrix-vector product disappears: each output element is an accumulation of additions and subtractions of activation values, gated by the sign of the weight. This removes the multiplier, which is the largest and most power-hungry component of a digital multiply-accumulate unit, from the inner loop, and it changes what the ideal hardware looks like [10, 11]. It also shifts the binding constraint. Once weights occupy roughly one bit, activation precision and the memory traffic of the key-value (KV) cache dominate, which is why the frontier of the field has moved toward four-bit and lower activations and toward three-bit KV caches [12, 13].

1.3 Related surveys

The literature has grown from a handful of papers in 2023 to a steady stream through 2026, spanning native pre-training recipes, post-training routes toward one bit, optimization theory, inference kernels, and a first wave of custom accelerators (Fig. 2). Three existing surveys touch this area. Gong et al. [14] review low-bit LLMs broadly, from eight-bit integer down to binary, with a systems-and-algorithms structure and a mid-2024 horizon. Liu et al. [15] survey the binarization of LLMs; this framing under-weights ternary representations, even though ternary is where native training has succeeded. Hao et al. [16] survey low-precision training across all numerical formats in IEEE TPAMI, treating 1-bit models as one subsection and not covering inference systems or hardware. None is scoped to the native 1-bit and 1.58-bit paradigm as a coherent object of study. And none covers the 2025 and 2026 wave: BitNet b1.58 2B4T, BitNet a4.8, and BitNet v2; GPU and CPU inference

kernels; 1-bit embedding, speech, and vision-language-action models; ternary mixture-of-experts; and the first lookup-table ASICs and compute-in-ROM accelerators.

This paper adds two things the existing surveys do not. First, a single common-axis comparison: every surveyed work is placed on a five-axis taxonomy (Section 4), and that classification is what lets otherwise incomparable results be tabulated together, reported accuracy in consistent columns and, more substantively, reported bit-widths re-stated under one effective-bits accounting, so that a “1.08-bit” post-training method and a genuine 1.58-bit native one can be read on the same scale (Section 8). Second, and centrally, an independent re-measurement.

On accuracy, we run a controlled test the vendor comparisons lack: a native ternary model against its FP16 counterpart at matched parameter count, tokenizer, and training data, from open artifacts. At the two-billion-parameter scale the two land within 0.6 points on a seven-task zero-shot suite, with the ternary model’s perplexity about 14% higher, downstream parity with a worse language-modeling loss. A post-hoc 4-bit quantization loses more of the suite than that, and a 1.58-bit model at the one-billion-parameter scale loses much more, a pattern the paper’s abstract already flags as suggestive rather than conclusive: two data points, one of them a checkpoint pair whose training match is undocumented (Section 10.2), cannot on their own settle whether scale is the real explanation.

On efficiency, the finding is sharper than the literature’s framing. We rebuild the reference runtime and run the flagship open 1-bit model against half-precision, 8-bit, and three 4-bit builds on one commodity x86 machine. Each of the three 4-bit builds decodes at least as fast as the ternary model and is smaller on disk; the k-quant format among them, the one we also measured for energy, is comparable on per-token energy, and on the two accuracy tasks scored through the same runtime it holds close. The ternary model’s one retained advantage is a 24-to-27% smaller resident set. On this hardware, then, a 1-bit model shows no observed advantage over a standard 4-bit quantization on throughput or on-disk size, and no more than parity on energy; its efficiency case at this scale holds against half precision and an 8-bit baseline but not against 4-bit, and a GPU and a second machine are untested. The reference runtime, moreover, does not run this model correctly as documented, applying the wrong feed-forward activation until a one-line source fix (Section 10.3), a sign that the open deployment path for these models is not yet mature. These are the paper’s principal findings.

1.4 Contributions

1. **An independent re-evaluation.** For accuracy we run three pairs from open matched checkpoints, controlled to varying degrees (Section 10.3): native ternary against FP16 at a matched 2.4B parameters (within 0.6 points on a seven-task mean), a 4-bit quantization against its full-precision source, and a 1B model in 1.58-bit and bf16 form whose training match to its comparator is undocumented; and we re-measure MMLU and GSM8K, which an earlier version of the study had left out. For efficiency we rebuild the reference bitnet.cpp runtime and measure decode and prefill throughput, resident memory, and on-disk size for the flagship open 1-bit model against half-precision, 8-bit, and three 4-bit builds, on one x86 laptop. We also measure per-token battery energy against the half-precision, 8-bit, and 4-bit k-quant builds on the same machine (Section 10.4). On that machine the paradigm’s advantage over half precision does not extend to any 4-bit build: each decodes at least as fast and is smaller on disk, and the k-quant format, the one we also checked for energy and for accuracy on two tasks, matches the ternary model on per-token energy and holds close on accuracy; the 1-bit model keeps only a 24-to-27% resident-memory edge. A by-product of the exercise is a symptom worth recording: the reference runtime, built from source as its documentation describes, does not run this model correctly, applying a half-precision-era feed-forward activation where the model needs a squared-ReLU, so that a practitioner following the documented path gets near-broken output until a one-line source change (Section 10.3). The open deployment path for these models is not yet mature.
2. **A common-axis comparison.** Reported accuracy is re-tabulated in consistent columns (still in each source’s own evaluation terms), and reported bit-widths are re-stated under one effective-bits accounting (nominal code width plus the amortized cost of scales, masks, and

codebooks), so that methods using incompatible setups can be read side by side (Section 8, Table 2).

3. **A curated review and taxonomy.** We survey 135 open-access works of the native 1-bit and 1.58-bit paradigm from January 2023 to June 2026, plus five foundational works from 2013 to 2022 for context, and classify every one on a five-axis taxonomy of weight representation, training paradigm, quantized components, optimization mechanism, and systems stack, with a secondary cut by modality (Section 4, Fig. 5, Table 1). This is a curated, arXiv-primary review, not a systematic one (Section 3.2).
4. **A research agenda** stated as concrete, testable open problems: the capacity ceiling and its dependence on the training-token budget, training stability at frontier parameter scale, the activation-precision floor, a convergence theory for straight-through training, the absence of a shipping processor with a ternary matrix-multiplication primitive, and the robustness and safety profile of 1-bit models (Section 11).

1.5 Scope

This paper covers the native paradigm, in which the sub-two-bit constraint is present during training, together with the post-training and quantization-aware branches that explicitly target 1.58 bits or fewer, the kernels and hardware that execute such models, and the theory that explains their behavior. General sub-four-bit quantization, pruning, distillation for speed, and long-context or serving-system work are cited only where they bear directly on the 1-bit case.

The corpus is open-access only: for every work a freely downloadable version exists on arXiv, in JMLR or PMLR, in the ACL Anthology, on OpenReview, on a gold open-access journal page, or in an official model or framework repository, and a local copy has been archived. This is a deliberate choice, so that every technical claim is re-checkable without a paywall and so that the re-measurement of Section 10 can use the exact released artifacts. One work met every other criterion but had no obtainable open-access copy and was set aside; its abstract, with those of the paywalled records the search surfaced, was checked against every major conclusion, and none would change (Section 3).

1.6 Organization

Fig. 1 maps the paper. Section 2 fixes notation and background. Section 3 states the scope and how the corpus was assembled. Section 4 presents the taxonomy. Sections 5, 6, and 7 cover architecture, training, and inference systems. Section 8 presents the common-axis comparison. Section 9 surveys applications and extensions. Section 10 reports the independent re-evaluation, the analytical core of the paper alongside Section 8. Section 11 sets out the open problems, and Section 12 concludes.

2 Background and Preliminaries

2.1 Number formats and effective bits per weight

A dense linear layer multiplies an input vector by a weight matrix to produce an output vector, and that weight matrix is stored in some numeric format. Full-precision training uses 32-bit floating point (FP32) or, in practice, the 16-bit BF16 or FP16 formats, each consisting of a sign, an exponent, and a mantissa. Integer quantization replaces each weight with an integer code multiplied by a shared floating-point scale; 8-bit and 4-bit integer codes are the common operating points. Binary quantization restricts the code to the two values -1 and +1, that is, one bit. Ternary quantization restricts the code to the three values -1, 0, and +1, which carries $\log_2 3 \approx 1.58$ bits of information and is usually stored in two bits, or in packed schemes that approach the information bound [1], [17].

The reported number of bits per weight is not always the nominal code width. Partial schemes keep a small fraction of salient columns in higher precision: BiLLM reports about 1.08 effective bits and PB-LLM a tunable figure above one [2], [3]. Structural-sparsity schemes push below one bit

Paper organization

§2 Background & preliminaries

- 2.1 Number formats & effective bits
- 2.2 Binary neural-network lineage
- 2.3 Straight-through estimator
- 2.4 Transformer-specific obstacles
- 2.5 Scaling behavior
- 2.6 Evaluation metrics

§3 Scope & corpus

- 3.1 Scope & eligibility
- 3.2 How the corpus was assembled
- 3.3 Effective bits & evidence basis
- 3.4 Limitations

§4 Taxonomy of binary & ternary LLMs

- 4.1 Weight representation
- 4.2 Training paradigm
- 4.3 Quantized components
- 4.4 Optimization mechanism
- 4.5 Systems stack
- 4.6 Secondary cut by modality
- 4.7 Reading the taxonomy

§5 Architectural design

- 5.1 The BitLinear primitive
- 5.2 Ternary versus binary in practice
- 5.3 Activation quantization
- 5.4 KV cache
- 5.5 Beyond the dense decoder
- 5.6 Matmul-free designs

§6 Training methodology & theory

- 6.1 The native training loop
- 6.2 Recipes
- 6.3 The STE and its alternatives
- 6.4 Distillation
- 6.5 Post-training routes to one bit
- 6.6 Scaling laws at one bit
- 6.7 Stability & smoothness

§7 Inference systems & hardware

- 7.1 The need for a software stack
- 7.2 bitnet.cpp & CPU kernels
- 7.3 GPU
- 7.4 Custom accelerators
- 7.5 Summary of the systems corpus

Paper organization (*continued*)

§8 Performance & benchmarking

- 8.1 The comparison problem
- 8.2 Perplexity & the parity claim
- 8.3 Downstream benchmarks
- 8.4 Reasoning & generation quality
- 8.5 Efficiency
- 8.6 A common-axis comparison
- 8.7 Summary of the evidence

§9 Applications & extensions

- 9.1 On-device & edge deployment
- 9.2 Embedding models
- 9.3 Speech
- 9.4 Vision-language & VLA models
- 9.5 Mixture-of-experts
- 9.6 Ecosystem beyond one laboratory
- 9.7 Environmental framing

§10 Independent re-evaluation

- 10.1 Rationale & scope
- 10.2 Setup
- 10.3 Accuracy results
- 10.4 Efficiency results
- 10.5 Summary of findings
- 10.6 Reproducibility audit
- 10.7 Threats

§11 Challenges & open problems

- 11.1 Capacity ceiling & token budget
- 11.2 Training stability at scale
- 11.3 The activation-precision floor
- 11.4 A theory of STE training
- 11.5 Hardware co-design
- 11.6 Reasoning & smoothness
- 11.7 Robustness, safety, security
- 11.8 Sub-1.58-bit representations
- 11.9 Evaluation & contamination

§12 Conclusion

Figure 1: Organization of the paper as a tree, read in two columns left to right. The shaded branches are this paper’s analytical contribution.

theme \ year	2023	2024	2025	2026 H1	cutoff
Foundations & native training	• BitNet - BitLinear, 1-bit from scratch • TSLD - first ternary QAT of a large generative LM	• BitNet b1.58 - ternary, FP16 parity from ~3B • Spectra / TriLM - open ternary suite 99M-3.9B • MatMul-free LM - removes the attention matmul	• BitNet b1.58 2B4T - first open native 2B model, 4T tok • BitNet a4.8 - 4-bit activations • BitNet v2 - Hadamard rotation, native A4	• Sparse-BitNet - N:M sparsity + ternary • Sherry - 1.25-bit packing • NanoQuant / LittleBit - sub-1-bit	Literature cutoff 30 June 2026 Later work tracked in the online living appendix
Post-training routes toward one bit	• PB-LLM - first LLM binarization for compression	• BiLLM (1.08-bit) • BinaryMoS • DB-LLM • OneBit • ARB-LLM • FBI-LLM	• PTQTP • PT2-LLM • Tequila • PTQ1.61 • ICQuant	• HARP • TWLA • BWLA - first W1A6 • influence-weighted Walsh rotations	
Systems & inference kernels	–	• bitnet.cpp / 1-bit AI Infra 1.1 - CPU ternary kernels	• Spectra 1.1 TriRun - GPU ternary kernel	–	
Hardware accelerators	–	–	• PIM-LLM • BitROM • TeLLMe • TerEffic	• TOM - compute-in-ROM • Platinum - LUT-ASIC • T-SAR - SIMD-ISA • Vec-LUT	
Applications & extensions	–	–	• BitVLA (robotics) • BitTTS • one-bit ASR • BitNet Distillation • MoTE (ternary MoE)	• Ternary Mamba • BitNet Text Embeddings • Fairy2i (real -> complex 2-bit)	
Theory & scaling analysis	–	• Scaling Laws for Precision • QID (undertrained-LLM favoring) • theory of 1-bit NNs	• QuEST (stable W1A1) • CAGE (curvature-aware STE) • ParetoQ (unified 1-4 bit)	• "Fitting is not enough" (smoothness) • extreme-low-bit reasoning failure modes	

Figure 2: Timeline of native 1-bit and 1.58-bit LLM research, January 2023 to June 2026, organized by theme and by year. Entries are curated highlights, not the full corpus; the full classification is Table 1.

by storing only the nonzero ternary positions [18, 19]. Packing choices for ternary weights trade a fraction of a bit against kernel speed [20], [21]. A consistent effective-bits accounting, defined as the code width plus the per-weight amortized cost of scales, masks, and codebooks, is therefore a prerequisite for comparing methods; Section 8 adopts one.

The benefit of a small integer code width is memory and bandwidth. A 2B-parameter model occupies four to five GB in BF16 and about 0.4 GB with ternary weights in a non-embedding accounting, and because the decode step is memory-bound, it speeds up in roughly the same proportion [9]; Section 10.4 measures the fuller resident-memory figure once the embedding and runtime buffers are counted too. The second benefit is arithmetic. With each weight restricted to -1, 0, or +1, the per-weight multiplication becomes a conditional add or subtract, so the inner product reduces to a masked accumulation [10].

2.2 The binary neural network lineage

Training networks with binary weights predates LLMs. BinaryConnect binarizes weights in the forward and backward pass while retaining a full-precision “shadow” copy in which gradients accumulate, and it observes a regularizing effect [22]. XNOR-Net binarizes both weights and activations so that a convolution becomes a bitwise XNOR followed by a population count, and it introduces per-filter scaling factors to reduce the approximation error [23]. The transfer to language models proved harder. BinaryBERT finds the loss landscape of a directly binarized BERT too rugged to optimize and instead initializes the model by splitting a trained half-width ternary network [24]. BiBERT identifies information loss in the forward pass and gradient-direction mismatch in the backward pass as the two failure modes, and addresses them with an information-preserving attention operator and a direction-matching distillation loss [25]. Three lessons carried into the LLM era: a full-precision shadow weight is what makes gradient-based training of a discrete variable work; ternary is markedly easier to optimize than binary; and distillation from a full-precision teacher is often what closes the last accuracy gap.

2.3 The straight-through estimator and its pathologies

The quantizer, which maps a full-precision weight to its quantized code, is piecewise constant, so its gradient is zero almost everywhere and training cannot back-propagate through it. The straight-through estimator (STE) replaces that gradient with the identity, often clipped to the quantizer’s input range, so that the forward pass sees the quantized value while the backward pass updates the shadow weight as if the quantizer were absent [26] (Fig. 3). Practically all native 1-bit training uses

a scaled STE. BitNet’s **BitLinear** layer centers the weights and divides by their mean absolute value before applying the sign function, quantizes activations to INT8 per token, and applies the STE through both [6, 7]. BitNet b1.58 uses an absolute-mean ternary rule [8], and small-model studies find the median a more outlier-robust divisor [27].

The STE is biased, and the bias manifests as slow convergence and instability that worsen as the bit-width decreases. Recent work treats the estimator itself as the object of study. CAGE adds a curvature-aware correction term derived from a multi-objective view of quantization-aware training (QAT) and proves convergence in the smooth non-convex setting [28]. QuEST fits the weight and activation distributions with a Hadamard normalization and an MSE-optimal grid, and it introduces a trust gradient that minimizes the error between the quantized-state gradient and the true gradient, reporting stable training down to one-bit weights and activations [29]. A further line augments the STE with zeroth-order information [30]. An orthogonal alternative removes the STE entirely: Direct Quantized Training keeps only low-precision weights and uses stochastic rounding to carry information between steps, eliminating the shadow copy and its memory cost [31]. For extreme fine-tuning specifically, PV-Tuning shows that the STE assumption is suboptimal and replaces it with a representation-agnostic coordinate-descent procedure [32].

2.4 Transformer-specific obstacles

Two properties of Transformers make sub-two-bit quantization harder than the convolutional case.

The first is activation outliers. A small number of channels in the attention output and in the feed-forward down-projection carry values one to two orders of magnitude larger than the rest. Quantizing weights to roughly one bit sharply increases sensitivity to these outliers, so most 1-bit-weight methods keep activations at INT8 [8]. Two routes go lower. The first is a hybrid scheme: BitNet a4.8 uses INT4 for the attention and feed-forward inputs but keeps 8 bits for the outlier-heavy intermediate states, which it sparsifies first, reaching an effective W1.58A4 configuration with only 55% of parameters active and a three-bit KV cache [12]. The second is rotation: the **H-BitLinear** layer of BitNet v2 applies an online Hadamard transform that spreads outlier energy across coordinates before quantization, which enables native four-bit activations [13]. Learnable and data-adaptive rotations generalize the fixed Hadamard [33], [34, 35], and the same idea drives several post-training routes toward one-bit weights.

The second obstacle is the placement of normalization relative to the quantized projection, which affects stability. **BitLinear** places a LayerNorm or SubLN before quantization, and inserting an additional RMSNorm has been reported to stabilize fine-tuning to 1.58 bits [36]. Continual schemes that begin in full precision and transition to 1.58-bit training part-way through pre-training must manage loss spikes at the switch; both retaining the optimizer state and phasing in the quantization strength mitigate them [37].

2.5 Scaling behavior

Whether the full-precision parity of BitNet b1.58 is a genuine scaling property or an artifact of limited training-token budgets is contested, and it is central to the open problems of Section 11. Precision-aware scaling laws model low-precision training as a reduction of the effective parameter count and predict that post-training quantization degrades further as the amount of pre-training data increases [4, 5]. Against this, native ternary suites show ternary models overtaking quantized and even full-precision models on a bits-for-bits basis beyond about 1B parameters, with gains that grow when training data rather than parameter count is scaled [1, 20]. ParetoQ unifies one-bit to four-bit QAT in a single framework and reports a learning transition between two and three bits, below which the representations reorganize substantially; ternary, two-bit, and three-bit quantization occupy a comparable size-accuracy frontier that generally beats four-bit and binary [38]. Theory has begun to catch up: a kernel-limit analysis establishes a scaling law for 1-bit networks [39], and precision-expressivity trade-offs are being formalized [40, 41].

2.6 Evaluation metrics

The corpus reports, with varying completeness, the following quantities. On the accuracy side: perplexity on WikiText-2 and C4; zero-shot accuracy on ARC-Easy, ARC-Challenge, HellaSwag, WinoGrande, PIQA, OpenBookQA, and BoolQ, usually reported as an average; and, for instruction-tuned models at the 2B scale, MMLU, GSM8K, and HumanEval or HumanEval+. On the systems side: decode throughput in tokens per second and time per output token; prefill latency and time to first token; peak resident memory; and energy per token or per inference. Cross-paper comparison is complicated by different harness versions, prompt formats, and few-shot counts, and, on the efficiency side, by different hardware, thread counts, and baselines, such as FP16 against INT8, or a general inference framework against a custom kernel. Section 8 states the effective-bits convention and the axes used to place these results on one scale, and Section 10 reports independently measured values for a subset.

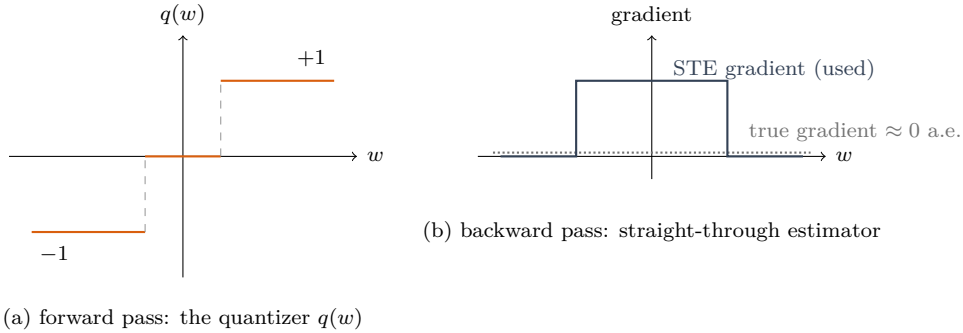


Figure 3: The straight-through estimator. (a) The forward-pass quantizer, here the ternary rule with thresholds at ± 0.5 of the normalized weight w , is piecewise constant with zero true gradient almost everywhere. (b) The backward pass instead propagates a clipped-identity gradient (dark), letting the full-precision shadow weight update despite the quantizer’s own zero gradient.

3 Scope and Corpus

3.1 Scope and eligibility

A work is in scope if it primarily concerns binary or ternary language models, or a method, kernel, or hardware design that explicitly targets them. The bit-width gate is on the **nominal or targeted** weight precision: a work is eligible if it aims at binary weights, ternary weights, or a partial or salient-split scheme built on a sub-two-bit backbone. The gate is deliberately not on the effective bits per weight of Section 3.3, because that figure, which folds in the amortized cost of scales, bitmaps, and codebooks, is itself one of the survey’s findings: several eligible works turn out to carry two to four effective bits once that cost is counted (BiLLM at about 2.88, STBLLM at about 4.13), and re-stating those figures on a common scale is the point of Table 2, not a reason to exclude the works. We include work first made public between January 2023 and June 2026 that makes an empirical or theoretical contribution and for which a freely downloadable full text exists; a local copy of every included work is archived. A unified study that sweeps a wider bit-width range, such as a one-to-four-bit quantization-aware-training comparison, is included for its binary and ternary results and cited only for those. Work whose method targets only three bits or more, or that has no language-model component, is cited only where it bears directly on the binary and ternary case; systems, analysis, and survey works that define no model of their own appear in the tables with the weight, paradigm, and component axes marked not applicable rather than forced into a category. Foundational work on binary networks and the straight-through estimator from 2013 to 2022 is discussed for context and marked separately from the in-window corpus.

The open-access restriction is deliberate. It lets any reader re-check every technical claim without a paywall, and it lets the re-evaluation of Section 10 run on the exact released artifacts. Its cost is the possible omission of paywalled work with no preprint. One work met every other criterion but had no obtainable open-access copy and was set aside; its abstract, together with those of the paywalled records the search surfaced, was checked against every principal conclusion of the paper, and none would change.

3.2 How the corpus was assembled

The search ran as four passes, all falling between 3 and 5 September 2026, and covers work public on or before 30 June 2026; the final of the four, a raw arXiv API re-harvest against the full query set, closed the window on 5 September. The primary source was arXiv, queried through its API over cs.CL, cs.LG, cs.AI, and cs.AR with a Boolean query over bit-width terms (“1-bit”, “1.58-bit”, “ternary”, “binariz*”, “sub-2-bit”), model terms, and technique terms (“BitNet”, “BitLinear”, “quantization-aware training”, “matmul-free”, and others), plus fifteen exact-name queries for known systems. This was supplemented by two iterations of backward and forward citation chasing on eight seed papers through the Semantic Scholar citation graph, by targeted web searches for named methods, and by monitoring Hugging Face Papers and official model and framework release logs. It was not an exhaustive multi-database search; for a fast-moving subfield that publishes predominantly as preprints, arXiv plus citation chasing captures the frontier, and the open-access criterion already excludes venue-only paywalled work.

Candidate records were de-duplicated, screened on title and abstract, then screened in full text, with the open-access copy retrieved in the same pass and a failure to retrieve treated as an exclusion. Screening removed off-topic records (most often a primary contribution outside language modeling, such as error-correcting-code transformers or high-energy-physics classification, with a 1-bit model as an incidental example), records whose primary method spanned a bit-width range above two bits, and records made public after the cutoff, which fall outside the surveyed corpus.

The flow below is the funnel for the final consolidation pass, a raw arXiv API re-harvest with no summarizer in the loop, together with the cumulative outcome across all four search passes. The 135 in-window primary studies are not spread evenly over the four search years; Fig. 4 gives the year-by-year count.

Corpus-assembly funnel. The final consolidation pass (a raw arXiv API re-harvest with no summarizer in the loop), shown with the cumulative outcome across all four search passes.

stage	count
Unique arXiv records, final raw re-harvest (core query + 15 name queries, cs.CL/LG/AI/AR, Jan 2023 to Jun 2026)	109
Already in the corpus from earlier passes	69
New, sent to full-text screening	40
New records included at full text	15
New records excluded: outside language modeling (EC-1)	16
New records excluded: bit-width range above two bits (EC-2)	7
New records excluded: duplicate or re-confirmed prior exclusion	2
Cumulative full-text eligible, all passes	136
Excluded, no obtainable open-access copy (IC-5): PT-BitNet	1
Delayed 20% re-screen (28 sampled, 26 confirmed): removed on second reading	1
Delayed 20% re-screen: re-included after an open-access copy was located	1
In-window primary studies included	135
Pre-2023 foundational works, classified in Table 1 for context	5
Rows in Table 1	140
Non-paper artifacts kept as dated snapshots (two model cards, one framework repository)	3
Post-cutoff records excluded from this version of the corpus	~5

The re-screen removed one W4A4 scaling-law study that fails the bit-width criterion and recovered one native-ternary encoding study once its open-access copy was found; both revisions are in the counts above. This process departs from the frozen protocol (`survey-protocol.md`) in three stated

ways: screening was by one reviewer rather than two, the 20% re-screen followed the first pass by a few days rather than the two-week washout the protocol specifies (Section 3.4), and there was no exhaustive multi-database harvest or public pre-registration.

3.3 Effective-bits accounting and evidence basis

Reported bit-widths in this literature are not comparable as stated, because partial and salient-split schemes count only the code and omit the group scales, bitmaps, and codebooks they also store. Throughout, the **effective bits per weight** is the nominal code width plus the per-weight amortized cost of those structures; where a source’s own strict accounting or an independent one exists, that figure is used and the basis is noted per row in Table 2.

Rather than collapse source quality into one graded number, each work carries an **evidence basis** made of two facts that a reader can check directly in Table 1’s “Evidence basis” column. The first is the **venue**: a peer-reviewed conference or journal, or “preprint” if the only version is on arXiv or under review. The second is whether the **central result has an independent check**, meaning our own reproduction in Section 10 or a separate open study that reaches the same finding; most works have neither. Artifact availability, whether an open model, code, or weights exist, is recorded as a free-text note per row where known, since it could not be verified uniformly across all 140 works.

For synthesis we use a single derived split. A source is **corroborated** if it is peer reviewed or has an independent check; otherwise it is a **single unreproduced preprint**. Forty-eight of the 140 works are corroborated. This split is about the reliability of a source and its artifacts, not about whether its headline claim is settled: a corroborated source can still make a disputed claim. The parity claim of the native line is the clearest example. Its sources are all corroborated, BitNet b1.58, BitNet b1.58 2B4T, both Spectra suites, and ParetoQ, which is also at NeurIPS, and they corroborate one another; our re-run of the released BitNet b1.58 2B4T checkpoint in Section 10 confirms that it loads, runs, and is competitive on a common suite. None of that tests parity at frontier scale, which remains an open problem (Section 11.1). Any statement in the synthesis that rests only on single unreproduced preprints is marked as such.

3.4 Limitations

The corpus is preprint-heavy, because the field publishes faster than journal review cycles; this is mitigated by the corroborated-versus-single-preprint split of Section 3.3, by anchoring to peer-reviewed versions where they exist (BitNet in JMLR; OneBit, BinaryMoS, PV-Tuning, ParetoQ, and LittleBit at NeurIPS; ARB-LLM and PT2-LLM at ICLR; BiLLM and LC-QAT at ICML; ICQuant at COLM), and by the re-evaluation of Section 10. The search is arXiv-primary rather than exhaustive, so open-access work in society and publisher venues with no preprint may be missed. Screening was done by one reviewer; a re-screen of a random 20% sample (28 studies) confirmed 26 of the 28 inclusion decisions (93% raw intra-rater agreement), and the two revisions it prompted are folded into the counts above. That re-screen followed the first pass by a few days rather than the two-week washout the protocol specified, so it measures the stability of the eligibility criteria more than blind intra-rater agreement. One automated citation list retrieved during the search contained unverifiable entries, which were discarded; every archived record was confirmed against its primary source. Finally, some relevant work will appear between the cutoff and publication; that is an inherent limitation of any fixed-cutoff survey.

4 A Taxonomy of Binary and Ternary LLMs

A survey of this area must impose structure, because the phrase “1-bit LLM” is used for at least three different research programs: training a model under a sub-two-bit constraint from scratch, compressing a trained model down toward one bit, and building hardware that executes ternary weights efficiently. We organize the field along five orthogonal axes (Table 1). Four of them, weight representation, training paradigm, quantized components, and systems stack, partition the corpus, so

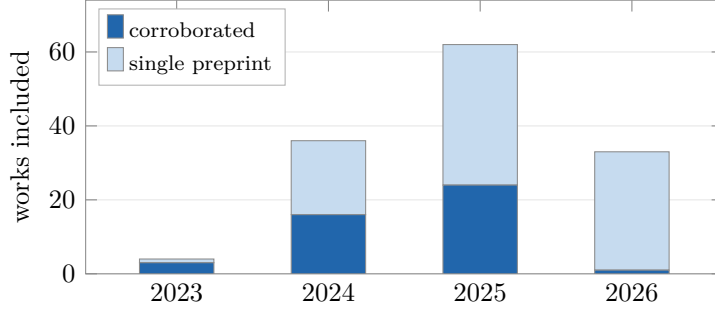


Figure 4: In-window corpus by year of first publication, stacked by evidence basis: corroborated means peer reviewed or with an independent check; the rest are single unreproduced preprints. 135 primary studies, January 2023 to June 2026 (2026 is a partial year, through the June cutoff). The falling corroborated share in recent years partly reflects review lag, not only falling scrutiny.

their category counts sum to the full 140 works; the fifth, optimization mechanism, is a non-exclusive tally, since one work can combine mechanisms. The axes are deliberately independent, so that a work’s position on one does not determine its position on another, and the most influential works are those that advance the state of the art on several axes at once.

4.1 Axis 1: weight representation

The first choice is the value set. Binary weights, drawn from $\{-1, +1\}$, give the largest nominal compression and the simplest kernel, but they cannot express the absence of a connection. Thirty-four surveyed works are primarily binary, and most of them are post-training or fine-tuning methods that apply the binary constraint to an already-trained model [2, 42, 43], [44]. Ternary weights, drawn from $\{-1, 0, +1\}$, add a zero state. This both restores sparsity, since roughly 40% to 60% of weights go to zero in practice, and makes optimization markedly easier in experiments. Sixty-two works are ternary, and this is where native from-scratch training has succeeded [8], [1, 11]; an early instance, predating BitNet, demonstrated the first ternary and binary sequence-to-sequence Transformers for summarization and machine translation [45]. Mixed or partial schemes, of which there are 21, keep a small set of salient columns, or a low-rank residual branch, in higher precision. Examples are the roughly 1.08 effective bits of BiLLM [2], the tunable fraction of PB-LLM [3], the dedicated high-precision expert branch of pQuant [46], and the saliency-graph-guided scaling of SAGE-PTQ [47]; each trades a fraction of a bit for a substantial accuracy recovery. A small but distinct line uses complex-valued weights in $\{\pm 1, \pm i\}$, which costs two bits of storage but supports multiplication-free accumulation [48, 49]. The remaining 23 works, kernels, accelerators, theory, and surveys, define no weight format of their own and are counted separately in Fig. 5.

4.2 Axis 2: training paradigm

The second choice is when the low-bit constraint enters. Native from-scratch training, in 27 works, places a quantizer in BitLinear from the first optimization step; this is the defining paradigm of the survey and the one that has demonstrated full-precision parity at scale [6, 8, 9], [1]. Continual quantization-aware pre-training, in 2 works, begins in full precision and transitions to 1.58-bit training part-way through, which reduces total cost but introduces a loss spike at the switch that must be managed [37, 50]. Quantization-aware fine-tuning, in 37 works, inserts the quantizer into a fine-tuning or short continued-training loop over a pretrained full-precision model [43, 51], [52]. Post-training quantization, in 30 works, uses only a calibration set and no gradient updates [2, 42], [53–56]. At a fixed bit-width, the accuracy ordering is generally native, then quantization-aware training, then post-training quantization, and the cost ordering is the reverse; the practical question each paper answers is where on that trade-off a given deployment sits. The remaining 44 works, systems papers,

analyses, and surveys with no training procedure of their own, carry no value on this axis and are counted separately in Fig. 5.

4.3 Axis 3: quantized components

The third axis is what is quantized beyond the weight matrices. Weight-only methods, in 80 works, leave activations at FP16 or INT8 and capture the memory-bandwidth benefit without changing the precision of the compute path. Weight-and-activation methods, in 30 works, quantize activations as well, typically starting at INT8 and pushing toward INT4; this is where the outlier problem of Section 2.4 becomes acute and where hybrid and rotation methods are needed [12, 13, 34], [35]. Two works quantize only activations, through sparsification. A small frontier additionally quantizes the KV cache to three bits [12], [13] and couples activation or weight sparsification to quantization [21, 57–59]. The other 28 works have no model of their own and carry no value on this axis.

4.4 Axis 4: optimization mechanism

The fourth axis is how the discrete constraint is optimized. The default is a scaled straight-through estimator, with absolute-mean or absolute-max normalization, sometimes the median, applied through both weights and activations; roughly 20 works use this in essentially unmodified form. A group of about six works treats the estimator as the object of study, through curvature-aware corrections [28], a trust-region gradient [29], or zeroth-order augmentation [30]. Distillation-driven methods, roughly 13, close the accuracy gap to a full-precision teacher through autoregressive, logit, attention, or feature losses [44], [51, 52, 60]. Reconstruction and calibration methods, roughly 20, minimize a layer-output error over calibration data, often with salient-weight handling and Hessian information, and are the workhorse of post-training quantization toward one bit [2, 42], [53, 55, 56, 61], [47, 62]. Rotation and incoherence methods, roughly seven, apply a structured orthogonal change of basis before quantization to spread outlier energy; examples are the fixed Hadamard transform of BitNet v2 [13] and its learned or data-adaptive generalizations [33, 34], [63]. Decomposition and factorization methods, roughly ten, represent the weight as a product or sum of low-rank or binary factors [43], [19, 53, 64]. Stochastic rounding without any STE is a minority alternative that removes the shadow-weight memory cost [31]. Finally, sparsity-coupled optimization ties an N:M or top-K mask to the quantizer [18, 21, 58], [57].

4.5 Axis 5: systems stack

The fifth axis records how far down the stack a work reaches, and it partitions the 140 works into four exclusive bins: algorithm only, contributes a kernel, is a hardware design, or not applicable. Ninety-five works are algorithm only. Twenty-five contribute a kernel: the I2_S and TL1 or TL2 lookup kernels of bitnet.cpp [65, 66], the fused ternary CPU kernels of FairyFuse [67], the matrix-vector engine of RSR-core [68], the in-register SIMD lookup of T-SAR [69], the CPU table-lookup renaissance of T-MAC [70], the 2-bit CPU and GPU microkernels of [71], and the GPU kernels of QuEST and Spectra 1.1 [20, 29]. This bin includes the five works that reach all the way to a hardware design as well, namely BitNet b1.58, BitNet b1.58 2B4T, MatMul-free LM, Sparse-BitNet, and BMT-BAT [8], [9, 11, 58, 72]; they are counted once here, not separately, and closing the gap between them and the rest of the field is one of the open problems of Section 11. Sixteen works are a hardware design with no algorithmic contribution of their own: lookup-table ASICs [73, 74], the design-space-exploration LUT-accelerator framework of [75], compute-in-ROM designs [76, 77], processing-in-memory designs [78], [79], the fault-tolerant compute-in-memory design of ReTern [80], and edge-FPGA accelerators [81], [82–84], together with the silicon-validated ternary accelerator VitaLLM [85]. The four remaining works are surveys with no systems contribution of their own.

4.6 A secondary classification by modality

Seventy-two works target text-decoder LLMs, and this is where the technical core of the survey sits. A fast-growing minority applies the same machinery elsewhere: to 1-bit text-embedding models [86, 87], [88]; to speech, in the form of 1.58-bit text-to-speech [89] and one-bit automatic speech recognition [90]; to vision-language and vision-language-action models [91], [92, 93]; to ternary mixture-of-experts [94]; to state-space models [60], [95]; and, further afield, to reinforcement-learning agents [96] and autonomous-system trajectory prediction [97]. These branches are surveyed in Section 9.

4.7 Reading the taxonomy

The value of the taxonomy is comparative. It shows that native training has concentrated on ternary weights while binary has remained largely a post-training target (Fig. 6); that activation quantization and the systems stack are where the field is currently moving fastest; and that the most influential works, namely the BitNet line, Spectra, MatMul-free LM, ParetoQ, and Sparse-BitNet, are precisely those that advance the state of the art on three or more axes at once. Table 1 gives the full classification, and Fig. 5 depicts it as a tree. Cut by year instead of pooled, binary’s share of new work peaked at a third in 2024 and had fallen to under a tenth by 2026, while ternary stayed the largest or second-largest category throughout (Fig. 7).

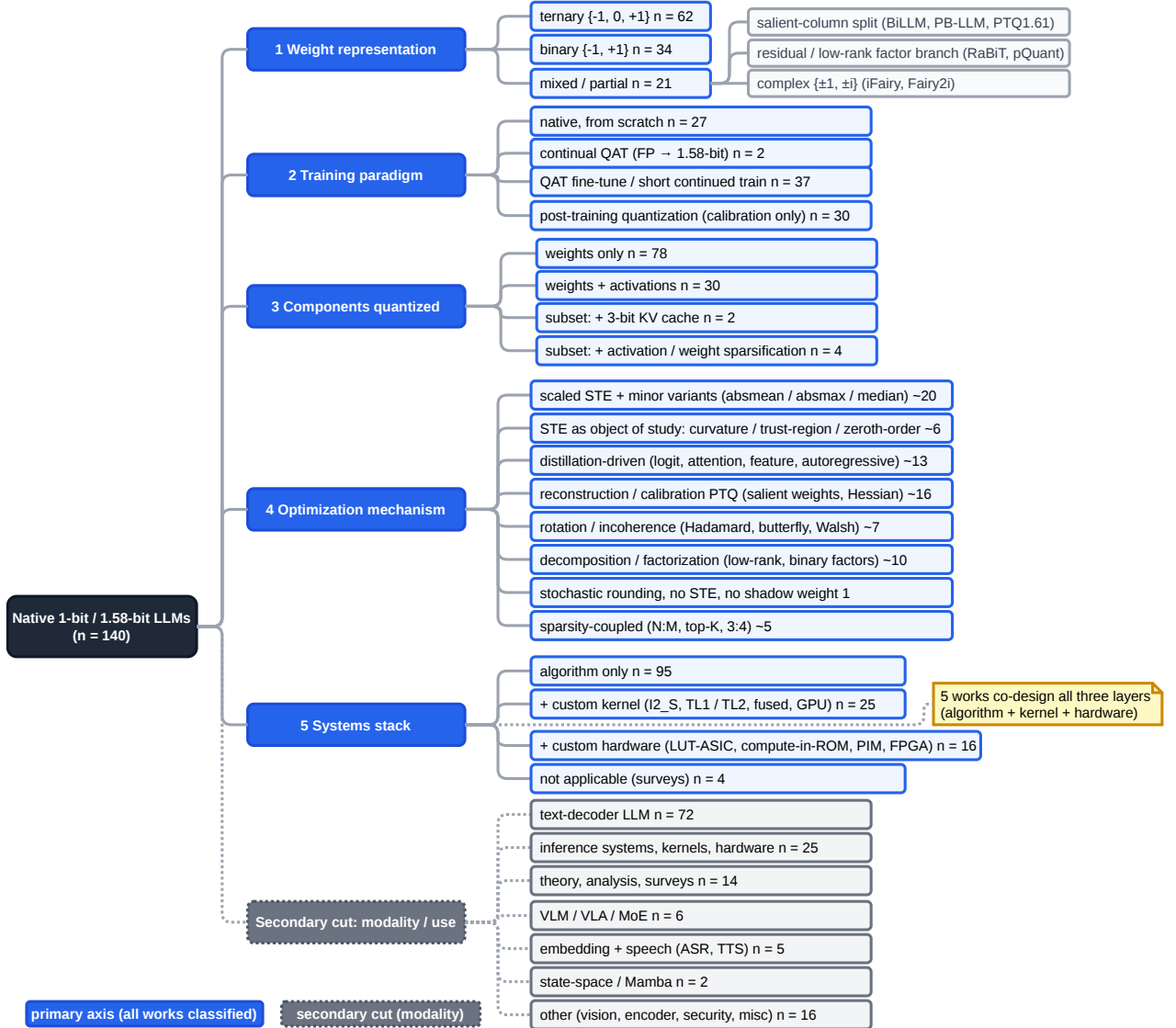


Figure 5: Five-axis taxonomy of the surveyed corpus (135 in-window works plus five pre-2023 foundational works, listed in Table 1). Axes 1, 2, 3, and 5 partition all 140 works; axis 4 (optimization mechanism) is a non-exclusive tally and does not sum to 140. The modality cut at the bottom is a secondary partition.

	native	cont.QAT	QAT-FT	PTQ
binary	6	1	9	16
ternary	18	1	16	4
mixed-partial	3	0	11	7

Figure 6: Weight representation against training paradigm, restricted to the three weight types and four paradigms that carry a training procedure (systems, analysis, and survey works are excluded). Native training concentrates on ternary (18 of 27 native works); post-training quantization concentrates on binary (16 of 27 PTQ works).

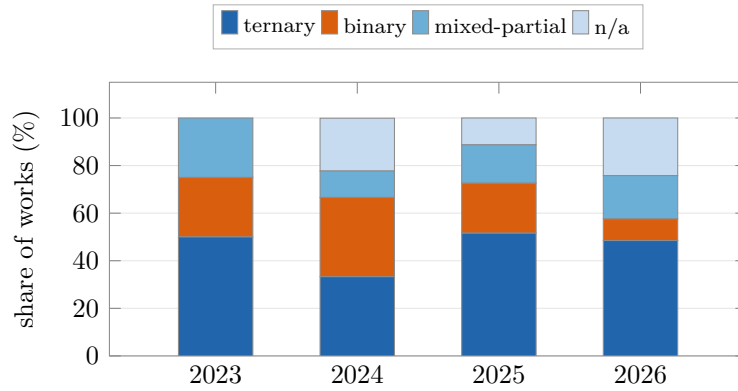


Figure 7: Weight-representation share by year, each bar normalized to 100% of that year’s works. Binary peaked at a third of new work in 2024 and fell to under a tenth by 2026; ternary stayed largest or second-largest throughout. 2023 (four works) and 2026 (partial year) should be read cautiously.

Table 1: Classification of the surveyed corpus on the taxonomy axes. Weight type, training paradigm, and quantized components each partition the corpus; their counts sum to 140. *Evidence basis* gives venue (*preprint* if not peer reviewed) and independent-check status (*ext. repro* for an external one); it is not a graded score. In the cells: *n/a* = axis does not apply; *NA* = not applicable (activation-bit/systems-layer); *NR* = not reported; *var* = a range rather than one value.

Reference	Year	Weight type	Training paradigm	Components quantized	Act. bits	Systems layer	Modality	Evidence basis
Bengio et al. [26]	2013	binary	background	W	NA	algorithm	background	preprint
Courbariaux et al. [22]	2015	binary	native-scratch	W	32	algorithm	vision	NeurIPS
Rastegari et al. [23]	2016	binary	native-scratch	W+A	1	algo+kernel	vision	ECCV
Bai et al. [24]	2021	binary	QAT-finetune	W+A	8	algorithm	text-encoder	ACL
Qin et al. [25]	2022	binary	QAT-finetune	W+A+emb	1	algorithm	text-encoder	ICLR
Kim et al. [98]	2023	ternary	QAT-finetune	W	16	algorithm	text-decoder	NeurIPS
Wang et al. [6]	2023	binary	native-scratch	W+A	8	algo+kernel	text-decoder	JMLR
Shang et al. [3]	2023	mixed-partial	PTQ	W	16	algorithm	text-decoder	ICLR
Ansar et al. [99]	2024	binary	native-scratch	W	16	algorithm	text-encoder	IEEE-TSUSC
Chen et al. [100]	2024	binary	PTQ	W	16	algorithm	text-decoder	ICLR
Chen et al. [101]	2024	mixed-partial	QAT-finetune	W	16	algorithm	text-decoder	preprint
Daliri et al. [39]	2024	binary	analysis	W	NA	algorithm	theory	preprint
Dong et al. [18]	2024	binary	PTQ	W	16	algo+kernel	text-decoder	ICLR25
Du et al. [51]	2024	mixed-partial	QAT-finetune	W	16	algorithm	text-decoder	ACL
Das et al. [102]	2024	n/a	analysis	W	8	NA	security	preprint
Gong et al. [14]	2024	n/a	survey	NA	NA	NA	survey	Neural-Networks
Huang et al. [2]	2024	binary	PTQ	W	16	algorithm	text-decoder	ICML
Jo et al. [103]	2024	binary	QAT-finetune	W	16	algorithm	text-decoder	NeurIPS
Kaushal et al. [1]	2024	ternary	native-scratch	W	16	algorithm	text-decoder	preprint + ext. repro
Zhao et al. [31]	2024	ternary	native-scratch	W	8	algorithm	text-decoder	ACML
Kumar et al. [5]	2024	n/a	analysis	W+A	var	algorithm	theory	ICLR25
Li et al. [42]	2024	binary	PTQ	W	16	algorithm	text-decoder	ICLR25
Ma et al. [8]	2024	ternary	native-scratch	W+A	8	algo+kernel+hw	text-decoder	preprint + ext. repro
Ma et al. [44]	2024	binary	native-scratch	W	16	algorithm	text-decoder	preprint
Malekar et al. [10]	2024	n/a	analysis	NA	NA	hardware	analysis	preprint
Malinovskii et al. [32]	2024	mixed-partial	QAT-finetune	W	16	algorithm	text-decoder	NeurIPS(oral)
Nielsen and Schneider-Kamp [27]	2024	ternary	native-scratch	W+A	8	algorithm	small-LM/ vision	preprint
Nielsen et al. [104]	2024	ternary	native-scratch	W	16	algorithm	MLP/GNN/ encoder	preprint

Reference	Year	Weight type	Training paradigm	Components quantized	Act. bits	Systems layer	Modality	Evidence basis
Ouyang et al. [4]	2024	n/a	analysis	W	var	algorithm	theory	preprint
Sundaram and Iyer [92]	2024	ternary	native-scratch	W+A	8	algorithm	VLM	preprint
Tang et al. [60]	2024	binary	native-scratch	W	16	algorithm	SSM	TMLR
Wang et al. [65]	2024	ternary	systems	NA	NA	kernel+hw	systems	preprint
Wang et al. [12]	2024	ternary	native-scratch	W+A+KV+ sparsify	4	algo+kernel	text-decoder	preprint
Wang et al. [57]	2024	ternary	native-scratch	A-sparsify	8	algorithm	text-decoder	preprint
Xu et al. [43]	2024	binary	QAT-finetune	W	16	algorithm	text-decoder	NeurIPS
Zhu et al. [11]	2024	ternary	native-scratch	W+A	8	algo+kernel+ hw	text-decoder	preprint
Dehghankar et al. [105]	2024	n/a	systems	W	NA	algorithm	systems	ICML25
Chen et al. [87]	2024	ternary	QAT-finetune	W	16	algorithm	embedding	preprint
Fan and Tao [106]	2024	n/a	analysis	NA	NA	NA	analysis- security	preprint
Kawamura et al. [89]	2025	ternary	QAT-finetune	W	16	algorithm	TTS	Interspeech
Ye et al. [107]	2025	binary	PTQ	W+A	6	algorithm	text-decoder	preprint
Dung et al. [108]	2025	binary	PTQ	W	16	algorithm	text-decoder	preprint
Huang et al. [109]	2025	ternary	QAT-finetune	W	16	algorithm	text-decoder	preprint
Vaidhya et al. [20]	2025	ternary	native-scratch	W	16	algo+kernel	text-decoder	preprint + ext. re- pro
Ardakani et al. [110]	2025	ternary	systems	W+A	8	kernel+hw	edge-systems	CVPRW
Lee et al. [19]	2025	binary	QAT-finetune	W	16	algorithm	text-decoder	NeurIPS
Liu et al. [15]	2025	n/a	survey	NA	NA	NA	survey	preprint
Ma et al. [9]	2025	ternary	native-scratch	W+A	8	algo+kernel+ hw	text-decoder	preprint + Sec.10
Wang et al. [94]	2025	ternary	native-scratch	W-experts	16	algorithm	VLM/MoE	preprint
Nielsen et al. [37]	2025	ternary	continual-QAT	W	16	algorithm	text-decoder	preprint
Yan et al. [54]	2025	ternary	PTQ	W	16	algorithm	text-decoder	ICLR26
Xiao et al. [53]	2025	ternary	PTQ	W	16	algorithm	text-decoder	preprint
Liu et al. [38]	2025	mixed- partial	QAT-finetune	W	16	algorithm	theory/text- decoder	NeurIPS + ext. repro
Panferov et al. [29]	2025	binary	QAT-finetune	W+A	1	algo+kernel	text-decoder	preprint
Oh et al. [69]	2025	ternary	systems	NA	NA	kernel+hw	systems	DATE26
Tabesh et al. [28]	2025	mixed- partial	QAT-finetune	W+A	3	algorithm	text-decoder	MLSys26
Qiao et al. [81]	2025	ternary	systems	NA	NA	hardware	systems	preprint
Yin et al. [83]	2025	ternary	systems	NA	NA	hardware	systems	preprint
Tu et al. [50]	2025	binary	continual-QAT	W	16	algorithm	text-decoder	preprint
Wang et al. [7]	2025	mixed- partial	native-scratch	W+A	8	algo+kernel	text-decoder	JMLR

Reference	Year	Weight type	Training paradigm	Components quantized	Act. bits	Systems layer	Modality	Evidence basis
Wang et al. [66]	2025	ternary	systems	NA	NA	kernel	systems	ACL
Wang et al. [13]	2025	ternary	native-scratch	W+A+KV	4	algo+kernel	text-decoder	preprint
Wang et al. [91]	2025	ternary	native-scratch	W+A	8	algorithm	VLA	preprint
Song et al. [111]	2025	binary	PTQ	W+A	1	algorithm	text-decoder	preprint
Boza and Macko [112]	2025	binary	PTQ	W	16	algorithm	text-decoder	TMLR
Wu et al. [52]	2025	ternary	QAT-finetune	W+A	8	algorithm	text-decoder	preprint
Zhang et al. [76]	2025	ternary	systems	NA	NA	hardware	systems	ASP-DAC
Gu et al. [113]	2025	binary	PTQ	W	16	algorithm	text-decoder	preprint
Zhang and Huang [114]	2025	ternary	systems	NA	NA	hardware	systems	preprint
Frantar et al. [115]	2025	n/a	analysis	W	var	algorithm	theory	preprint
Chen et al. [116]	2025	binary	PTQ	W	16	algorithm	text-decoder	preprint
Yang and Aamodt [30]	2025	mixed-partial	QAT-finetune	W	16	algorithm	text-decoder	NeurIPS
Chen et al. [117]	2025	ternary	QAT-finetune	W	16	algorithm	text-decoder	preprint
Hao et al. [16]	2025	n/a	survey	W+A	var	algorithm	survey	IEEE-TPAMI
Tran and Nguyen [118]	2025	binary	PTQ	W	16	algorithm	text-decoder	ICLR
Malekar et al. [78]	2025	ternary	systems	NA	NA	hardware	systems	preprint
Yan et al. [119]	2025	binary	PTQ	W	16	algo+kernel	text-decoder	preprint
Zhao et al. [120]	2025	mixed-partial	PTQ	W	16	algorithm	text-decoder	ACL
Xia et al. [121]	2025	binary	PTQ	W	16	algorithm	text-decoder	ECAI
Huang et al. [74]	2025	ternary	systems	NA	NA	hardware	systems	preprint
Crulis et al. [122]	2025	ternary	QAT-finetune	W	16	algorithm	VLM	preprint
Li et al. [90]	2025	binary	QAT-finetune	W	16	algorithm	ASR	Interspeech
Connor et al. [88]	2025	ternary	QAT-finetune	W	16	algorithm	embedding	preprint
Huang et al. [21]	2026	ternary	QAT-finetune	W	16	algo+kernel	text-decoder	preprint
Li et al. [86]	2026	ternary	native-scratch	W	8	algorithm	embedding	preprint
Dade et al. [123]	2026	ternary	systems	NA	NA	kernel	systems	preprint
Savdhariya [17]	2026	ternary	systems	W	NA	algorithm	storage	preprint
Song et al. [124]	2026	mixed-partial	PTQ	W+A	4	algorithm	text-decoder	preprint
Nargund and Shukla [125]	2026	ternary	native-scratch	W	16	algorithm	text-decoder	preprint
Zhang et al. [58]	2026	ternary	native-scratch	W+sparsify	8	algo+kernel+hw	text-decoder	preprint
Zhao et al. [34]	2026	ternary	PTQ	W+A	4	algorithm	text-decoder	preprint
Zuo et al. [67]	2026	ternary	systems	NA	NA	kernel	systems	preprint
Wang et al. [126]	2026	ternary	PTQ	W	16	algorithm	text-decoder	preprint
Chakrabarti et al. [40]	2026	n/a	analysis	W	var	algorithm	theory	preprint
Zhang [41]	2026	n/a	analysis	W	var	algorithm	theory	preprint

Reference	Year	Weight type	Training paradigm	Components quantized	Act. bits	Systems layer	Modality	Evidence basis
Wang et al. [127]	2026	mixed-partial	QAT-finetune	W	16	algorithm	text-decoder	preprint
Pizzo [128]	2026	ternary	native-scratch	W	16	algorithm	text-decoder	preprint
Chong et al. [129]	2026	binary	PTQ	W	16	algorithm	text-decoder	preprint
You et al. [130]	2026	binary	QAT-finetune	W	16	algorithm	text-decoder	preprint
Zagitov et al. [33]	2026	n/a	PTQ	W+A	2	algorithm	text-decoder	preprint
Pavlov [63]	2026	n/a	PTQ	W	2	algorithm	text-decoder	preprint
Zhang et al. [46]	2026	mixed-partial	QAT-finetune	W	16	algorithm	text-decoder	preprint
Wang et al. [64]	2026	mixed-partial	QAT-finetune	W	16	algorithm	text-decoder	ICML
Alimaskina et al. [131]	2026	n/a	analysis	W	2	algorithm	reasoning	preprint
Zhao et al. [35]	2026	binary	PTQ	W+A	6	algorithm	text-decoder	preprint
Xu et al. [132]	2026	n/a	analysis	W	var	algorithm	theory	preprint
Wang et al. [49]	2025	mixed-partial	PTQ	W	16	algorithm	text-decoder	preprint
Maskey et al. [133]	2026	n/a	QAT-finetune	W	16	algorithm	text-decoder	preprint
Li et al. [61]	2025	n/a	PTQ	W	16	algorithm	text-decoder	COLM
Wang et al. [134]	2025	n/a	systems	NA	NA	hardware	security	EuroSys
Shan et al. [73]	2025	ternary	systems	NA	NA	hardware	systems	ASP-DAC
Li et al. [135]	2025	ternary	systems	NA	NA	kernel	systems	MobiSys
Guan et al. [77]	2026	ternary	systems	NA	NA	hardware	systems	preprint
Qiao et al. [82]	2025	ternary	systems	NA	NA	hardware	systems	preprint
Jørgensen [136]	2025	n/a	survey	NA	NA	algorithm	survey	preprint
Aman et al. [137]	2025	mixed-partial	native-scratch	W	16	algorithm	VLM	preprint
Chen et al. [138]	2025	mixed-partial	QAT-finetune	W	16	algorithm	text-decoder	preprint
Zhang et al. [139]	2026	mixed-partial	QAT-finetune	W+A	16	algorithm	text-decoder	preprint
Dehghankar and Asudeh [68]	2026	n/a	systems	NA	NA	kernel	systems	preprint
Grainge et al. [140]	2025	ternary	QAT-finetune	W	16	algorithm	vision	IEEE-RAL
Wang et al. [48]	2025	mixed-partial	native-scratch	W	16	algorithm	text-decoder	preprint
Ganesaraja et al. [95]	2026	ternary	QAT-finetune	W	16	algorithm	SSM	preprint
Zhang et al. [93]	2025	ternary	QAT-finetune	W	16	algorithm	VLM	preprint
Steinmetz et al. [36]	2025	ternary	QAT-finetune	W	16	algorithm	text-decoder	preprint
Ortega et al. [79]	2024	n/a	systems	NA	NA	hardware	systems	preprint
Liu et al. [45]	2023	ternary	QAT-finetune	W+A	8	algorithm	text-seq2seq	preprint

Reference	Year	Weight type	Training paradigm	Components quantized	Act. bits	Systems layer	Modality	Evidence basis
Wei et al. [70]	2024	ternary	systems	NA	NA	kernel	systems	preprint
Ji et al. [72]	2024	binary	QAT-finetune	W+A	NR	algo+kernel+hw	text-decoder	preprint
Xu et al. [55]	2024	mixed-partial	PTQ	W	16	algorithm	text-decoder	preprint
Edalati et al. [56]	2024	binary	PTQ	W	16	algorithm	text-decoder	preprint
Malhotra and Gupta [80]	2025	ternary	systems	W+A	8	hardware	systems	preprint
Xiao et al. [62]	2025	mixed-partial	PTQ	W	16	algorithm	text-decoder	preprint
Georganas et al. [71]	2025	n/a	systems	NA	NA	kernel	systems	preprint
Qi et al. [59]	2025	ternary	systems	A-sparsify	8	algorithm	text-decoder	preprint
Zhang et al. [84]	2025	ternary	systems	NA	NA	hardware	systems	preprint
Sajid et al. [96]	2026	ternary	QAT-finetune	W+A	8	algorithm	RL	preprint
Geens et al. [75]	2026	ternary	systems	NA	NA	hardware	systems	preprint
Lin and Chang [85]	2026	ternary	systems	NA	NA	hardware	systems	preprint
Kang et al. [97]	2026	ternary	QAT-finetune	W	16	algorithm	robotics	preprint
Abdalla et al. [47]	2026	mixed-partial	PTQ	W	16	algorithm	text-decoder	preprint

5 Architectural Design

5.1 The BitLinear primitive

The standard native 1-bit LLM is built from a single substitution: the standard linear layer is replaced by a **BitLinear** layer that normalizes its input, quantizes the weight to a low-bit code with a shared scale, quantizes the activation, and back-propagates through both quantizers with a straight-through estimator [6], [7]. In the original BitNet, the weight rule is a sign function applied after subtracting the mean and dividing by the mean absolute value, and the activation rule is per-token absolute-max quantization to INT8, offset for non-linearities that produce non-negative outputs. BitNet b1.58 changes the weight rule to an absolute-mean ternary quantizer: it divides by the mean absolute value, rounds to the nearest of $\{-1, 0, +1\}$, and clamps. This yields the $\{-1, 0, +1\}$ value set and drives roughly half of the weights to zero [8]. Small-model studies report that using the median in place of the mean in the scale makes the quantizer less sensitive to a few large weights and can improve accuracy at very small scales [27].

A recurring architectural detail is the placement of normalization. **BitLinear** applies its normalization, LayerNorm in BitNet and SubLN in later variants, before quantization, so that the quantizer sees a controlled input distribution. Several works find that an additional RMSNorm inserted at the layer boundary stabilizes the transition to 1.58-bit weights during fine-tuning [36], and BitNet Distillation combines SubLN with a short continued-pretraining phase before task distillation [52].

5.2 Ternary versus binary in practice

The move from binary to ternary is the single largest architectural lever in the native setting. Binary from-scratch training is possible: FBI-LLM demonstrates a fully binary LLM that matches half precision by relying heavily on an autoregressive distillation loss [44], and QuEST reaches stable one-bit weights and activations with a trust-gradient estimator and Hadamard-normalized fitting [29]. In each case, however, more machinery, such as distillation, careful initialization, or a better gradient estimator, is needed to reach the loss that ternary attains directly. The zero state of ternary weights doubles as sparsity, unstructured until a method imposes a pattern on it (Section 5.5), which is why ternary models compose naturally with N:M sparse tensor cores [58] and with packing schemes that store four ternary weights in five bits [21] or approach the 1.6-bit information bound [20].

5.3 Activation quantization

Once weights occupy roughly one bit, activation precision becomes the binding constraint. Most native models keep activations at INT8 [8], [9]. Two routes go lower. BitNet a4.8 uses a hybrid scheme: INT4 activations for the inputs to attention and the feed-forward network, but the outlier-heavy intermediate states, namely the attention output and the feed-forward down-projection input, are first sparsified with a top-K mask and then quantized to INT8. The effective configuration is W1.58A4 with only 55% of parameters active and a three-bit KV cache, and the feed-forward network uses a squared-ReLU gate that produces about 80% activation sparsity [12]. BitNet v2 instead rotates: an online Hadamard transform inside a modified **H-BitLinear** layer spreads outlier energy across coordinates so that a plain INT4 or FP4 quantizer suffices for all activations natively [13]. Post-training work has converged on the same rotation idea with learned or data-adaptive transforms, including butterfly-structured orthogonal processors [33], Kronecker-structured rotations [34], orthogonal-Kronecker transforms with proximal-SVD refinement that reach W1A6 [35], and influence-weighted Walsh rescaling [63]. Fig. 8 draws the three datapaths side by side: the plain **BitLinear** layer, the Hadamard-rotated **H-BitLinear** layer, and the hybrid split-and-sparsify path of BitNet a4.8. The current practical floor for activations alongside roughly one-bit weights is four bits natively; post-training quantization mostly lands at six bits, though TWLA reports four through the same Kronecker-rotation family on a single preprint benchmark [34], not yet corroborated elsewhere. Whether A2 is reachable at all is an open question (Section 11).

5.4 KV cache

The KV cache dominates memory for long contexts. BitNet a4.8 and BitNet v2 both quantize it to three bits with no reported accuracy cost [12], [13]. Dedicated KV-cache quantization for ternary models, using signed-digit lookup schemes and sub-one-bit residual quantization of keys and values, is an active topic in 2026, although much of it falls after the cutoff of this survey.

5.5 Beyond the dense decoder

For mixture-of-experts models, MoTE trains more low-precision experts rather than fewer high-precision ones. Starting from a dense checkpoint, it up-cycles the feed-forward network into one shared full-precision expert and many ternary routed experts, keeping the active-parameter count fixed while eliminating the memory overhead of a full-precision expert pool [94]. Q-Sparse shows that full activation sparsity composes with both BitNet b1.58 and mixture-of-experts, and derives an inference-optimal scaling law for the combination [57].

For state-space models, Bi-Mamba binarizes a Mamba model from scratch with an autoregressive distillation loss and matches half precision at up to 2.7B parameters, opening a low-bit path for linear-complexity architectures [60]; Ternary Mamba applies grouped quantization-aware training at W1.58A16 [95].

For sub-1.58-bit representations, Sparse-BitNet observes that ternary models tolerate N:M structured sparsity far better than full-precision models, since their weights are already about 42% zero, and applies dynamic N:M sparsification jointly with 1.58-bit quantization, backed by custom sparse tensor-core kernels [58]. Sherry regularizes this to a 3:4 pattern that packs to a power-of-two-aligned 1.25 bits [21]. Structural binarization below one bit [18] and latent-factorization schemes that reach about 0.1 bits per weight [19] are the current extreme.

5.6 One-bit attention and matrix-multiplication-free designs

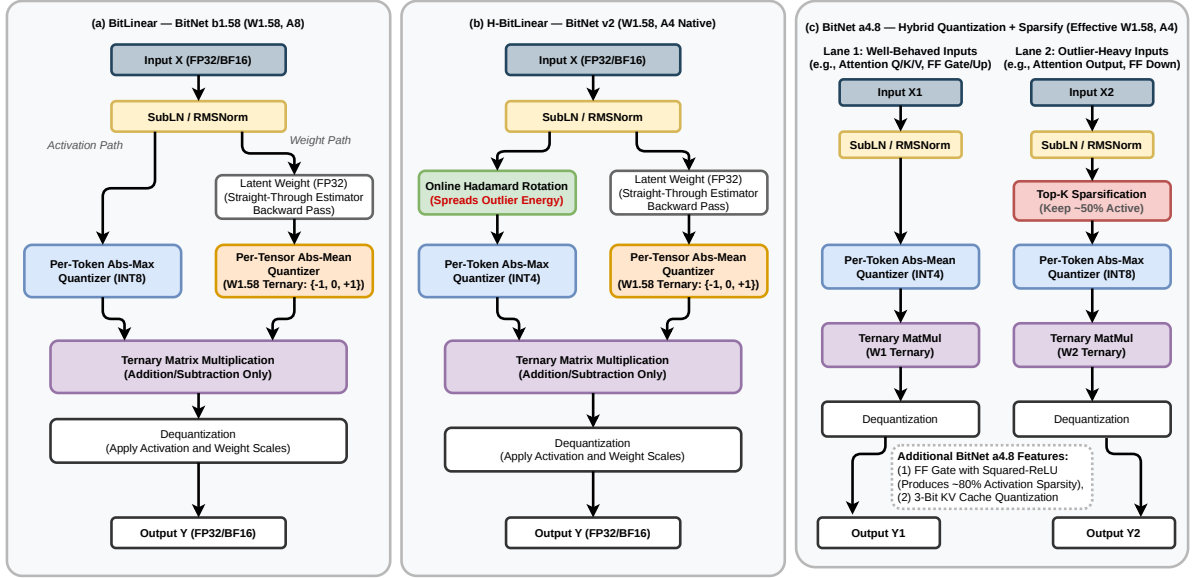
BitNet quantizes only the projection layers and leaves the attention score computation in higher precision, which caps the achievable speedup [10]. MatMul-free LM removes the remaining matrix multiplications by replacing self-attention with a gated linear-attention recurrence over ternary BitLinear layers, so that the whole model consists of additions and element-wise operations; it matches an optimized Transformer baseline up to 2.7B parameters and has been mapped to FPGA and neuromorphic hardware [11]. One-bit query-key attention and fully additive accumulation through complex $\{\pm 1, \pm i\}$ codebooks [48], [49] are further points on this line.

6 Training Methodology and Theory

6.1 The native training loop

Native 1-bit training keeps a full-precision shadow weight in the optimizer and derives the quantized weight from it on the fly in every forward pass. The forward pass uses the quantized weight; the backward pass computes gradients with respect to the quantized weight and applies them to the shadow weight through the straight-through estimator, usually clipped to the active range of the quantizer [6, 8]. At the end of training, the shadow weights are quantized once and discarded, leaving a model that stores about 1.58 bits per weight and, with an appropriate kernel, replaces its matrix multiplications with integer additions. The cost of this scheme is training-time memory: the shadow weights and their optimizer state are full precision, adding the memory footprint of a full-precision copy of the model on top of the quantized forward pass, and the compression saving is realized entirely at inference time. Direct Quantized Training questions the shadow copy itself, keeping only low-precision weights and using stochastic rounding to carry sub-quantum information between steps; it reports that ternary-only training is feasible and that widening to eight-bit weights matches BitNet b1.58 at reduced training memory [31].

Evolutions of BitLinear: From INT8 to INT4 Native to Hybrid Quantization/Sparsification
(All Variants Use Ternary Weights and Add/Sub MatMul)



Note: Across all variants, the Ternary Matrix Multiplication is identical (add/sub only), while activation preparation evolves to handle quantization with increasing efficiency and sparsification.

Figure 8: Three BitLinear datapath variants on a common template. Blue is the activation quantizer, orange the weight quantizer, and purple the ternary matrix multiplication, identical across all three. (a) BitLinear (BitNet b1.58): INT8 activations. (b) H-BitLinear (BitNet v2): an online Hadamard rotation (green) enables INT4 activations. (c) BitNet a4.8: well-behaved inputs go directly to INT4; outlier-heavy inputs are top-K sparsified (red) then quantized to INT8. All three retain a full-precision latent weight trained with a straight-through estimator.

6.2 Recipes

The published recipes share a common shape. There is a two-stage schedule for the learning rate and weight decay, with higher values early, when the quantized landscape is rough, and lower values later; a warmup on the quantization strength; and, for instruction-tuned models, a standard supervised fine-tuning stage followed by direct preference optimization, applied while the weights remain ternary [9]. Data scale matters more than parameter scale for ternary models: the Spectra suites find that TriLM performance improves faster with training tokens than with parameters, and Spectra 1.1 trains to 1.2T tokens on that basis [1], [20]. Continual schemes trade a small amount of accuracy for a large training saving by running most of pre-training in full precision and only the final phase in 1.58-bit; the loss spike at the transition is reduced by retaining the optimizer state across the switch and by phasing in the quantization strength gradually [37, 50].

6.3 The straight-through estimator and its alternatives

The straight-through estimator is biased: the gradient it supplies is not the gradient of the expected quantized loss, and the mismatch grows as the bit-width falls, producing slow convergence and instability. Recent work treats the estimator as a design object, along several lines.

The first line is curvature-aware estimation. CAGE frames quantization-aware training as a multi-objective optimization that simultaneously minimizes the task loss and the quantization error, derives a Pareto-optimality condition, and adds a curvature-dependent correction term to the STE gradient, with a convergence proof in the smooth non-convex setting; it matches the accuracy of a prior W4A4 method at W3A3 [28].

The second line is trust-region estimation. QuEST fits the continuous weight and activation distributions with a Hadamard normalization and an MSE-optimal grid, and then uses a trust gradient that explicitly minimizes the distance between the quantized-state gradient and the unknown true gradient; this is what allows it to train stably at one-bit weights and activations [29].

The third line adds zeroth-order information: a finite-difference estimate of the true gradient,

added to the STE gradient, reduces its bias at low bit-width [30]. A fourth line replaces the uniform quantizer with a non-uniform one; a k-means codebook improves low-bit quantization-aware training by placing levels where the weight mass is [133].

A final line moves beyond the STE for fine-tuning. PV-Tuning shows that the STE is suboptimal for extreme fine-tuning specifically, and replaces it with a representation-agnostic coordinate descent that alternates updates of the discrete assignment and the continuous parameters, achieving the first Pareto-optimal 2-bit Llama-2 [32]. LC-QAT makes vector-quantized training differentiable by parameterizing each codeword as a linear map of a discrete vector, which avoids the codebook lookup in the training forward pass [64].

6.4 Distillation

Distillation from a full-precision teacher is the most common way to close the last accuracy gap. FBI-LLM trains a binary model from scratch with an autoregressive distillation loss and finds that a pretrained initialization is not required [44]; Bi-Mamba does the same for state-space models [60]. BitDistiller pairs quantization-aware training with a confidence-aware Kullback-Leibler objective in a self-distillation loop for sub-four-bit weights [51]; token-scaled logit distillation was the first method to bring ternary quantization-aware training of large generative models to within one perplexity point of the teacher [98]. BitNet Distillation adds SubLN, a short continued-pretraining phase, and combined logit-and-attention distillation to convert a full-precision LLM into a 1.58-bit task model [52]. LBLLM stages the process into a post-training-quantization initialization, layer-wise weight distillation, and finally learning of the activation scales [124].

6.5 Post-training routes toward one bit

The post-training-quantization branch, 30 works on the training-paradigm axis of Section 4.2, approaches one bit without gradient updates. The techniques cluster into a few families. The first is salient- weight handling with binary residual approximation, as in the roughly 1.08 effective bits of BiLLM [2] and the partial binarization of PB-LLM [3]. The second is alternating refinement of the binarization parameters, as in ARB-LLM, which was the first binary post-training method to beat a half-precision model of equal size [42]. The third is ternary-specific quantizers with iterative grid fitting and column reordering [53, 54]. The fourth is explicit outlier handling, whether by index coding at a 0.3-bit overhead [61] or by learned distribution transforms [107], [35]. The fifth is output-alignment objectives that correct the error accumulation and representation-space anisotropy from which naive output-matching post-training quantization suffers in the one-bit regime [108].

6.6 Scaling laws at one bit

Whether native 1.58-bit models genuinely match full precision, or only appear to do so under limited training-token budgets, is the central empirical dispute in the field. Precision-aware scaling laws model low-precision training as a reduction of the effective parameter count and predict that post-training-quantization degradation increases with pre-training data, eventually making additional data harmful for a model that will be quantized [5]. The quantization-induced degradation study reaches a compatible conclusion from more than 1,500 checkpoints: degradation grows with the number of training tokens, so that projections to 100-trillion-token models are pessimistic for low-bit quantization [4]. On the other side, the native ternary suites show TriLMs overtaking quantized and full-precision models on a bits-for-bits basis beyond about 1B parameters [1, 20], and the unified 1-bit to 4-bit study of ParetoQ finds a learning transition between two and three bits, with ternary, two-bit, and three-bit quantization sharing a size-accuracy frontier that beats four-bit and binary [38]. A partial reconciliation is that the two camps measure different quantities, namely post-training quantization of a fixed model against training under the constraint, and that the crossover point, not its existence, is what matters in practice. Theory is beginning to bear on this: a kernel-limit analysis proves a scaling law for 1-bit networks [39], and precision-expressivity trade-offs are being formalized for quantized transformers [40, 41].

6.7 Training stability and smoothness

Beyond the loss value, extremely quantized models show a degradation of smoothness: a rapid reduction in the number of plausible next tokens within a prediction neighborhood, which sparsifies the decoding tree and hurts generation quality independently of perplexity. A smoothness-preserving term added to either post-training quantization or quantization-aware training recovers additional accuracy [132]. Tequila identifies deadzone trapping, in which ternary weights become stuck at the $\{-1,0\}$ or $\{0,+1\}$ boundary and receive only noise gradients, and repurposes the trapped weights as dynamic biases so that they receive a meaningful signal, at near-zero inference overhead [109]. HGF stabilizes 1.58-bit training with a per-layer gated low-rank correction [128]. These are early steps toward the training stability that a native 1-bit model an order of magnitude larger than today’s 2B, 4T-token ceiling would require (Section 11).

7 Inference Systems and Hardware

7.1 The need for a software stack

A ternary weight matrix stored at two bits already reduces memory traffic by roughly eight times relative to BF16, but a naive kernel still dequantizes to floating point and runs an ordinary matrix multiplication, capturing none of the arithmetic saving. Realizing the add-and-subtract inner product requires either bit-serial computation or lookup tables, and both need dedicated kernels. The dominant cost in ternary inference is the mixed-precision matrix multiplication of ternary weights against INT8 activations, so this is what the kernels target [66].

7.2 bitnet.cpp and CPU kernels

bitnet.cpp is the reference CPU stack for BitNet b1.58 and ternary models [65, 66]. It provides two kernel families. The I2_S kernel packs ternary weights as two-bit integers with a shared scale and performs lossless integer inference. The TL1 and TL2 kernels are ternary lookup tables: TL1 precomputes the products of a small block of ternary weights against the possible activation values and indexes into them, and TL2 refines the packing to reduce the spatial footprint of the table. Reported speedups over an FP16 baseline are 2.37 to 6.17 times on x86 and 1.37 to 5.07 times on ARM, with energy reductions of 72% to 82% on x86 and 55% to 70% on ARM, and the headline demonstration is a 100B-parameter BitNet b1.58 model running at five to seven tokens per second, comparable to human reading speed, on a single CPU [65]. A 2026 update adds parallel kernels with configurable tiling for a further 1.15 to 2.1 times [141]. Independent CPU frameworks pursue the same goal by different means: Litespark uses hand-tuned SIMD code [123], FairyFuse uses fused ternary kernels that avoid materializing intermediates [67], T-SAR repurposes the SIMD register file for in-register lookup-table generation with a small instruction-set change [69], and RSR-core implements the Redundant Segment Reduction algorithm as production CPU and CUDA kernels, with a reported 62-times CPU speedup on ternary LLMs [68]. A more general theoretical result gives an $O(n^2/\log n)$ matrix-vector algorithm for binary and ternary matrices by preprocessing the fixed weight matrix into an index [105].

7.3 GPU

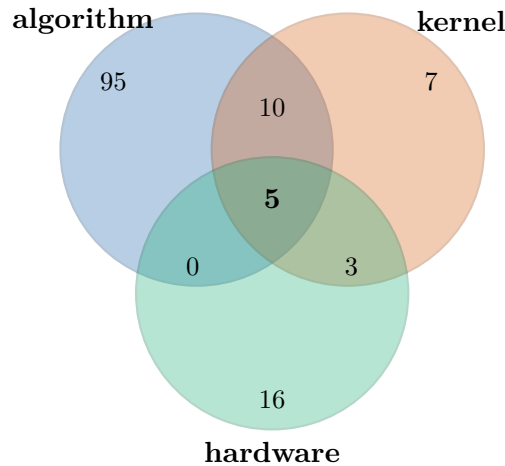
GPUs are optimized for dense floating-point matrix multiplication, so ternary weights initially offered little benefit on them. This has changed. An official GPU kernel for BitNet shipped in 2025 [141]; the TriRun kernel of Spectra 1.1 accelerates end-to-end ternary inference by up to five times over a floating-point baseline by packing two ternary weights per byte [20]; and QuEST provides GPU kernels for its W1 through W4 models [29]. The matrix-multiplication-free design sidesteps the question altogether by removing the operation that GPUs are built for [11].

7.4 Custom accelerators

Ternary weights change what the ideal processor looks like, and a first wave of designs explores the space. Lookup-table ASICs precompute partial products and index into them. Platinum uses offline-generated construction paths and adaptive bit-serial or ternary-weight execution, reporting a 73-times speedup over a spiking baseline and a 2.15-times speedup over a 16-thread CPU implementation in under one square millimetre of chip area [73]; TENET is a sparsity-aware, lookup-table-centric edge architecture [74]; and Vec-LUT vectorizes the table lookup for parallelism [135]. Compute-in-ROM designs store the rarely updated ternary base model as standard-cell logic, exploiting the zero state to eliminate area for zero-valued weights, with an SRAM layer of low-rank adapters for on-device adaptation: BitROM targets billion-parameter models [76] and TOM reports 3,306 tokens per second on a BitNet-2B model with dynamic power gating of inactive ROM banks [77]. Processing-in-memory designs [78, 79] and edge-FPGA accelerators with table-lookup matrix multiplication for both the prefill and decode phases [81–83] complete the picture. The fault tolerance of compute-in-memory ternary LLMs, which can exploit the natural redundancy of the zero state, is itself a research topic.

7.5 Summary of the systems corpus

Three observations stand out. First, the CPU story is mature: several independent frameworks now beat FP16 by factors of two to six with lossless accuracy, and the claim that a capable model can run without a GPU is genuine. Second, the GPU and custom-silicon work is early: the accelerator papers report simulator or single-chip results rather than shipping parts, and no commercial processor exposes a ternary matrix-multiplication primitive. Third, only five works in the entire corpus co-design across algorithm, kernel, and hardware (Fig. 9); the rest optimize one layer while assuming the others. Closing that gap, by building an add-and-subtract matrix-multiplication unit with a matching memory layout and instruction-set extension, co-designed with the training recipe, is the clearest systems opportunity (Section 11). Section 10 reports our own CPU measurements against the figures cited above.



(4 survey works with no systems-stack contribution are not pictured)

Figure 9: How far down the stack the corpus reaches: co-occurrence of algorithm, kernel, and hardware contributions; areas are illustrative, not proportional to count. Ninety-five works are algorithm only; only five co-design across all three layers.

8 Performance and Benchmarking

8.1 The comparison problem

Results in this literature are difficult to align. Papers use different evaluation harnesses and harness versions, different prompt formats and few-shot counts, and different perplexity corpora, namely WikiText-2 against C4 validation; on the efficiency side they differ in hardware, thread count, kernel, and baseline, whether FP16 against INT8 or a general framework against a custom kernel. The reported number of bits per weight also varies in what it includes (Section 2.1). We therefore report each claim in the terms the source used, group claims by what is actually comparable, and flag cases in which a direct comparison is not sound. Table 1 collects the headline figures; the common-axis re-tabulation is Table 2; and our own independent measurements are in Section 10.

8.2 Perplexity and the parity claim

The central claim of the native line is parity with a full-precision model of equal size and training-token budget. BitNet b1.58 reports this from roughly 3B parameters: at 3B, its WikiText-2 and C4 perplexity matches a reproduced FP16 LLaMA trained on the same 100B RedPajama tokens, its zero-shot average matches from the same scale, and the gap to FP16 narrows as size grows to 7B [8, 12]. The Spectra suites make the stronger claim that a ternary model can match a full-precision model of equal parameter count: TriLM 3.9B matches FloatLM 3.9B across their benchmark set, despite a higher perplexity and a size 5.9 times smaller in bits [1]. Spectra 1.1 extends this to 1.2T tokens and finds that ternary models gain more from data than from parameters [20]. The counter-evidence, discussed in Section 6.6, is that post-training-quantization degradation grows with training tokens, so these parity results, all measured at four trillion tokens or fewer, may not extrapolate [4, 5].

For post-training routes toward one bit, perplexity is the standard yardstick and the progression is clear: BiLLM reached 8.41 on LLaMA2-70B at roughly 1.08 effective bits [2]; DB-LLM cut two-bit perplexity from 9.64 to 7.23 [100]; the rotation-and-column-bitmap variant of ARB-LLM was the first binary post-training method to beat a half-precision model of the same size [42]; and PTQTP and PT2-LLM now bring training-free ternary within reach of 1.58-bit quantization-aware-training accuracy in about an hour, rather than the 10 to 14 GPU-days that training requires [53, 54].

8.3 Downstream benchmarks

The common zero-shot suite is ARC-Easy, ARC-Challenge, HellaSwag, WinoGrande, PIQA, OpenBookQA, and BoolQ, run through a standard harness and averaged; native-line papers report this suite, while post-training-quantization papers more often report perplexity plus a subset. For instruction-tuned models at the 2B scale, MMLU, GSM8K, and HumanEval+ are added. BitNet b1.58 2B4T reports a zero-shot average of 54.19 against 55.23 for a full-precision Qwen2.5-1.5B, with the 1-bit model ahead on GSM8K (58.38 against 56.79, a lead Section 10.3 reproduces) and ARC-Challenge (49.91) and behind on MMLU (53.17 against 60.25, a deficit Section 10.3 also reproduces), and ahead of an INT4-AWQ Qwen2.5-1.5B on the average [9]. The unified study of ParetoQ is the most useful single reference point: across 1-bit to 4-bit quantization-aware training, ternary, two-bit, and three-bit quantization occupy a common size-accuracy frontier that beats both four-bit and binary, with a sharp learning transition between two and three bits below which the representations reorganize, and a 600M ternary model beats the prior 3B ternary state of the art [38].

8.4 Reasoning and generation quality

Two findings from 2025 and 2026 complicate the downstream picture. First, extreme quantization degrades output smoothness, that is, the number of plausible next tokens in a prediction neighborhood, independently of perplexity, which sparsifies the decoding tree and hurts generation quality; a smoothness-preserving term recovers accuracy in both post-training quantization and quantization-aware training [132]. Second, two-bit reasoning models fail not by losing the answer but by failing

to commit to it, which inflates the trace length and negates the per-token speedup; lightweight FP16 planning and a loop-rescue mechanism recover most of the gap [131]. Whether native 1.58-bit training suffers the same pathologies as post-hoc two-bit quantization is an open question.

8.5 Efficiency

Efficiency is where the numbers are largest and least standardized. On memory, the non-embedding footprint of BitNet b1.58 2B4T is 0.4 GB against 2.6 GB for a size-comparable FP16 model [9], and the size in bits of a TriLM is about six times smaller than that of an FP16 model with the same parameter count [1]. On latency, the same report gives 29 ms CPU decode against 65 ms, and BitNet b1.58 at 3B is 2.71 times faster than FP16 LLaMA on GPU, rising to 4.1 times at 70B as the linear layers come to dominate [8]. On throughput, BitNet b1.58 70B supports about eleven times the batch size and about 8.9 times the throughput of FP16 LLaMA 70B on two A100 GPUs [8]. On energy, the arithmetic saving for the matrix multiplication alone is about 71 times on a 7-nanometre process, and the end-to-end decode energy for the 2B model is 0.028 J against 0.186 J to 0.649 J for full-precision comparators [8, 9]. On the kernel side, bitnet.cpp reports 2.37 to 6.17 times over FP16 on x86 and 1.37 to 5.07 times on ARM, with 72% to 82% and 55% to 70% energy reductions and a 100B model at reading speed on one CPU [65, 66], though our own run of the 2B model on an x86 laptop puts the decode speedup over FP16 at about 2 times, below that range (Section 10.4); the TriRun GPU kernel of Spectra 1.1 gives a 4.9-times end-to-end speedup for a 70B model on a single L40S GPU and up to about 78 times on the ternary layer in high-batch settings [20]; and LittleBit reports an 11.6-times inference speedup over FP16 at 0.1 bits per weight, compressing Llama2-13B below 0.9 GB [19].

8.6 A common-axis comparison

Table 2 places the field on one set of axes: representation, weight and activation configuration, a consistent effective-bits figure (the nominal code width plus the amortized cost of scales, masks, and codebooks), the headline accuracy in the terms of the source, and the headline efficiency claim, each with its evidence basis (venue and whether the result has an independent check, per Section 3.3). All numbers are author-reported; Section 10 substitutes independent measurements for a subset.

Three points become visible only once the numbers are aligned. First, the reported number of bits often understates real storage. The headline figures for salient-split binary post-training quantization, such as 1.08 for BiLLM and about 1.1 for ARB-LLM, count only the code; a strict accounting that includes group-wise scales and bitmaps puts BiLLM at about 2.88 and STBLLM at about 4.13 effective bits per weight [129]. Native ternary, by contrast, is a genuine 1.58 bits, or between 1.25 and 2.0 bits depending on the packing; Fig. 10 plots the reported and effective figures side by side wherever both are known. Second, the “1-bit post-training quantization” line and the “native ternary” line are not at the same operating point. Post-training methods that claim roughly one bit almost always keep something in higher precision: salient columns (PTQ1.61 at 1.61, PB-LLM at 1.70), a residual branch (RaBiT at a nominal two bits and an effective 2.02), or higher-precision activations (DBellQuant and BWLA at A6). Genuine one-bit weights and activations exist only in native training, and only below about 1B parameters [29]. Third, the strongest cross-method claim is comparative rather than absolute. The unified study of ParetoQ is the reference point: ternary, two-bit, and three-bit quantization occupy a common size-accuracy frontier that beats four-bit and binary [38]. Most results from 2025 and 2026, including CAT-Q reaching an accuracy comparable to BitNet v2 with a factor of 100,000 fewer tokens through post-training quantization [126], HESTIA matching 100B-token native training with 10B tokens [127], and LC-QAT matching quantization-aware training with 1% to 10% of the data [64], concern closing the cost gap at roughly fixed accuracy rather than pushing accuracy itself.

Splitting Table 2 by evidence basis (Section 3.3) shows that the central claims do not rest on the least reliable sources. Twenty-eight of the 57 rows are corroborated, that is peer reviewed or with an independent check, and every source behind the parity claim, namely BitNet b1.58, BitNet b1.58 2B4T, both Spectra suites, and ParetoQ, is in that set: Spectra, ParetoQ, and the original BitNet

b1.58 corroborate one another, ParetoQ is also at NeurIPS, and Section 10 re-runs the released BitNet b1.58 2B4T checkpoint and adds a controlled test of the parity claim on Spectra’s matched TriLM and FloatLM pair, which lands within 0.6 points on a seven-task mean at 2.4B parameters with a higher perplexity. That bears on the claim at this scale; whether it holds at the frontier is a separate question that Section 11.1 lists as open. The other 29 rows are single unreproduced preprints, most of them post-training-quantization results claiming the lowest bit counts (BTC-LLM at 0.8, HBLLM at 1.08, DBellQuant and BWLA at 1.0), together with the bitnet.cpp infrastructure report, which moved into this group once we found that its reference runtime does not compute this model correctly as built (Section 10). Any statement in this section that rests only on single unreproduced preprints is marked as such, following the convention set in Section 3.3.

8.7 Summary of the benchmarking evidence

Three statements are defensible. Native ternary training is reported, at up to about 7B parameters and 4T tokens, to match full precision on standard benchmarks; those figures are author-reported rather than independently re-run at that scale (Section 10 checks a smaller one), and whether parity holds at the frontier is untested and disputed. Post-training routes have closed most of the gap to 1.58-bit quantization-aware-training accuracy at a fraction of the cost, but only down to a two-bit equivalent; genuine one-bit post-training quantization still trails. The efficiency gains on CPU are large and independently plausible; the GPU and custom-silicon gains are reported from kernels and simulators rather than deployed systems, and Section 10 tests the CPU claims directly.

Table 2: Reported headline results (author-reported; a subset is independently re-measured elsewhere in this paper). Effective bits is the nominal code width unless marked *effective* (overhead-adjusted) or *var* (a range), following the accounting in Section 8.6. *Evidence basis* is venue plus independent-check status; it is not a graded score. Blank cells mean not reported.

Reference	Model	Scale	Configuration	Eff. bits	WikiText-2 ppl.	Zero-shot avg.	Memory	Latency / speed	Evidence basis
Ma et al. [8]	BitNet b1.58	3B	W1.58A8 100B tok	1.58	matches FP16 @3B	matches FP16 @3B	3.55x less GPU mem @3B	2.71x faster @3B; 4.1x @70B	preprint + ext. repro
Ma et al. [9]	BitNet b1.58 2B4T	2B	W1.58A8 4T tok	1.58		54.19 (Qwen2.5-1.5B 55.23)	0.4 GB non-emb	29 ms CPU de-code	preprint + Sec.10
Wang et al. [12]	BitNet a4.8	7B	W1.58A4 hybrid	1.58	$\approx$ b1.58	$\approx$ b1.58 (std err 1.06%)		faster (INT4 kernels)	preprint
Wang et al. [13]	BitNet v2	7B	W1.58A8 / A4 H-BitLinear	1.58	minimal b1.58	vs +0.16/0.49/0.61% avg over b1.58 @1.3/3/7B			preprint
Kaushal et al. [1]	TriLM (Spectra)	3.9B	ternary 300B tok	1.58	higher than FP	ppl matches FloatLM 3.9 B across benchmarks	5.9x smaller bitsize	max speedup plateau 10x (vs 4x INT4)	preprint + ext. repro
Vaidhya et al. [20]	TriLM (Spectra-1.1)	≤ 3.9 B	ternary up to 1.2T tok	1.58		beats Spectra @ matched size (MMLU)		TriRun GPU 4.9x e2e @70B L40S; 78x layer high-batch	preprint + ext. repro
Liu et al. [38]	ParetoQ ternary	600M	unified 1-4 bit QAT	1.58		beats prior SoTA 3B ternary	1/5 params of 3B SoTA	2-bit best for mem+speed	NeurIPS + ext. repro
Huang et al. [2]	BiLLM	LLaMA2-70B	1.08-bit PTQ	2.88	8.41			binarize 7B in 0.5h / 1 GPU	ICML
Chen et al. [100]	DB-LLM	7-13B	2-bit dual-binary PTQ	2.0	9.64 $\rightarrow$ 7.23			+20% compute reduction at same bit-width	ICLR
Xu et al. [43]	OneBit	LLaMA 7-13B	1-bit QAT	1.0		$\geq 81\%$ of FP16 perf			NeurIPS
Ma et al. [44]	FBI-LLM	130M/1.3B/7B	binary from scratch	1.0	$\approx$ matches FP magnitude				preprint

Reference	Model	Scale	Configuration	Eff. bits	WikiText-2 ppl.	Zero-shot avg.	Memory	Latency / speed	Evidence basis
Li et al. [42]	ARB-LLM_RC	7-70B	binary PTQ	1.1	< FP16 same size	first binary PTQ to surpass FP16 same size			ICLR25
Dong et al. [18]	STBLLM	7-70B	<1-bit N:M	4.13	< other binary	beats other <1-bit binary	less memory	custom kernel	CUDA ICLR25
Jo et al. [103]	BinaryMoS	1-13B	binary QAT	1.0		beats binary and 2-bit methods	$\approx$ static-binary size		NeurIPS
Panferov et al. [29]	QuEST	≤ 800 M	W1A1.. W4A4	var				GPU kernels provided	preprint
Tang et al. [60]	Bi-Mamba	780M/1.3B/2.7B	1-bit SSM from scratch	1.0	$\approx$ FP16	$\approx$ FP16	drastic vs orig Mamba		TMLR
Huang et al. [109]	Tequila	≤ 3 B	ternary QAT	1.58		+ >4% ARC vs SoTA; <1% gap to FP rivals 1.58-bit QAT		3.0x inference speedup	preprint
Xiao et al. [53]	PTQTP	0.6-70B	dual trit-plane PTQ	var				4.63x decode vs FP16	preprint
Yan et al. [54]	PT2-LLM	7-65B	ternary PTQ	1.58	$\approx$ 2-bit PTQ	competitive with SoTA 2-bit PTQ	lower memory than 2-bit PTQ	end-to-end pre-fill+decode speedup	ICLR26
Zhao et al. [34]	TWLA	7-70B	W1.58A4 PTQ	1.58				significant inference acceleration	preprint
Lee et al. [19]	LittleBit	7-70B	0.1 bits/weight	0.1		beats 0.7-BPW methods @ 0.1 BPW	31x memory (Llama2-13B <0.9 GB)	11.6x inference speedup vs FP16	NeurIPS
Wang et al. [65]	bitnet.cpp	up to 100B	ternary CPU kernels	1.58	lossless (claimed)			x86 2.37-6.17x; ARM 1.37-5.07x; 100B @ 5-7 tok/s	preprint
Wang et al. [66]	bitnet.cpp (paper)	ternary	I2_S + TL1/TL2	1.58	lossless			1 CPU up to 6.25x vs FP16; 2.32x vs low-bit	ACL
Wu et al. [52]	BitNet Distillation	≤ 7 B	FP $\rightarrow$ 1.58-bit task	1.58		$\approx$ FP task performance	≈ 10 x memory	≈ 2.6 x speedup	CPU (re-reported) preprint

Reference	Model	Scale	Configuration	Eff. bits	WikiText-2 ppl.	Zero-shot avg.	Memory	Latency / speed	Evidence basis
Kim et al. [98]	TSLD	1.3-6.7 B	ternary QAT	1.58	< 1.0 ppl	improved CSQA + arithmetic			NeurIPS
Zhao et al. [31]	Direct Quantized Training	<1B	ternary/8-bit STE	no 8.0	8-bit == BitNet b1.58		reduced training memory		ACML
Wang et al. [91]	BitVLA	2B	W1.58 + distilled vision enc	1.58		retains task competence	≈ BitNet 2B4T footprint		preprint
Kawamura et al. [89]	BitTTS	<1B	1.58-bit QAT weight-index	+ 1.58		beats same-size unquantized (MOS)	83% size reduction		Interspeech
Wang et al. [94]	MoTE	3B up-cycled	ternary routed experts	1.58		+4.3% avg vs MoE-LLaVA @3.4GB expert mem	fixed active params		preprint
Wang et al. [7]	BitNet b1 / b1.58	125 M-6.7B	W1A8 / W1.58A8	var	competitive vs 8-bit PTQ + FP16	b1.58 matches half-precision			JMLR-26
Kumar et al. [5]	(analysis)	<=1.7 B	precision-aware scaling	var					ICLR25
Ouyang et al. [4]	(analysis)	7-405B proj	QiD scaling	var					preprint
Malinovskii et al. [32]	PV-Tuning	7-70B	1-2 bit VQ fine-tune	2.0	SOTA 1-2 bit; first Pareto-optimal 2-bit Llama-2		same data; kernel-compatible	calib	NeurIPS(oral)
Du et al. [51]	BitDistiller	1-34B	uniform 2-3 bit QAT+KD	2.0	2-bit +3.54% vs LLM-QAT; +12.43% vs best PTQ		fewer resources	data/	ACL
Chen et al. [101]	EfficientQAT	7-70B	uniform 2-4 bit QAT	2.0		2-bit Llama2-70B 69.48 vs 72.41 FP (<3pt)	2-bit 70B on 1x A100-80GB in 41h		preprint
Tabesh et al. [28]	CAGE	Llama-3.2-3 B	W3A3 / W4A4 QAT	3.0		halves quant error vs QuEST/MXFP4 (GSM8K/HS/WG); W3A3 = prior W4A4	optimizer-agnostic		MLSys26
Zhao et al. [120]	PTQ1.61	LLaMA-7B	mixed-partial PTQ	1.61	ppl 12.50				ACL

Reference	Model	Scale	Configuration		Eff. bits	WikiText-2 ppl.	Zero-shot avg.	Memory	Latency / speed	Evidence basis
Song et al. [111]	binary W+A PTQ	7B	W(1+1)A(1-4) PTQ		2.0	W2A4 LLaMA-7B ppl 8.58 / LLaMA2-7B 8.89 (FP 5.68/5.47)				preprint
Boža and Macko [112]	DBF	Llama2-7B	binary PTQ	factors	2.0	beats single-matrix binarization; competitive w/ leading quant			2-3.5x @2-bit speedup	TMLR
Chen et al. [116]	HBLLM	OPT / LLaMA	1-bit PTQ (Haar wavelet)		1.08	LLaMA2-13B ppl 6.71 @1.08 avg bits; 1.22-2.48x FP16 ppl				preprint
Tran and Nguyen [118]	Multi-Boolean	OPT / LLaMA	K Boolean kernels		var	close to FP16; beats binarization + 2-bit				ICLR
Chong et al. [129]	NanoQuant	L2/L3/G3/Q3	sub-1-bit PTQ		var	competitive w/ higher-BPW binary PTQ; approaches binary QAT				preprint
You et al. [130]	RaBiT	Llama/Gemma/Qwen	binary QAT (free)	residual (matmul-free)	2.02	5.78 (QTIP-VQ 5.86)	ZS avg 61.5% (QTIP 59.0%); Qwen3-4B 66.7%			preprint
Pizzo [128]	HGF	Tiny Stories / 1.2-3 B	ternary LoRA	+ gated	1.58	recovers $\approx 55\%$ of BitNet $\rightarrow$ FP16 gap			+12-15% memory	preprint
Wang et al. [127]	HESTIA	Llama-3.2 1B/3B	ternary hard QAT	soft-to-	1.58		1B ZS +5.39% (0.519 to 0.547 vs Tequila); 3B +4.34%			preprint

Reference	Model	Scale	Configuration	Eff. bits	WikiText-2 ppl.	Zero-shot avg.	Memory	Latency / speed	Evidence basis
Wang et al. [126]	CAT-Q	1.7B-235B incl MoE	ternary PTQ	1.58	W1.58 $\approx$ BitNet v1/v2 @100B tok using 512 calib samples				preprint
Gu et al. [113]	BTC-LLM	LLaMA 7-65B	sub-1-bit PTQ (binary codebook)	0.8	binary baseline 6.06 ppl LLaMA2-7B; 0.7-bit 11.02 ppl @22x mem; +5.0% over STBLLM @0.8-bit			1.6x speedup	preprint
Ye et al. [107]	DBellQuant	OPT/LLaMA/Qwen/DS	binary W + 6-bit A PTQ	1.0	LLaMA2-13B ppl 14.39 (BiLLM 21.35 weight-only); LLaMA2-7B 21.69 @A6 / 23.04 @A4				preprint
Zhao et al. [35]	BWLA	7-70B	binary W + 6-bit A PTQ	1.0	first high-acc W1AX pure PTQ; Qwen3-32B ppl 11.92 @A6 (SoTA 38); +>70% on 5 ZS			3.26x speedup	preprint
Zagitov et al. [33]	HARP	Llama 1B-70B	any + learned rotation PTQ	2.0	best gains @2-bit; improves ppl over fixed RHT across scales			Llama2-7B @2-bit 128 tok/s (RHT 142; FP16 61)	preprint
Li et al. [61]	ICQuant	Llama2/3	any + index-coded outliers PTQ	2.3		2.3 b/wt beats 4-bit RTN; large gains at 2-bit			COLM
Wang et al. [64]	LC-QAT	1.7-8B	2-bit VQ-QAT	2.0	matches/beats SQ+VQ QAT with 0.1-10% of training data			custom CUDA 1.68x vs FP16	ICML

Reference	Model	Scale	Configuration	Eff. bits	WikiText-2 ppl.	Zero-shot avg.	Memory	Latency / speed	Evidence basis
Zhang et al. [58]	Sparse-Bit Net	Bit Net-2B	ternary + N:M sparsity	1.58	at 2:4 (50% sparsity): Bit Net +5.7% ppl vs BF16 + 18.8%			up to 1.30x train+ inference (6:8 op)	preprint
Wang et al. [57]	Q-Sparse	300M-7 B	ternary/FP + top- K sparsify	var					preprint
Wang et al. [48]	iFairy	700M / 1.3B	complex +-1,+i 2- bit native	2.0	700M avg ppl 11.13 (BitNet b1.58 11.51 repro); FP i Fairy 700M 10.08 < FP16 LLaMA 12.33				preprint
Wang et al. [49]	Fairy2i	LLa MA-2 7 B	pretrained real → complex 2-bit	2.0	restores LLa MA-2 7B near-FP; beats real bi- nary/ternary			mult-free accumu- lation	preprint
Song et al. [124]	LBLLM	LLa MA	binary W(1+1) + A4, 3-stage distill	2.0	surpasses So TA binariza- tion @W2A4; beats CBQ W4A4				preprint

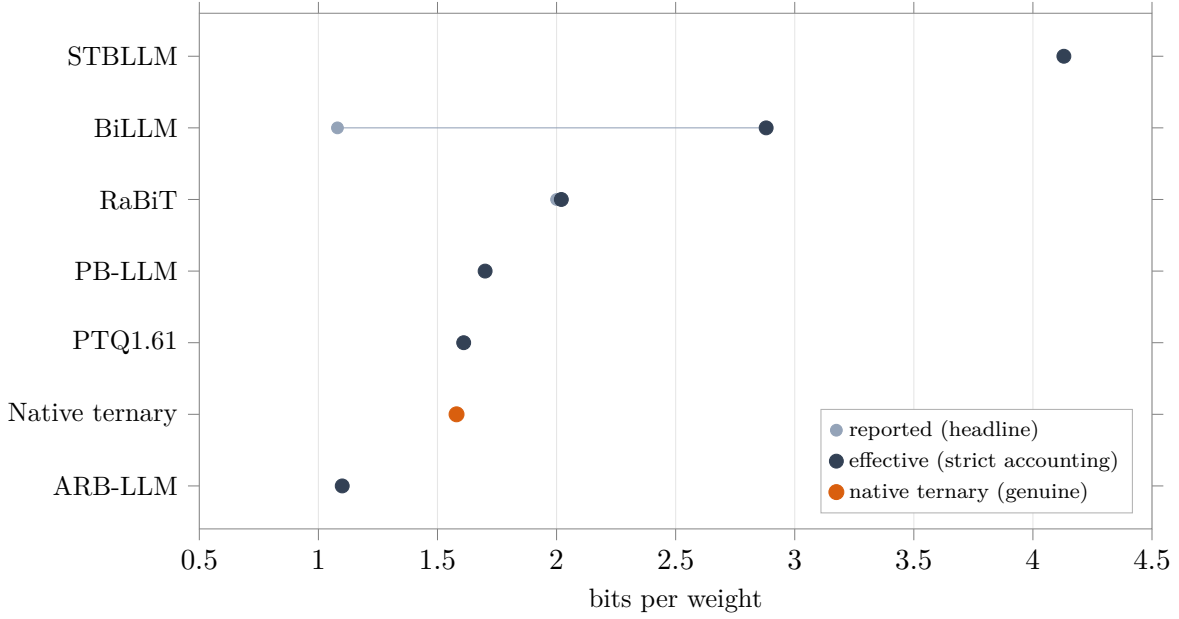


Figure 10: Reported versus effective bits per weight. Where a strict accounting exists, the headline figure (grey) understates real storage against the effective one (dark): BiLLM’s 1.08 becomes 2.88, RaBiT’s nominal two bits becomes 2.02. ARB-LLM, PTQ1.61, PB-LLM, and STBLLM report only one figure. Native ternary (orange) has no such accounting gap between a deflated headline and its real storage cost: 1.58 is both nominal and effective. That is a separate question from the packed format a given kernel actually uses on disk, which for `I2_S` is two bits per weight for hardware convenience, not an overhead the accounting here is missing.

9 Applications and Extensions

The native 1-bit machinery was built for text-decoder LLMs, but the period from 2024 to 2026 saw it carried into a widening set of modalities and deployment settings. This section surveys those branches. Each is small today, but each is growing, and together they are the fastest-moving part of the field.

9.1 On-device and edge deployment

The unifying theme of the area is running capable models without a data center. Concrete demonstrations include BitNet b1.58 2B4T with open CPU and GPU inference [9], a 100B model at reading speed on one CPU [65], an inference characterization on a Raspberry Pi with BitNet Q1.58 against k-quantized baselines [110], and analyses of when commodity CPUs beat GPUs for on-device serving [114]. The security of the deployed weights is a separate concern: TZ-LLM protects on-device model parameters inside an Arm TrustZone enclave, using pipelined restoration to hide decryption latency [134].

9.2 Embedding models

Embedding models are attractive ternary targets, because the downstream operation of approximate nearest-neighbor search tolerates quantized vectors well. A ternary-weight fine-tuning framework with self-taught distillation preserves retrieval quality while reducing memory and latency, and composes with approximate-nearest-neighbor indices [87]; Ultra-Quantisation applies 1.58-bit encodings directly to the embedding-search problem [88]; and BitNet Text Embeddings is a native 1-bit embedding-model family, at 0.27B and 0.6B parameters, released with the `I2_S` kernel path of `bitnet.cpp` [86, 141].

9.3 Speech

BitTTS quantizes both the acoustic model and the vocoder of a text-to-speech system to 1.58-bit with quantization-aware training, together with a weight-indexing scheme that stores five ternary weights per INT8; the result is an 83% size reduction while improving quality over an unquantized model of the same size [89]. On the recognition side, an extremely low-bit Conformer is trained toward one bit with co-training and stochastic precision [90].

9.4 Vision-language and vision-language-action models

LLaVaOLMoBitnet1B was the first ternary multimodal LLM, pairing a full-precision vision encoder with a ternary OLMo-Bitnet backbone [92]. BitVLA extends this to robotics: it builds on BitNet b1.58 2B4T, then compresses the vision encoder to 1.58-bit weights with eight-bit activations through a quantize-then-distill stage before robotic fine-tuning, retaining task competence at a fraction of the memory [91]. TernaryCLIP compresses a CLIP vision-language model to ternary weights with distilled knowledge [93]; TeTRA ternarizes a vision-transformer backbone and binarizes its embedding head for compact visual place recognition on drones [140]; and BitMar explores low-bit multimodal fusion with an episodic memory for edge devices [137].

9.5 Mixture-of-experts

MoTE trains many ternary routed experts rather than few full-precision ones, keeping the active-parameter budget fixed while removing the memory overhead of a full-precision expert pool; it is competitive with a full-precision MoE-LLaVA at equal expert memory, and the gap widens as the memory budget tightens [94].

9.6 Ecosystem beyond a single laboratory

The paradigm is no longer confined to one laboratory. The Falcon-Edge family from TII, comprising Falcon3-1.58bit models at 1B and 3B in base and instruction-tuned variants, is trained with ternary weights during pre-training and ships a fine-tuning library [142]; the Spectra suites from NolanoOrg release more than 500 intermediate TriLM checkpoints from 99M to 3.9B parameters [1], [20]; and independent groups have produced binary and ternary variants across the OPT, Llama, Qwen, and Mistral families [42], [18, 53].

9.7 Environmental framing

Several works motivate 1-bit LLMs explicitly on energy and carbon grounds [6, 99], and the reported 70% to 96% energy reductions for CPU inference [65, 143] are the strongest lever for that argument. A rigorous life-cycle account, one that includes the higher training-time cost of native 1-bit models, has not yet been published and is noted here as a gap.

10 Independent Re-Evaluation

10.1 Rationale and scope

The efficiency figures that motivate this paradigm are almost all author-reported and no third party has re-checked them, including the efficiency table for BitNet b1.58 2B4T of 0.4 GB, 29 ms, and 0.028 J [143] and the 2.37-to-6.17-times bitnet.cpp speedups [65], all quoted against a half-precision baseline. This section re-measures a small, well-defined subset of those claims on commodity hardware, adding the 8-bit and 4-bit baselines a practitioner would actually compare against. It verifies published claims; it does not train models or ablate methods.

10.2 Setup

For Arm A (the accuracy arm), we use EleutherAI’s lm-evaluation-harness at commit `b954108` (pinned 4 September 2026), on a laptop RTX 4080 with 12 GB of memory shared with other workloads, with the batch size capped so that memory use stays within budget. The primary pairing is BitNet b1.58 2B4T [9], about 2.4 billion parameters in total and 2.0 billion outside the token embedding, against Qwen2.5-1.5B-Instruct, about 1.5 billion parameters, the full-precision model the BitNet 2B4T card itself compares against. That pairing is **not a controlled baseline**: the two differ in parameter count, architecture, tokenizer, and training data. To address that we add three comparisons from open checkpoints, in decreasing order of how tightly matched they are. The cleanest is Spectra’s TriLM and FloatLM at 2.4 billion parameters, identical tokenizer, data, and 300-billion-token budget, ternary trained from scratch against FP16 trained from scratch [1]. Next, Qwen2.5-1.5B-Instruct in bf16 against a 4-bit (nf4) quantization of the same weights, which isolates the cost of post-hoc 4-bit compression. Least tightly matched is Falcon3-1B in its 1.58-bit and its bf16 form: TII trains the 1.58-bit model with ternary weights from pre-training and releases a bf16 checkpoint alongside it [142], but does not document whether that bf16 checkpoint is a dequantized copy of the same run or the earlier full-precision Falcon3-1B, so this pair is matched in model family and parameter count but not demonstrably in training data or schedule. All checkpoints are the instruction-tuned variants where one exists, evaluated without a chat template as the harness default; the tables abbreviate Qwen2.5-1.5B-Instruct to Qwen2.5-1.5B. The tasks are the seven-task zero-shot suite (ARC-Easy, ARC-Challenge, HellaSwag, WinoGrande, PIQA, OpenBookQA, and BoolQ) reported as plain accuracy, plus WikiText-2 word perplexity and MMLU at zero shot. Each model is run once, with a HellaSwag determinism check.

MMLU is run at zero shot rather than the card’s five: BitNet’s native Hugging Face layer compiles per input shape, and the fifty-seven MMLU subjects each carry a different five-shot context length, which thrashes the compilation cache; zero shot keeps a fixed shape and runs at the speed of the rest of the suite. GSM8K, dropped from an earlier version of this study, is re-measured here (Section 10.3), but on its own terms: it needs multi-step generation and the model’s chat template, so it is run at five shot with the template applied and generation to 256 tokens, and its numbers are not comparable to the loglikelihood, template-free suite above. A build note: on Windows, that native layer only compiles at all with a Triton build matching the installed PyTorch; without it it runs eagerly and even the zero-shot suite is several times slower.

For Arm B (the efficiency arm), we build the reference bitnet.cpp runtime from source and run every model through that one binary: its `I2_S` ternary kernel serves the BitNet checkpoint, and the `llama.cpp` code path it is built on serves the Qwen2.5-1.5B baselines at five quantization levels: FP16, Q8_0, and three 4-bit formats, Q4_0, Q4_K_S, and the Q4_K_M k-quant that is the common default for local inference. The FP16 baseline here is an FP16 GGUF of the same Qwen checkpoint whose BF16 weights the accuracy arm loads; the C++ runtime has no BF16 path, and the two formats are numerically close, so the accuracy and efficiency baselines are the same model in the nearest format each runtime supports and are not compared against each other. The 4-bit comparison is on this one x86 machine only; a GPU, a second CPU, and quantizers outside the GGUF family such as AWQ or GPTQ would each be needed to generalize it further. Every measurement shares a runtime, a machine, and a harness. The machine is an Intel Core i9-14900HX laptop, eight performance cores and sixteen efficiency cores, on AC power. We sweep threads over 1, 2, 4, 8, and 16, take prefill at 128, 512, and 2048 tokens and decode at 128 tokens, and repeat every point five times through the bundled `llama-bench`. Peak resident memory is sampled from the process working set during a decode run. Energy is measured in a separate pass on battery, where the smart-battery controller exposes whole-system power to the operating system as a discharge rate (it reads zero on AC, and this machine has no running-average power-limit counter); we sample that rate during a thirty-second idle window and during a sustained decode, and divide mean power by the decode rate. The energy pass covers the ternary model and the half-precision, 8-bit, and 4-bit k-quant baselines. On battery the processor runs at a lower power limit, and that limit tightens as the battery drains, so the pass is slower than the plugged-in sweep, absolute energy figures drift across runs, and only within-run

model-to-model ratios are compared.

Three notes on getting the runtime to work. It does not link as a shared library on this toolchain, because the core ggml library references an `I2_S` quantizer symbol that lives only in the CPU backend; a static build resolves it. Its BitNet feed-forward activation is set to `SiLU` where BitNet b1.58 2B4T uses a gated squared-ReLU, which we changed and rebuilt (Section 10.3); the throughput figures are the same either way. And the TL2 lookup-table kernel, which needs a per-shape code-generation step shipped only for older BitNet sizes, was not exercised, so the ternary figures are `I2_S` only. We built microsoft/BitNet at commit `0b341e5`, whose `3rdparty/llama.cpp` submodule was at commit `390c307`; the one-line fix changes `LLM_FFN_SILU` to `LLM_FFN_RELU_SQR` in `src/models/bitnet.cpp`.

10.3 Accuracy results

Table 3 reports, for the primary pairing, plain accuracy on the seven-task zero-shot suite, its mean, WikiText-2 word perplexity at a 2048-token stride, and MMLU at zero shot. Both HellaSwag determinism checks reproduced to the last digit across two independent runs. Accuracy is the unnormalized figure (`acc`); we report it rather than the length-normalized `acc_norm` so that one metric is used across all seven tasks. For HellaSwag the two diverge widely, `acc_norm` is 67.6 for BitNet and 68.3 for Qwen, because the endings vary in length and an unnormalized log-likelihood favors the shorter ones; this is why the HellaSwag column reads near 50 while the vendor’s length-normalized figure is near 68, and why the C++ runtime’s own HellaSwag scorer (Table 6), which averages log-probability per ending token, lands between the two at about 62 to 66. The metric, not the runtime, accounts for the gap.

Table 3: Independent accuracy re-measurement of the primary pairing: unnormalized accuracy (`acc`) on the seven-task zero-shot suite with its mean, WikiText-2 word perplexity at a 2048-token stride, and MMLU at zero shot. One run per model, with a HellaSwag determinism check. HellaSwag’s unnormalized `acc` reads near 50 for both models by design (an unnormalized log-likelihood favors shorter endings, discussed below); the length-normalized `acc_norm` both vendors and papers usually quote is near 68 for both.

model	ARCe	ARCc	HS	WG	PIQA	OBQA	BoolQ	mean	ppl	MMLU
BitNet b1.58 2B4T (W1.58, 2.4B)	76.8	47.0	50.7	72.5	77.0	30.8	79.0	61.98	16.67	52.07
Qwen2.5-1.5B-Instruct (BF16, 1.5B)	76.7	43.6	50.9	62.7	76.3	31.8	78.0	60.00	13.48	60.03

Two comparisons matter. First, against the vendor figure: the BitNet b1.58 2B4T model card reports a headline “zero-shot average” of 54.19, below our 61.98 on the seven-task suite and below its own reference Qwen. That figure averages a broader set that includes MMLU and GSM8K, and most of the difference is MMLU. Our zero-shot MMLU, 52.07 for BitNet against 60.03 for Qwen, closely tracks the card’s five-shot values (53.17 and 60.25) under a different protocol, and matches its roughly eight-point deficit: the ternary model is competitive on commonsense reasoning but clearly behind on world knowledge, and any average that folds MMLU in inherits that gap. A single vendor “average” is not portable across evaluation setups; per-task numbers are the only safe unit of comparison.

Second, the controlled comparisons. The BitNet-versus-Qwen pairing is not size- or training-matched, so we ran three pairs from open artifacts that are, each in its own right (Section 10.2):

Three things follow (Fig. 11 plots the three pairs’ seven-task means against each other directly). The Spectra pair, the cleanest test, puts native ternary within 0.6 points of FP16 on the seven-task mean at 2.4 billion parameters, with perplexity 14% higher: downstream parity alongside a worse language-modeling loss, the exact shape the native-training literature reports for itself (Section 8.2), now measured under a matched protocol rather than inferred. Both Spectra models score near chance on MMLU, so that column is uninformative for this pair; the 300-billion-token budget simply does not buy world knowledge at 2.4 billion parameters. Second, post-hoc 4-bit quantization of Qwen costs 2.7 points on the mean and 2.0 on MMLU, more than native ternary loses at a comparable scale, consistent with the ParetoQ finding that ternary quantization-aware training sits ahead of four-bit post-training quantization on the size-accuracy frontier (Section 8.3).

Table 4: Three pairs from open checkpoints, in decreasing order of how tightly matched they are: change in seven-task mean, WikiText-2 perplexity, and zero-shot MMLU for the ternary or 4-bit member against its full-precision counterpart.

pair (most to least matched)	ternary or 4-bit	full precision	Δ mean	ppl	Δ MMLU
Spectra 2.4B, native ternary vs FP16 (300B tok)	TriLM 51.49	FloatLM 52.05	-0.56	14.90 vs 13.06	-2.2
Qwen2.5-1.5B, nf4 4-bit vs bf16	57.33	60.00	-2.67	14.59 vs 13.48	-2.0
Falcon3-1B, 1.58-bit vs bf16 (checkpoint relation undocumented)	45.69	57.22	-11.53	32.82 vs 18.85	-18.4

Third, and this is the part that most constrains how far the parity result generalizes, the Falcon3-1B pair loses heavily: 11.5 points on the seven-task mean, a 74% higher perplexity, and MMLU driven to chance. This pair cannot separate a scale effect from a training-recipe one, both because it is at one-billion rather than multi-billion parameters and because its two checkpoints are not demonstrably matched in training data (Section 10.2). What it does establish is a clean negative: the 1.58-bit format at the one-billion-parameter scale does not inherit the parity that native ternary shows at 2.4 billion, though with the checkpoint match unresolved, that negative could reflect the pair’s mismatch as readily as it could reflect scale. Taken together, the two pairs that are matched closely enough to compare are consistent with parity being a property of native ternary pre-training carried out at multi-billion scale rather than a general consequence of constraining weights to 1.58 bits, but two pairs, one of them confounded, do not establish that on their own; whether it holds above the roughly 2B, 4T-token ceiling at which it has been measured is the open question of Section 11.1. The uncontrolled BitNet-versus-Qwen result, a slight lead for the 2B ternary model on the seven-task mean and a deficit on perplexity and MMLU, sits consistently between the Spectra and the 4-bit pairs.

GSM8K is the one benchmark the seven-task suite leaves out, because it needs multi-step generation rather than a single loglikelihood; we re-measure it separately, at five shot with each model’s chat template and greedy generation to 256 tokens (Table 5). The two extraction rules that the standard harness reports are worth keeping apart: flexible-extract takes the last number in the output, strict-match accepts only the canonical answer line. For BitNet and the full-precision Falcon3-1B the two nearly agree; for the other models they diverge sharply, the model reaching a right answer but not ending in the canonical form, so the flexible figure is the one that tracks arithmetic ability and the one we compare on. On it, three things follow. First, the vendor’s GSM8K claim reproduces in direction: BitNet b1.58 2B4T scores 61.9 against our own Qwen2.5-1.5B’s 57.5, a 4.4-point lead where the card reports a 1.6-point one (58.38 against 56.79), so the one benchmark on which the card puts the 1-bit model ahead holds up under an independent run. Second, the 4-bit nf4 quantization of Qwen costs 7.8 points on GSM8K, against 2.7 on the seven-task mean: arithmetic is more fragile under post-hoc quantization than commonsense reasoning is. Third, the controlled 1B and 2.4B pairs behave as they did elsewhere, the 1.58-bit Falcon3-1B losing 17 points against its bf16 form and both Spectra models sitting at the GSM8K noise floor, uninformative for that pair exactly as MMLU was.

We separately checked the accuracy cost of the quantizations used in the efficiency arm, running WikiText-2 perplexity and a HellaSwag subset (the first 1000 of 10042 items) through the C++ runtime’s own batched evaluators (`llama-perplexity`, whose HellaSwag mode averages log-probability per ending token) so the three Qwen levels are measured against each other under one scorer. This is a narrow check, two metrics rather than the full seven-task suite, and it is not a substitute for running that suite on the 4-bit model, which the runtime’s harness does not support. Quantizing the baseline is cheap on these two tasks: the 8-bit build costs 0.1% on perplexity, and the 4-bit k-quant costs 4.5% on perplexity (9.25 against 8.85). On the HellaSwag subset the observed differences were

Table 5: GSM8K exact-match accuracy at five shot, with generation and the chat template for the instruction-tuned models above the rule (the Spectra base models below the rule carry no template, are run template-free, and sit at the GSM8K noise floor rather than being meaningfully tested). flexible-extract takes the last number in the output; strict-match accepts only the canonical answer line. These numbers use a different protocol from the loglikelihood suite of Table 3 and are not folded into any mean.

model	flexible-extract	strict-match
BitNet b1.58 2B4T	61.9	59.6
Qwen2.5-1.5B-Instruct, bf16	57.5	35.0
Qwen2.5-1.5B-Instruct, nf4 4-bit	49.7	36.8
Falcon3-1B-Instruct, bf16	44.7	42.6
Falcon3-1B-Instruct, 1.58-bit	27.7	7.4
Spectra TriLM 2.4B, no template	3.0	1.9
Spectra FloatLM 2.4B, no template	1.9	1.3

small, 0.5 points for the 8-bit build and 0.3 for the 4-bit k-quant against half precision’s 65.3%, within the marginal confidence intervals; we did not run a paired significance test, so this rules out a large accuracy drop but not a small one.

Table 6: Accuracy cost of the GGUF quantizations used in the efficiency arm, scored through the C++ runtime’s own batched evaluators so the three Qwen levels are measured alike: WikiText-2 perplexity and a 1000-item HellaSwag subset.

model	WikiText-2 ppl	HellaSwag (n=1000)
Qwen2.5-1.5B, FP16	8.85	65.3%
Qwen2.5-1.5B, Q8_0	8.86	65.8%
Qwen2.5-1.5B, Q4_K_M	9.25	65.6%

The BitNet checkpoint needed a source fix to the runtime before it produced sensible output at all. Built from the project as documented (commit [0b341e5](#), `llama.cpp` submodule at [390c307](#), Section 10.2), the runtime’s BitNet graph selects a SiLU feed-forward activation, whereas BitNet b1.58 2B4T uses a gated squared-ReLU (its released configuration sets the feed-forward activation to squared-ReLU); this is an open issue in the project. The fix is a one-line change to the activation selector in the BitNet model source, from SiLU to squared-ReLU (`LLM_FFN_SILU` to `LLM_FFN_RELU_SQR` in `src/models/bitnet.cpp`). Left uncorrected, WikiText-2 perplexity through the runtime is 86.9 and HellaSwag is 38.7%, far below any working model. Changing that one line and rebuilding brings token-level WikiText-2 perplexity to 12.99 and HellaSwag to 62.4%, both in line with the transformers implementation of the same weights (16.67 word-level perplexity, HellaSwag near 62 to 68 depending on the metric); the throughput and memory figures of Section 10.4 are unchanged, since a squared-ReLU costs no more than a SiLU. The released weights are therefore sound; the defect we traced is in the reference runtime, not the checkpoint, though it is not the only rough edge we found there (below), and we did not catch it until the results were scrutinized. One report questions the checkpoint itself, reading its 1,187,801,280-byte size as a truncated re-upload with empty feed-forward tensors [144]; this is the shared-embedding layout, in which the model keeps no separate output projection, and a checkpoint with zeroed feed-forward weights could not reach a perplexity in the low teens under any activation, so that reading is inconsistent with the result here. The file used is pinned to Hugging Face revision [a1f2f1c765812aa8af3f6eda4a313707064bba15](#), SHA-256 [4221b252fdd5fd25e15847adfeb5ee88886506ba50b8a34548374492884c2162](#). A separate, minor issue remains: the GGUF omits the `tokenizer.ggml.pre` field, so the runtime warns and falls back to a default pre-tokenizer; we did not run a matched-tokenization check to isolate its effect, but the activation-corrected perplexity and HellaSwag scores landing in the expected range suggest it is small. The two runtime perplexities that can be stated, 12.99 from `llama.cpp` and 16.67 from transformers (the accuracy table above), are not directly comparable, since one is a token-level and the other a word-level figure on different tokenizations; both are in the range the model card implies.

10.4 Efficiency results

Table 7, and Fig. 12, report decode throughput, prefill throughput, on-disk size, and peak resident memory. The Qwen baseline is shown at four 4-bit and low-bit settings, Q4_0, Q4_K_S, Q4_K_M, and Q8_0, plus FP16, so the 4-bit comparison does not hinge on one format. Throughput is reported at a common four-thread operating point: that is BitNet’s own decode optimum on this hybrid-core part, and holding it fixed keeps the comparison on one setting. It is not the global optimum for every model, so we also give each model its own best point below, and the ordering is the same either way. Prefill is the 512-token point; decode generates 128 tokens; each throughput figure is the mean of five repeats, with the standard deviation shown. Disk size is exact; peak resident memory is a single sample taken at eight threads and is essentially thread-independent.

Table 7: Independent efficiency re-evaluation on one x86 laptop at a common four-thread operating point: decode and prefill throughput (mean of five repeats, standard deviation in parentheses), on-disk size, and peak resident memory, for the BitNet I2_S checkpoint and Qwen2.5-1.5B at five GGUF quantization levels.

configuration	decode tok/s	prefill tok/s	disk (GiB)	peak RAM (GB)
BitNet b1.58 2B4T, I2_S	28.6 (1.6)	108.9 (5.2)	1.10	1.22
Qwen2.5-1.5B, Q4_0	36.3 (1.1)	116.1 (0.8)	0.99	1.66
Qwen2.5-1.5B, Q4_K_S	28.7 (4.8)	109.4 (5.4)	1.00	1.64
Qwen2.5-1.5B, Q4_K_M	33.8 (0.3)	107.6 (2.7)	1.04	1.60
Qwen2.5-1.5B, Q8_0	24.1 (0.1)	72.4 (3.1)	1.76	1.64
Qwen2.5-1.5B, FP16	14.4 (0.3)	70.9 (0.8)	3.31	3.02

Against FP16 the vendor’s direction reproduces: the ternary model decodes 1.99 times as fast, close to the roughly 2.2-times ratio the card’s own table implies. The memory claim does not carry over as stated: the card’s 0.4 GB against 2.6 GB, a 6.5-times advantage, is a weights-only, non-embedding accounting, whereas the running process needs 1.22 GB against FP16’s 3.02 GB once the 128k-vocabulary embedding, the key-value cache, and the compute buffers are counted, a 2.5-times advantage. Absolute decode is 35 ms per token against the card’s 29 ms, consistent given a different processor.

A comparison closer to deployment is against the 4-bit builds people actually run locally, and against every one of them, on this machine, the ternary model has no decode-speed or disk-size advantage and only a narrow memory one. At the common four-thread point it decodes no faster than any of the three 4-bit formats and slower than two of them: 28.6 tokens per second against 36.3 for Q4_0, 33.8 for Q4_K_M, and 28.7 for Q4_K_S, whose four-thread run was noisy and which reaches 33.8 at eight threads. All three 4-bit builds are smaller on disk (0.99 to 1.04 GiB against 1.10) and tie or beat it on prefill. The ternary model does keep a resident-memory edge, 1.22 GB against 1.60 to 1.66 GB, about 24 to 27%. Giving each model its own best thread count does not change the ordering: BitNet’s fastest decode is 28.6 at four threads, while Q4_0 reaches 36.3 at four and the two k-quants reach 33.8 to 34.2 at eight; BitNet’s fastest 512-token prefill is 112 at eight threads against 116 to 132 for the 4-bit formats at their best. Its 2.4B parameters, only 2.0B of them outside the token embedding (Section 10.2), are why: a naive estimate from parameter count and nominal bit-width alone, 2.4B at roughly 1.6 bits, predicts under half a gigabyte, well below the measured 1.10 GiB. The gap is the token-embedding table, which is not ternary-packed and is a larger share of this smaller model’s parameters than Qwen’s embedding is of its own 1.5B; that overhead, not a straightforward bit-width trade-off, is most of why the two on-disk sizes land so close together. Against Q8_0 the picture is intermediate: 1.19 times on decode, 0.74 times on memory. Decode throughput for every model peaks at four or eight threads on this hybrid-core part and falls off at 16 as the efficiency cores and memory bandwidth become the limit; prefill for every model keeps scaling to 8 or 16 threads. The 2.37-to-6.17-times bitnet.cpp speedups quoted elsewhere [65] are measured against FP16 at 3B parameters and above; this 2B, x86 point sits below that band, and against a 4-bit baseline on this machine it inverts for all three formats tested. Whether it also inverts on a GPU or a different CPU is not tested here.

The battery pass covers the ternary model and the 4-bit k-quant (Q4_K_M), the 8-bit build, and half precision, run three times across two battery-discharge sessions. On battery the processor runs at a lower power limit and that limit tightens as the battery drains, so whole-system energy per token drifts across runs, the ternary model measuring 1.9, 2.4, and 2.8 J per token in the three, and only the model-to-model ratios within a run are stable. Those ratios are consistent. In the mid-range run the ternary model and the k-quant both cost about 2.4 J per token, the 8-bit build about 3.8, and half precision about 5.3; across all three runs the ternary model sat at 0.88 to 1.00 times the energy of the k-quant, 0.59 to 0.69 times the 8-bit build, and 0.40 to 0.46 times half precision. The energy ranking follows the decode-throughput ranking exactly: the ternary model and the k-quant emit tokens fastest, so they amortize the machine’s large idle draw, 29 to 49 W depending on battery state, over the most tokens. Against the 4-bit k-quant, then, the ternary model has no energy-per-token advantage on this machine, the same result the decode-speed and disk-size comparisons give; its advantage is a little over half off against half precision and roughly a third off against the 8-bit build, and that is a speed effect, not a lower compute cost. An idle-subtracted compute-energy figure did not reproduce across runs, since the idle draw itself varied by almost 20 W between sessions and on a draining battery some model runs drew less than the bracketing idle windows, so only whole-system energy is reported. None of these whole-laptop figures is comparable to the model card’s 0.028 J, which is a CPU-side, non-embedding number.

10.5 Summary of findings

On the accuracy arm, five things held. Every released checkpoint loaded and ran under a single pinned harness with no code changes beyond a Windows Triton pin. Every determinism check reproduced exactly. The controlled Spectra pair put native ternary within 0.6 points of FP16 on the seven-task mean at 2.4 billion parameters, with a 14% higher perplexity, so the “downstream parity, worse language-modeling loss” pattern of the native-training literature reproduces under a matched protocol, not just in the uncontrolled vendor pairing. Our zero-shot MMLU, 52.07 for BitNet b1.58 2B4T against 60.03 for Qwen, closely tracks the card’s five-shot values under a different protocol and matches its roughly eight-point world-knowledge deficit. And GSM8K, re-measured with generation and the chat template, reproduces in direction the one benchmark on which the card leads: BitNet b1.58 2B4T scores 61.9 to our Qwen’s 57.5 on flexible-extract (Section 10.3).

What the controlled comparisons also show is a cost, and a boundary on the claim. Post-hoc 4-bit quantization of Qwen loses 2.7 points on the seven-task mean and 7.8 on GSM8K, more than native ternary loses at a similar scale. The 1.58-bit Falcon3-1B pair loses far more, 11.5 points on the mean, 17 on GSM8K, and essentially all of MMLU; that pair does not isolate scale from training recipe (Section 10.3), but it does show that constraining weights to 1.58 bits at the one-billion-parameter scale does not by itself deliver parity. Parity, where we see it, is a property of native ternary pre-training at multi-billion scale, and even there it is downstream parity with a worse language-modeling loss, not equivalence. Whether it survives above the measured regime is still untested (Section 11.1).

One vendor number is not directly comparable rather than reproduced or refuted: the BitNet b1.58 2B4T model card’s single headline “zero-shot average” of 54.19. Our seven-task suite gives 61.98, but the card’s figure is an average over a different and broader task set, taken with the chat template we did not apply, so the two cannot be placed side by side. It is a concrete instance of a vendor summary statistic that a reader cannot check without reconstructing the exact task list and prompt format.

On the efficiency arm the headline result is that, on this commodity x86 machine, a standard 4-bit quantization of the full-precision baseline matches or beats the flagship 1-bit model on every axis a practitioner would weigh except one. Against all three 4-bit builds the ternary model shows no decode-speed advantage (it is slower than two of them and ties the third) and no on-disk-size advantage; against the 4-bit k-quant it shows no energy-per-token advantage either, the two within 0 to 12% across three battery runs (Section 10.4). Its one retained edge is a 24-to-27% smaller resident set. On the two accuracy tasks that could be scored through the same runtime the k-quant stays close (Section 10.3). All of this holds whether threads are fixed at four or tuned per model.

The vendor’s own comparisons, made only against FP16, do reproduce in direction: the roughly 2-times decode advantage over FP16 holds, the 6.5-times memory advantage becomes 2.5-times once memory is counted as resident set rather than non-embedding weights, and whole-system energy per token is about 0.43 times FP16, tracking the decode-speed ratio. Against Q8_0 the decode edge is 1.2-times and the energy ratio about 0.6-times, again a speed effect. So the paradigm’s efficiency case at this scale holds against half precision and against Q8_0, but not against any of the tested 4-bit builds; it is also a case about these released checkpoints and this one machine, not a general property of the representation, since the comparison pairs a 2.4B ternary model against a 1.5B full-precision one and cannot separate the two. We did not test a GPU or a second machine; Section 10.7 discusses how far this is likely to carry.

Two further findings came out of the efficiency arm. The 4-bit k-quant of the baseline costs 4.5% on perplexity and, on the 1000-item HellaSwag subset, a difference of 0.3 points that we did not test for significance (Section 10.3), so the faster, smaller k-quant stays close in accuracy on the tasks checked. And the reference CPU runtime, built from the project as documented, does not produce correct output for BitNet b1.58 2B4T until a source constant is changed: its BitNet graph applies a SiLU feed-forward activation where the model uses gated squared-ReLU, a known open issue. Uncorrected, WikiText-2 perplexity is 86.9 and HellaSwag 38.7%; the one-line fix brings these to 12.99 and 62.4% (Section 10.3). The released weights are sound. The throughput and memory figures of Section 10.4 are unchanged by the fix, since a squared-ReLU is no more costly than a SiLU and the activation is a negligible fraction of the per-token work.

Two benchmarks that an earlier version of this study left out are now re-measured. MMLU, dropped because five-shot evaluation thrashed BitNet’s per-shape compilation cache, is re-measured at zero shot; GSM8K, which needs generation and a chat template rather than the loglikelihood path used for everything else, is re-measured at five shot with the template and reported separately (Section 10.3). Both reproduce the corresponding vendor figures. Nothing in the accuracy arm now rests on an unverified vendor number.

10.6 Reproducibility audit

For the checkpoints in the accuracy arm we record, as a by-product, what license governs reuse, whether the weights load under a stock harness, and whether our numbers land near the vendor’s. The controlled-comparison checkpoints of Section 10.3, the Spectra TriLM and FloatLM pair, Falcon3-1B in bf16, and the nf4 quantization of Qwen, all loaded through the stock `AutoModelForCausalLM` path with no special handling and are not repeated in Table 8.

Table 8: Reproducibility audit for the accuracy-arm checkpoints: governing license, whether the weights load under a stock harness, and whether our measurements land near the vendor’s.

checkpoint / artifact	license	loads and runs	reproduces
BitNet b1.58 2B4T, bf16 weights (transformers)	MIT	yes	per-task yes, headline average no
BitNet b1.58 2B4T, released I2_S GGUF (bitnet.cpp)	MIT	only after a source patch	yes once patched (ppl 12.99)
Falcon3-1B-Instruct-1.58bit	Falcon LLM License 2.0	yes	no public per-task baseline
Qwen2.5-1.5B-Instruct	Apache-2.0	yes	full-precision reference

The BitNet rows are the point of this audit. The bf16 weights load as a native transformers model type, so the model card’s `trust_remote_code` instruction is stale, and on Windows the layer compiles only against a matched Triton build; run that way the model is competitive (Section 10.3). The released GGUF, which is the only artifact that delivers the efficiency the whole approach is for, loads but returns near-broken output from the reference runtime as built, because that runtime applies the wrong feed-forward activation to this model (Section 10.3); a one-line source change fixes it, but a practitioner following the documented build would not know to make it. The released weights

themselves are sound. Falcon3 and Qwen load through the stock `AutoModelForCausalLM` path with no special handling. License terms vary across the corpus, with MIT, Apache, and vendor-specific licenses all appearing, which matters for any downstream reuse. A full audit across every released model in the corpus is beyond this paper’s scope.

10.7 Threats

Both arms run on a single laptop. For accuracy the GPU is shared with other workloads, which we mitigate with a fixed batch size and off-peak scheduling; for efficiency the machine runs under its default thermal and power policy, so the absolute throughput figures are specific to that part and not a controlled-environment measurement. The energy figures are coarser still: the discharge-rate sensor is noisy, the on-battery power limit falls as the battery drains so absolute per-token energy drifts across runs, and the numbers are whole-system rather than package or component; they support the within-run model-to-model ratios (Section 10.4) but not an absolute claim, and we do not report an idle-subtracted figure because it did not reproduce. The author-reported baselines cannot be matched exactly, and benchmark contamination is a risk in heavily fine-tuned small models, which we note per model.

The efficiency findings are for a CPU datapath and may not transfer to a GPU, and the transfer is worth reasoning about because the strong claim, that a 4-bit build is at least as good, could plausibly flip. On a GPU neither ternary nor 4-bit weights map to a native matrix-multiply instruction: both are unpacked to a supported floating-point or integer type and run on the tensor cores, or go through a custom kernel. The decode-time benefit that low-bit weights capture on CPU is memory-bandwidth, and that benefit exists on GPU too and accrues to a 4-bit kernel as much as to a ternary one, so the CPU result that a 4-bit kernel captures it is a reasonable prior for the GPU case. Two things could still separate them there. A ternary-specific GPU kernel that exploits the zero state or a sub-1.6-bit packing (Section 7.3) might pull ahead of the tuned 4-bit kernels now standard for served inference, in a way the general-purpose CPU path does not expose. And prefill and high-batch decode are compute-bound rather than bandwidth-bound, so their ordering need not follow the single-stream decode ordering measured here. We tested none of this, and a GPU comparison against a production 4-bit kernel is the obvious next step.

Every part of this reproduction is pinned to a specific commit or file rather than a moving target: the evaluation harness at commit `b954108` (Section 10.2); microsoft/BitNet at commit `0b341e5` with its `llama.cpp` submodule at `390c307` and the one-line `LLM_FFN_SILU-to-LLM_FFN_RELU_SQR` activation fix (Section 10.3); and the BitNet GGUF checkpoint by Hugging Face revision and SHA-256 (Section 10.3). The Qwen GGUF builds were produced with `llama-quantize` from one pinned FP16 GGUF, and the nf4 quantization with `bitsandbytes` 0.50.2. Every accuracy and efficiency number reported in this section is the direct output of that pinned pipeline on the machine described in Section 10.2, run once per configuration except where a repeat count is stated.

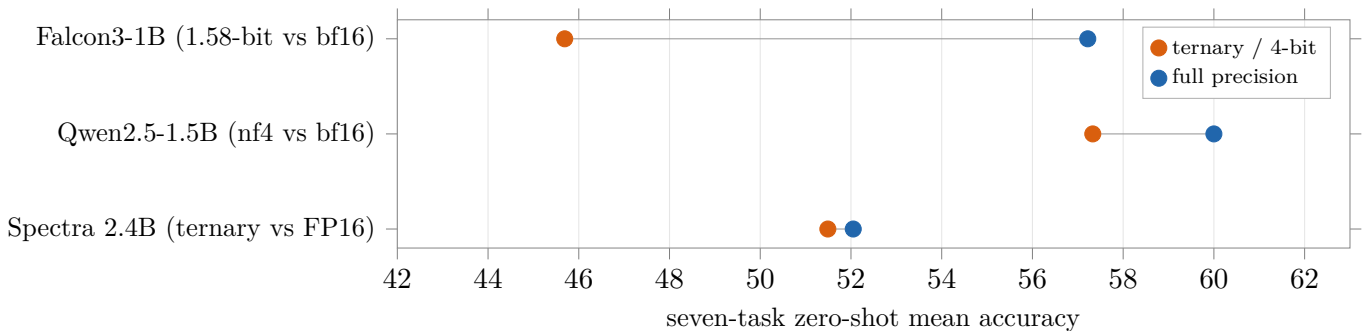


Figure 11: The three controlled pairs of Table 4: seven-task zero-shot mean for each ternary or 4-bit model (orange) against its full-precision counterpart (blue), in decreasing order of match quality. The Spectra pair lands within 0.6 points; the least-matched Falcon3-1B pair loses 11.5.

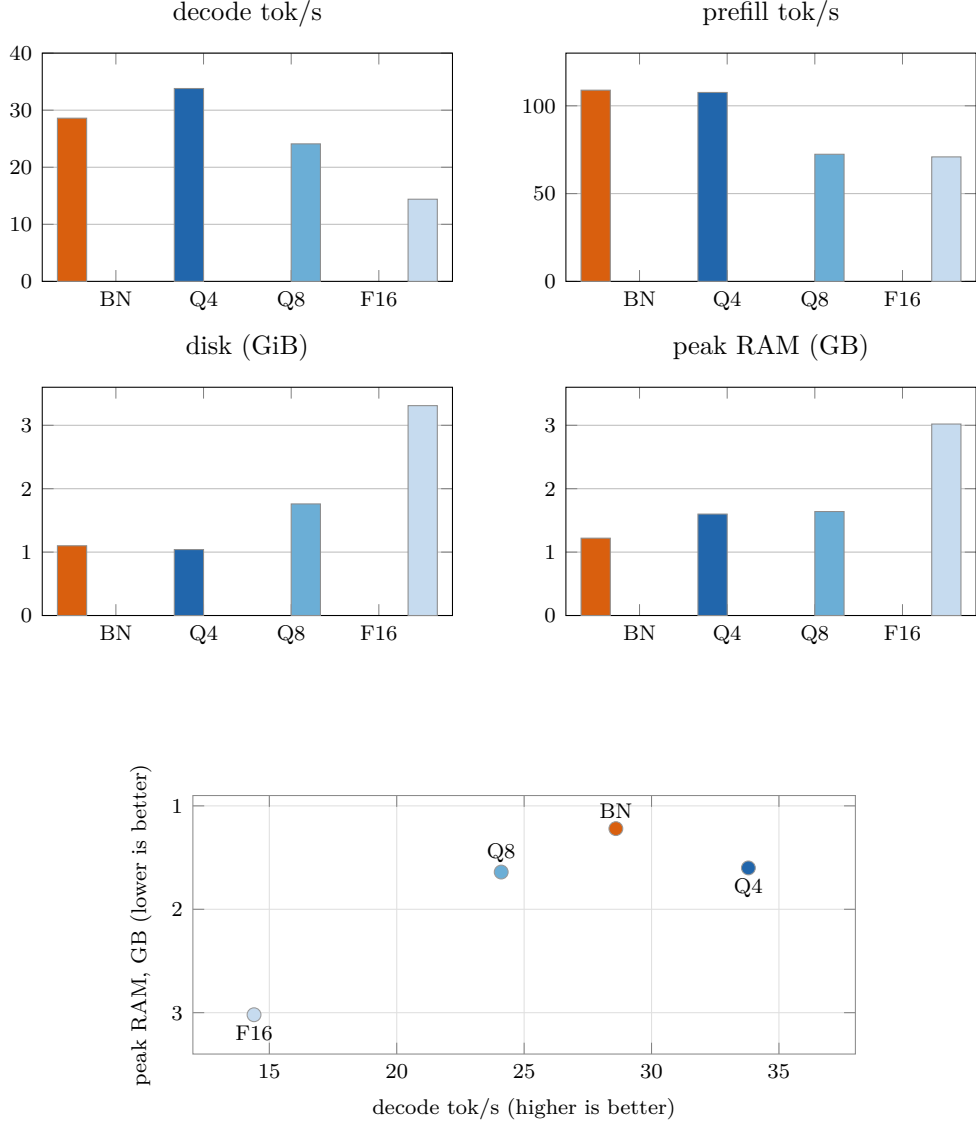


Figure 12: Independent efficiency re-evaluation on one x86 laptop: decode throughput (mean of five repeats at a common four-thread point), on-disk size (exact), and peak resident memory (a single sample at eight threads, essentially thread-independent). BN (orange) is the BitNet b1.58 2B4T I2_S checkpoint; Q4, Q8, and F16 (blues) are Qwen2.5-1.5B-Instruct at Q4_K_M, Q8_0, and FP16. The bottom panel plots decode throughput against peak memory: BitNet and Q4 share the Pareto frontier, while Q8 and F16 are dominated on both axes.

11 Challenges and Open Problems

Each of the following is stated as a gap that could be closed, not as a wish list.

11.1 The capacity ceiling and its dependence on the token budget

The first open problem is whether the full-precision parity of BitNet b1.58 is a property that survives to frontier scale and token budgets, or an artifact of the two- to four-trillion-token regime in which it has been measured. The two bodies of evidence point in opposite directions, though, as Section 6.6 notes, they measure different quantities: precision-aware scaling laws and the quantization-induced degradation study are about post-training quantization of a fixed model, and find that its degradation grows with training tokens, projecting that a model quantized to low bits after being trained on 20 to 100 trillion tokens would fall progressively further behind its FP16 source [4, 5]. These findings

motivate testing whether an analogous degradation appears in native ternary training itself, at frontier token counts; they do not, by themselves, establish that it will. The native ternary suites and ParetoQ report the opposite trend within their range: ternary improves on a bits-for-bits basis as data scales, and a 600M ParetoQ ternary model beats a 3B prior-state-of-the-art ternary model [1, 20, 38]. Our controlled re-run of Spectra’s matched pair (Section 10.3) confirms downstream parity at 2.4B parameters and 300B tokens, with a 14% higher perplexity, but that is well inside the measured regime. These findings have not been reconciled, because no native 1-bit model has been trained at both a frontier parameter count and a frontier token count. This is the single most important experiment that the field is missing, and it is expensive but not prohibitively so.

11.2 Training stability at frontier scale

The second open problem is whether native 1.58-bit pre-training becomes unstable, with loss spikes or divergence, once the model is much larger than today’s and the token budget grows with it, and whether the current mitigations only delay that instability. The largest reported native run is 4T tokens at 2B parameters [9], so every stability claim in the literature is at or below that scale; a 20B or 200B native model on tens of trillions of tokens is untried. Continual quantization-aware training already documents loss spikes at the full-precision-to-1.58-bit transition and partial fixes, namely optimizer-state retention and a quantization warmup [37, 50], and the deadzone-trapping analysis of Tequila and the gated low-rank correction of HGF address related instabilities [109, 128]. Whether these mitigations compose into a stable run an order of magnitude larger is unknown.

11.3 The activation-precision floor

The third open problem is whether four bits is the practical floor for activations alongside roughly one-bit weights, or whether rotation together with sparsification can reach two-bit activations. Native A4 is solved by hybrid quantization and sparsification [12] and by an online Hadamard transform [13]; post-training quantization mostly reaches A6 [35, 107], with one single-preprint report of A4 through rotation [34] not yet corroborated; and QuEST demonstrates one-bit weights and activations but only below 1B parameters [29]. Nobody has demonstrated W1.58A2 at multi-billion scale with acceptable accuracy. Given that activation traffic now dominates once weights occupy roughly one bit, this is where the next efficiency step must come from.

11.4 A theory of straight-through training

The fourth open problem is the absence of a general convergence theory for straight-through training of 1-bit Transformers; only special cases are understood. CAGE proves convergence for its curvature-corrected estimator in the smooth non-convex setting [28]; a kernel-limit analysis proves a scaling law for 1-bit networks as width grows [39]; and precision-expressivity trade-offs are being formalized [40, 41]. What is missing is an account that connects the estimator bias to the observed instabilities and predicts which schedules and estimators avoid them.

11.5 Hardware co-design

The fifth open problem is that no shipping processor exposes a ternary matrix-multiplication primitive, and only five works in the corpus co-design across algorithm, kernel, and hardware [8, 9], [11, 58, 72]. The accelerator literature on lookup-table ASICs, compute-in-ROM, processing-in-memory, and edge FPGAs [73, 77, 78, 82], [84, 85] reports simulator and single-chip results. An add-and-subtract matrix-multiplication unit with a matching memory layout and a small instruction-set extension, co-designed with the training recipe and the sparsity pattern, does not exist. This gap is why the efficiency case does not close on today’s hardware: our re-evaluation (Section 10) finds that on a commodity CPU the flagship ternary model has no throughput, footprint, or per-token energy advantage over a 4-bit baseline, because the ternary matrix multiplication still runs on a general-purpose datapath that a 4-bit kernel uses at least as well. The Sparse-BitNet result, that ternary models tolerate structured

sparsity far better than full-precision models [58], suggests that the co-design target should be sparse ternary rather than dense ternary.

11.6 Reasoning, generation quality, and smoothness

The sixth open problem concerns generation quality. Extreme quantization degrades output smoothness, that is, the diversity of plausible next tokens, independently of perplexity, which sparsifies the decoding tree [132]; and two-bit reasoning models fail by not committing to an answer, which inflates the trace length [131]. Whether native 1.58-bit training, as opposed to post-hoc two-bit quantization, suffers the same pathologies, and whether smoothness-preserving objectives fix them at scale, is open.

11.7 Robustness, safety, and security

The seventh open problem is that the robustness and security of ternary models are under-studied. GenBFA shows that three bit-flips can collapse a quantized billion-parameter LLM [102]; MatMul-free LM is reported as more robust to word-level adversarial attacks than a standard Transformer, with a mixed picture at the character level [106]; and the fault tolerance of compute-in-memory ternary LLMs is beginning to be studied. Whether ternary weights are systematically more or less fragile than FP16, with respect to bit-flips, adversarial inputs, backdoors, and membership inference, is not established, and the deployment story of models on phones and in enclaves makes the question urgent.

11.8 Sub-1.58-bit and alternative representations

The eighth open problem is whether a representation past ternary is the better place to push. Structural binarization below one bit [18], latent factorization to about 0.1 bits [19], 1.25-bit regularized sparsity [21], and complex $\{\pm 1, \pm i\}$ codebooks [48, 49] all push past 1.58 bits. Whether any of these is a better operating point than ternary, once kernel and hardware support is accounted for, is unresolved.

11.9 Evaluation and contamination

The ninth open problem is that the field lacks a standard evaluation harness and a contamination control of its own. Native 1-bit models are small and heavily fine-tuned, which raises the risk of benchmark contamination; different papers use different harness versions, prompt formats, and few-shot counts (Section 8). A shared, versioned evaluation protocol for 1-bit LLMs would make the literature comparable.

12 Conclusion

Native 1-bit and 1.58-bit large language models have moved, in three years, from a single proof of concept to a coherent research program, with an open 2B-parameter model, a CPU inference stack, a first wave of custom-silicon proposals, and extensions into embeddings, speech, vision-language-action models, mixture-of-experts, and state-space models; the defining idea is not aggressive compression but a change of arithmetic, since when every weight is ternary the matrix multiplication becomes an addition, and the questions that follow, how to train through the discrete constraint, how far activation precision can drop, and what the ideal hardware looks like, are different in kind from those of four-bit quantization.

We surveyed the 135 open-access works of that program on a five-axis taxonomy and re-tabulated their reported numbers in consistent columns under one effective-bits accounting, then re-evaluated the two central claims ourselves. In a controlled comparison from open matched checkpoints, native ternary against FP16 at the same 2.4-billion parameters, tokenizer, and training data, the two came within 0.6 points on a seven-task zero-shot mean, with the ternary model’s perplexity about 14% higher: downstream parity with a worse language-modeling loss. Post-hoc 4-bit quantization of a

full-precision model lost more of that suite, and a 1.58-bit model at the one-billion-parameter scale, on a checkpoint pair whose training match is undocumented, lost far more; that pattern is consistent with, but too thin a basis on its own to establish, a scale dependence for the parity result, and whether it holds above the measured regime is still open.

The efficiency result cuts the other way: on the one x86 machine we tested, the flagship 1-bit model held no advantage in decode speed, on-disk size, or per-token energy over an ordinary 4-bit quantization of the full-precision baseline, and stayed only level with it on the accuracy tasks we could check, leaving a 24-to-27% smaller resident set as the only advantage that remained. The paradigm’s efficiency argument at this scale therefore stands on the comparison with half precision and with an 8-bit baseline, not with the tested 4-bit builds, and whether it recovers against those on a GPU, or against a production 4-bit kernel generally, is the open question that Section 10.7 lays out; that the reference CPU runtime does not even run the flagship model correctly as documented, without a one-line source fix, says the same thing from the software side, since the open deployment path is not yet mature.

The paradigm’s promise is thus a change of arithmetic that current commodity software and hardware do not yet cash in, and the open problems that remain are concrete: whether native 1.58-bit training reaches full-precision parity at frontier scale and token budgets is untested and is the most important missing experiment in the field, and training stability at frontier scale, the activation-precision floor below four bits, a convergence theory for straight-through training, and, above all, the absence of any shipping processor with a ternary matrix-multiplication primitive are the barriers between the current results and a future in which 1-bit models are a default rather than a curiosity. The evidence so far is that the paradigm is real; whether it becomes standard depends on co-design work that, for the most part, has not yet been done.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used Claude (Anthropic) to assist with literature search support, drafting and editing manuscript text, generating figures from author-specified content and data, and scripting the independent reproduction experiments in Section 10. The reported measurements were produced by running the cited third-party software and released model checkpoints directly, not by the AI assistant. All AI-assisted output was reviewed and verified by the authors, who take full responsibility for the content and conclusions of this manuscript.

References

- [1] Ayush Kaushal, Tejas Vaidhya, Arnab Kumar Mondal, Tejas Pandey, Aaryan Bhagat, and Irina Rish. Spectra: Surprising effectiveness of pretraining ternary language models at scale, 2024. URL <https://arxiv.org/abs/2407.12327>. arXiv:2407.12327.
- [2] Wei Huang, Yangdong Liu, Haotong Qin, Ying Li, Shiming Zhang, Xianglong Liu, Michele Magno, and Xiaojuan Qi. Billm: Pushing the limit of post-training quantization for llms. In *International Conference on Machine Learning (ICML)*, 2024. doi: 10.48550/arXiv.2402.04291. URL <https://arxiv.org/abs/2402.04291>. Also arXiv:2402.04291.
- [3] Yuzhang Shang, Zhihang Yuan, Qiang Wu, and Zhen Dong. Pb-llm: Partially binarized large language models. In *International Conference on Learning Representations (ICLR)*, 2023. doi: 10.48550/arXiv.2310.00034. URL <https://arxiv.org/abs/2310.00034>. Also arXiv:2310.00034.
- [4] Xu Ouyang, Tao Ge, Thomas Hartvigsen, Zhisong Zhang, Haitao Mi, and Dong Yu. Low-bit quantization favors undertrained llms: Scaling laws for quantized llms with 100t training tokens, 2024. URL <https://arxiv.org/abs/2411.17691>. arXiv:2411.17691.

- [5] Tanishq Kumar, Zachary Ankner, Benjamin F. Spector, Blake Bordelon, Niklas Muennighoff, Mansheej Paul, Cengiz Pehlevan, Christopher Ré, and Aditi Raghunathan. Scaling laws for precision. In *International Conference on Learning Representations (ICLR)*, 2024. doi: 10.48550/arXiv.2411.04330. URL <https://arxiv.org/abs/2411.04330>. Also arXiv:2411.04330.
- [6] Hongyu Wang, Shuming Ma, Li Dong, Shaohan Huang, Huaijie Wang, Lingxiao Ma, Fan Yang, Ruiping Wang, Yi Wu, and Furu Wei. Bitnet: Scaling 1-bit transformers for large language models. *Journal of Machine Learning Research*, 2023. doi: 10.48550/arXiv.2310.11453. URL <https://arxiv.org/abs/2310.11453>. Also arXiv:2310.11453.
- [7] Hongyu Wang, Shuming Ma, Lingxiao Ma, Lei Wang, Wenhui Wang, Li Dong, Shaohan Huang, Huaijie Wang, Jilong Xue, Ruiping Wang, Yi Wu, and Furu Wei. Bitnet: 1-bit pre-training for large language models. *Journal of Machine Learning Research*, 26(125):1–29, 2025. URL <http://jmlr.org/papers/v26/24-2050.html>.
- [8] Shuming Ma, Hongyu Wang, Lingxiao Ma, Lei Wang, Wenhui Wang, Shaohan Huang, Li Dong, Ruiping Wang, Jilong Xue, and Furu Wei. The era of 1-bit llms: All large language models are in 1.58 bits, 2024. URL <https://arxiv.org/abs/2402.17764>. arXiv:2402.17764.
- [9] Shuming Ma, Hongyu Wang, Shaohan Huang, Xingxing Zhang, Ying Hu, Ting Song, Yan Xia, and Furu Wei. Bitnet b1.58 2b4t technical report, 2025. URL <https://arxiv.org/abs/2504.12285>. arXiv:2504.12285.
- [10] Jinendra Malekar, Mohammed E. Elbittity, and Ramtin Zand. Matmul or no matmul in the era of 1-bit llms, 2024. URL <https://arxiv.org/abs/2408.11939>. arXiv:2408.11939.
- [11] Rui-Jie Zhu, Yu Zhang, Steven Abreu, Ethan Sifferman, Tyler Sheaves, Yiqiao Wang, Dustin Richmond, Sumit Bam Shrestha, Peng Zhou, and Jason K. Eshraghian. Scalable matmul-free language modeling, 2024. URL <https://arxiv.org/abs/2406.02528>. arXiv:2406.02528.
- [12] Hongyu Wang, Shuming Ma, and Furu Wei. Bitnet a4.8: 4-bit activations for 1-bit llms, 2024. URL <https://arxiv.org/abs/2411.04965>. arXiv:2411.04965.
- [13] Hongyu Wang, Shuming Ma, and Furu Wei. Bitnet v2: Native 4-bit activations with hadamard transformation for 1-bit llms, 2025. URL <https://arxiv.org/abs/2504.18415>. arXiv:2504.18415.
- [14] Ruihao Gong, Yifu Ding, Zining Wang, Chengtao Lv, Xingyu Zheng, Jinyang Du, Jinyang Guo, Xianglong Liu, Haotong Qin, Michele Magno, Yang Yong, Shiqiao Gu, and Dahua Lin. A survey of low-bit large language models: Basics, systems, and algorithms. *Neural Networks*, 2024. doi: 10.48550/arXiv.2409.16694. URL <https://arxiv.org/abs/2409.16694>. Also arXiv:2409.16694.
- [15] Liangdong Liu, Zhitong Zheng, Cong Wang, Tianhuang Su, and Zhenyu Yang. Binary neural networks for large language model: A survey, 2025. URL <https://arxiv.org/abs/2502.19008>. arXiv:2502.19008.
- [16] Zhiwei Hao, Jianyuan Guo, Li Shen, Yong Luo, Han Hu, Guoxia Wang, Dianhai Yu, Yonggang Wen, and Dacheng Tao. Low-precision training of large language models: Methods, challenges, and opportunities. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025. doi: 10.48550/arXiv.2505.01043. URL <https://arxiv.org/abs/2505.01043>. Also arXiv:2505.01043.
- [17] Maharshi Savdhariya. Nativeternary: A self-delimiting binary encoding for ternary neural network weights, 2026. URL <https://arxiv.org/abs/2604.03336>. arXiv:2604.03336.

- [18] Peijie Dong, Lujun Li, Yuedong Zhong, Dayou Du, Ruibo Fan, Yuhan Chen, Zhenheng Tang, Qiang Wang, Wei Xue, Yike Guo, and Xiaowen Chu. Stblm: Breaking the 1-bit barrier with structured binary llms. In *International Conference on Learning Representations (ICLR)*, 2024. doi: 10.48550/arXiv.2408.01803. URL <https://arxiv.org/abs/2408.01803>. Also arXiv:2408.01803.
- [19] Banseok Lee, Dongkyu Kim, Youngcheon You, and Youngmin Kim. Littlebit: Ultra low-bit quantization via latent factorization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. doi: 10.48550/arXiv.2506.13771. URL <https://arxiv.org/abs/2506.13771>. Also arXiv:2506.13771.
- [20] Tejas Vaidhya, Ayush Kaushal, Vineet Jain, Francis Couture-Harpin, Prashant Shishodia, Majid Behbahani, Yuriy Nevmyvaka, and Irina Rish. Spectra 1.1: Scaling laws and efficient inference for ternary language models, 2025. URL <https://arxiv.org/abs/2506.23025>. arXiv:2506.23025.
- [21] Hong Huang, Decheng Wu, Qiangqiang Hu, Guanghua Yu, Jinhai Yang, Jianchen Zhu, Xue Liu, and Dapeng Wu. Sherry: Hardware-efficient 1.25-bit ternary quantization via fine-grained sparsification, 2026. URL <https://arxiv.org/abs/2601.07892>. arXiv:2601.07892.
- [22] Matthieu Courbariaux, Yoshua Bengio, and Jean-Pierre David. Binaryconnect: Training deep neural networks with binary weights during propagations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015. doi: 10.48550/arXiv.1511.00363. URL <https://arxiv.org/abs/1511.00363>. Also arXiv:1511.00363.
- [23] Mohammad Rastegari, Vicente Ordonez, Joseph Redmon, and Ali Farhadi. Xnor-net: Imagenet classification using binary convolutional neural networks, 2016. URL <https://arxiv.org/abs/1603.05279>. arXiv:1603.05279.
- [24] Haoli Bai, Wei Zhang, Lu Hou, Lifeng Shang, Jing Jin, Xin Jiang, Qun Liu, Michael Lyu, and Irwin King. Binarybert: Pushing the limit of bert quantization. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2021. doi: 10.48550/arXiv.2012.15701. URL <https://arxiv.org/abs/2012.15701>. Also arXiv:2012.15701.
- [25] Haotong Qin, Yifu Ding, Mingyuan Zhang, Qinghua Yan, Aishan Liu, Qingqing Dang, Ziwei Liu, and Xianglong Liu. Bibert: Accurate fully binarized bert. In *International Conference on Learning Representations (ICLR)*, 2022. doi: 10.48550/arXiv.2203.06390. URL <https://arxiv.org/abs/2203.06390>. Also arXiv:2203.06390.
- [26] Yoshua Bengio, Nicholas Léonard, and Aaron Courville. Estimating or propagating gradients through stochastic neurons for conditional computation, 2013. URL <https://arxiv.org/abs/1308.3432>. arXiv:1308.3432.
- [27] Jacob Nielsen and Peter Schneider-Kamp. Bitnet b1.58 reloaded: State-of-the-art performance also on smaller networks, 2024. URL <https://arxiv.org/abs/2407.09527>. arXiv:2407.09527.
- [28] Soroush Tabesh, Mher Safaryan, Andrei Panferov, Alexandra Volkova, and Dan Alistarh. Cage: Curvature-aware gradient estimation for accurate quantization-aware training. In *Proceedings of Machine Learning and Systems (MLSys)*, 2025. doi: 10.48550/arXiv.2510.18784. URL <https://arxiv.org/abs/2510.18784>. Also arXiv:2510.18784.
- [29] Andrei Panferov, Jiale Chen, Soroush Tabesh, Roberto L. Castro, Mahdi Nikdan, and Dan Alistarh. Quest: Stable training of llms with 1-bit weights and activations, 2025. URL <https://arxiv.org/abs/2502.05003>. arXiv:2502.05003.
- [30] Ningfeng Yang and Tor M. Aamodt. Improving the straight-through estimator with zeroth-order information. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. doi: 10.48550/arXiv.2510.23926. URL <https://arxiv.org/abs/2510.23926>. Also arXiv:2510.23926.

- [31] Kaiyan Zhao, Tsuguchika Tabaru, Kenichi Kobayashi, Takumi Honda, Masafumi Yamazaki, and Yoshimasa Tsuruoka. Direct quantized training of language models with stochastic rounding. In *Asian Conference on Machine Learning (ACML)*, 2024. doi: 10.48550/arXiv.2412.04787. URL <https://arxiv.org/abs/2412.04787>. Also arXiv:2412.04787.
- [32] Vladimir Malinovskii, Denis Mazur, Ivan Ilin, Denis Kuznedelev, Konstantin Burlachenko, Kai Yi, Dan Alistarh, and Peter Richtarik. Pv-tuning: Beyond straight-through estimation for extreme llm compression. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. doi: 10.48550/arXiv.2405.14852. URL <https://arxiv.org/abs/2405.14852>. Also arXiv:2405.14852.
- [33] Artur Zagitov, Gleb Molodtsov, and Aleksandr Beznosikov. Harp: Hadamard-preconditioned adaptive rotation processor for extreme llm quantization, 2026. URL <https://arxiv.org/abs/2605.29843>. arXiv:2605.29843.
- [34] Zhixiong Zhao, Zukang Xu, Zhixuan Chen, Xing Hu, Zhe Jiang, and Dawei Yang. Twla: Achieving ternary weights and low-bit activations for llms via post-training quantization, 2026. URL <https://arxiv.org/abs/2606.13054>. arXiv:2606.13054.
- [35] Zhixiong Zhao, Zukang Xu, and Dawei Yang. Bwla: Breaking the barrier of w1ax post-training quantization for llms, 2026. URL <https://arxiv.org/abs/2605.00422>. arXiv:2605.00422.
- [36] Cody Steinmetz, Gavin Childress, Aaron Herbst, Gavin Jones, Jasdeep Singh, Eli Vang, and Keagan Weinstock. An extra rmsnorm is all you need for fine tuning to 1.58 bits, 2025. URL <https://arxiv.org/abs/2505.08823>. arXiv:2505.08823.
- [37] Jacob Nielsen, Peter Schneider-Kamp, and Lukas Galke. Continual quantization-aware pre-training: When to transition from 16-bit to 1.58-bit pre-training for bitnet language models?, 2025. URL <https://arxiv.org/abs/2502.11895>. arXiv:2502.11895.
- [38] Zechun Liu, Changsheng Zhao, Hanxian Huang, Sijia Chen, Jing Zhang, Jiawei Zhao, Scott Roy, Lisa Jin, Yunyang Xiong, Yangyang Shi, Lin Xiao, Yuandong Tian, Bilge Soran, Raghuraman Krishnamoorthi, Tijmen Blankevoort, and Vikas Chandra. Paretoq: Improving scaling laws in extremely low-bit llm quantization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. doi: 10.48550/arXiv.2502.02631. URL <https://arxiv.org/abs/2502.02631>. Also arXiv:2502.02631.
- [39] Majid Daliri, Zhao Song, and Chiwun Yang. Unlocking the theory behind scaling 1-bit neural networks, 2024. URL <https://arxiv.org/abs/2411.01663>. arXiv:2411.01663.
- [40] Sayak Chakrabarti, Toniann Pitassi, and Josh Alman. Every bit counts: A theoretical study of precision-expressivity tradeoffs in quantized transformers, 2026. URL <https://arxiv.org/abs/2602.02707>. arXiv:2602.02707.
- [41] Shao-Qun Zhang. On the expressive power of weight quantization in large language models, 2026. URL <https://arxiv.org/abs/2606.22249>. arXiv:2606.22249.
- [42] Zhiteng Li, Xianglong Yan, Tianao Zhang, Haotong Qin, Dong Xie, Jiang Tian, Zhongchao Shi, Linghe Kong, Yulun Zhang, and Xiaokang Yang. Arb-llm: Alternating refined binarizations for large language models. In *International Conference on Learning Representations (ICLR)*, 2024. doi: 10.48550/arXiv.2410.03129. URL <https://arxiv.org/abs/2410.03129>. Also arXiv:2410.03129.
- [43] Yuzhuang Xu, Xu Han, Zonghan Yang, Shuo Wang, Qingfu Zhu, Zhiyuan Liu, Weidong Liu, and Wanxiang Che. Onebit: Towards extremely low-bit large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. doi: 10.48550/arXiv.2402.11295. URL <https://arxiv.org/abs/2402.11295>. Also arXiv:2402.11295.

- [44] Liqun Ma, Mingjie Sun, and Zhiqiang Shen. Fbi-llm: Scaling up fully binarized llms from scratch via autoregressive distillation, 2024. URL <https://arxiv.org/abs/2407.07093>. arXiv:2407.07093.
- [45] Zechun Liu, Barlas Oğuz, and Yangyang Shi. Binary and ternary natural language generation, 2023. URL <https://arxiv.org/abs/2306.01841>. arXiv:2306.01841.
- [46] Wenzheng Zhang, Bingzheng Liu, Yang Hu, Xiaoying Bai, Wentao Zhang, and Bin Cui. pquant: Towards effective low-bit language models via decoupled linear quantization-aware training, 2026. URL <https://arxiv.org/abs/2602.22592>. arXiv:2602.22592.
- [47] Rayyan Abdalla, Amir Hussein, Min Wu, and Dinesh Manocha. Minimizing the hidden cost of scales: Graph-guided ultra-low-bit quantization for large language models, 2026. URL <https://arxiv.org/abs/2606.05429>. arXiv:2606.05429.
- [48] Feiyu Wang, Guoan Wang, Yihao Zhang, Shengfan Wang, Weitao Li, Bokai Huang, Shimao Chen, Zihan Jiang, Rui Xu, and Tong Yang. ifairy: the first 2-bit complex llm with all parameters in $\pm 1, \pm i$, 2025. URL <https://arxiv.org/abs/2508.05571>. arXiv:2508.05571.
- [49] Feiyu Wang, Xinyu Tan, Bokai Huang, Yihao Zhang, Guoan Wang, Peizhuang Cong, and Tong Yang. Fairy2i: Training complex llms from real llms, 2025. URL <https://arxiv.org/abs/2512.02901>. arXiv:2512.02901.
- [50] Zhijun Tu, Jian Li, Yuanyuan Xi, Siqi Liu, Chuanjian Liu, Hanting Chen, Jie Hu, and Yunhe Wang. Rethinking 1-bit optimization leveraging pre-trained large language models, 2025. URL <https://arxiv.org/abs/2508.06974>. arXiv:2508.06974.
- [51] Dayou Du, Yijia Zhang, Shijie Cao, Jiaqi Guo, Ting Cao, Xiaowen Chu, and Ningyi Xu. Bitdistiller: Unleashing the potential of sub-4-bit llms via self-distillation. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024. doi: 10.48550/arXiv.2402.10631. URL <https://arxiv.org/abs/2402.10631>. Also arXiv:2402.10631.
- [52] Xun Wu, Shaohan Huang, Wenhui Wang, Ting Song, Li Dong, Yan Xia, and Furu Wei. Bitnet distillation, 2025. URL <https://arxiv.org/abs/2510.13998>. arXiv:2510.13998.
- [53] He Xiao, Runming Yang, Qingyao Yang, Wendong Xu, Zhen Li, Yupeng Su, Zhengwu Liu, Hongxia Yang, and Ngai Wong. Ptqtp: Post-training quantization to trit-planes for large language models, 2025. URL <https://arxiv.org/abs/2509.16989>. arXiv:2509.16989.
- [54] Xianglong Yan, Chengzhu Bao, Zhiteng Li, Tianao Zhang, Kaicheng Yang, Haotong Qin, Ruobing Xie, Xingwu Sun, and Yulun Zhang. Pt2-llm: Post-training ternarization for large language models. In *International Conference on Learning Representations (ICLR)*, 2025. doi: 10.48550/arXiv.2510.03267. URL <https://arxiv.org/abs/2510.03267>. Also arXiv:2510.03267.
- [55] Yuzhuang Xu, Shiyu Ji, Qingfu Zhu, and Wanxiang Che. Crvq: Channel-relaxed vector quantization for extreme compression of llms, 2024. URL <https://arxiv.org/abs/2412.09282>. arXiv:2412.09282.
- [56] Ali Edalati, Alireza Ghaffari, Mahsa Ghazvini Nejad, Lu Hou, Boxing Chen, Masoud Asgharian, and Vahid Partovi Nia. Oac: Output-adaptive calibration for accurate post-training quantization, 2024. URL <https://arxiv.org/abs/2405.15025>. arXiv:2405.15025.
- [57] Hongyu Wang, Shuming Ma, Ruiping Wang, and Furu Wei. Q-sparse: All large language models can be fully sparsely-activated, 2024. URL <https://arxiv.org/abs/2407.10969>. arXiv:2407.10969.

- [58] Di Zhang, Xun Wu, Shaohan Huang, Yudong Wang, Hanyong Shao, Yingbo Hao, Zewen Chi, Li Dong, Ting Song, Yan Xia, Zhifang Sui, and Furu Wei. Sparse-bitnet: 1.58-bit llms are naturally friendly to semi-structured sparsity, 2026. URL <https://arxiv.org/abs/2603.05168>. arXiv:2603.05168.
- [59] Jiawen Qi, Chang Gao, Zhaochun Ren, and Qinyu Chen. Deltallm: A training-free framework exploiting temporal sparsity for efficient edge llm inference, 2025. URL <https://arxiv.org/abs/2507.19608>. arXiv:2507.19608.
- [60] Shengkun Tang, Liqun Ma, Haonan Li, Mingjie Sun, and Zhiqiang Shen. Bi-mamba: Towards accurate 1-bit state space models. *Transactions on Machine Learning Research*, 2024. doi: 10.48550/arXiv.2411.11843. URL <https://arxiv.org/abs/2411.11843>. Also arXiv:2411.11843.
- [61] Xinlin Li, Osama Hanna, Christina Fragouli, and Suhas Diggavi. Icqquant: Index coding enables low-bit llm quantization. In *Conference on Language Modeling (COLM)*, 2025. doi: 10.48550/arXiv.2505.00850. URL <https://arxiv.org/abs/2505.00850>. Also arXiv:2505.00850.
- [62] He Xiao, Qingyao Yang, Dirui Xie, Wendong Xu, Zunhai Su, Runming Yang, Haobo Liu, Wenyong Zhou, Zhengwu Liu, and Ngai Wong. Exploring layer-wise information effectiveness for post-training quantization in small language models, 2025. URL <https://arxiv.org/abs/2508.03332>. arXiv:2508.03332.
- [63] Gorgi Pavlov. Influence-inspired spectral rotations for extreme low-bit llm quantization, 2026. URL <https://arxiv.org/abs/2605.25203>. arXiv:2605.25203.
- [64] Haoyu Wang, Xingyu Yu, Haiyan Zhao, Fengxiang Wang, and Xu Han. Lc-qat: Data-efficient 2-bit qat for llms via linear-constrained vector quantization. In *International Conference on Machine Learning (ICML)*, 2026. doi: 10.48550/arXiv.2606.10531. URL <https://arxiv.org/abs/2606.10531>. Also arXiv:2606.10531.
- [65] Jinheng Wang, Hansong Zhou, Ting Song, Shaoguang Mao, Shuming Ma, Hongyu Wang, Yan Xia, and Furu Wei. 1-bit ai infra: Part 1.1, fast and lossless bitnet b1.58 inference on cpus, 2024. URL <https://arxiv.org/abs/2410.16144>. arXiv:2410.16144.
- [66] Jinheng Wang, Hansong Zhou, Ting Song, Shijie Cao, Yan Xia, Ting Cao, Jianyu Wei, Shuming Ma, Hongyu Wang, and Furu Wei. Bitnet.cpp: Efficient edge inference for ternary llms. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025. doi: 10.48550/arXiv.2502.11880. URL <https://arxiv.org/abs/2502.11880>. Also arXiv:2502.11880.
- [67] Fei Zuo, Xiaoyan Xi, Quanyi Zeng, Feiyu Wang, and Ho Fai Leung. Fairfuse: Multiplication-free llm inference on cpus via fused ternary kernels, 2026. URL <https://arxiv.org/abs/2604.20913>. arXiv:2604.20913.
- [68] Mohsen Dehghankar and Abolfazl Asudeh. Rsr-core: A high-performance engine for low-bit matrix-vector multiplication, 2026. URL <https://arxiv.org/abs/2603.27462>. arXiv:2603.27462.
- [69] Hyunwoo Oh, KyungIn Nam, Rajat Bhattacharjya, Hanning Chen, Tamoghno Das, Sanggeon Yun, Suyeon Jang, Andrew Ding, Nikil Dutt, and Mohsen Imani. T-sar: A full-stack co-design for cpu-only ternary llm inference via in-place simd alu reorganization. In *Design, Automation and Test in Europe (DATE)*, 2025. doi: 10.48550/arXiv.2511.13676. URL <https://arxiv.org/abs/2511.13676>. Also arXiv:2511.13676.
- [70] Jianyu Wei, Shijie Cao, Ting Cao, Lingxiao Ma, Lei Wang, Yanyong Zhang, and Mao Yang. T-mac: Cpu renaissance via table lookup for low-bit llm deployment on edge, 2024. URL <https://arxiv.org/abs/2407.00088>. arXiv:2407.00088.

- [71] Evangelos Georganas, Dhiraj Kalamkar, Alexander Heinecke, and Pradeep Dubey. Pushing the envelope of llm inference with ultra-low-bit quantized models, 2025. URL <https://arxiv.org/abs/2508.06753>. arXiv:2508.06753.
- [72] Yuhao Ji, Chao Fang, Shaobo Ma, Haikuo Shao, and Zhongfeng Wang. Co-designing binarized transformer and hardware accelerator for efficient end-to-end edge deployment, 2024. URL <https://arxiv.org/abs/2407.12070>. arXiv:2407.12070.
- [73] Haoxuan Shan, Cong Guo, Chiyue Wei, Feng Cheng, Junyao Zhang, Hai Li, and Yiran Chen. Platinum: Path-adaptable lut-based accelerator tailored for low-bit weight matrix multiplication. In *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2025. doi: 10.48550/arXiv.2511.21910. URL <https://arxiv.org/abs/2511.21910>. Also arXiv:2511.21910.
- [74] Zhirui Huang, Rui Ma, Shijie Cao, Ran Shu, Ian Wang, Ting Cao, Chixiao Chen, and Yongqiang Xiong. Tenet: An efficient sparsity-aware lut-centric architecture for ternary llm inference on edge, 2025. URL <https://arxiv.org/abs/2509.13765>. arXiv:2509.13765.
- [75] Robin Geens, Joran Heldens, Joren Dumoulin, and Marian Verhelst. Hardware generation and exploration of lookup table-based accelerators for 1.58-bit llm inference, 2026. URL <https://arxiv.org/abs/2604.25183>. arXiv:2604.25183.
- [76] Wenlun Zhang, Xinyu Li, Shimpei Ando, and Kentaro Yoshioka. Bitrom: Weight reload-free cirom architecture towards billion-parameter 1.58-bit llm inference. In *Asia and South Pacific Design Automation Conference (ASP-DAC)*, 2025. doi: 10.48550/arXiv.2509.08542. URL <https://arxiv.org/abs/2509.08542>. Also arXiv:2509.08542.
- [77] Hongyi Guan, Yijia Zhang, Wenqiang Wang, Yizhao Gao, Shijie Cao, Chen Zhang, and Ningyi Xu. Tom: A ternary read-only memory accelerator for llm-powered edge intelligence, 2026. URL <https://arxiv.org/abs/2602.20662>. arXiv:2602.20662.
- [78] Jinendra Malekar, Peyton Chandarana, Md Hasibul Amin, Mohammed E. Elbity, and Ramtin Zand. Pim-llm: A high-throughput hybrid pim architecture for 1-bit llms, 2025. URL <https://arxiv.org/abs/2504.01994>. arXiv:2504.01994.
- [79] Cristobal Ortega, Yann Falevoz, and Renaud Ayrignac. Pim-ai: A novel architecture for high-efficiency llm inference, 2024. URL <https://arxiv.org/abs/2411.17309>. arXiv:2411.17309.
- [80] Akul Malhotra and Sumeet Kumar Gupta. Retern: Exploiting natural redundancy and sign transformations for enhanced fault tolerance in compute-in-memory based ternary llms, 2025. URL <https://arxiv.org/abs/2506.01140>. arXiv:2506.01140.
- [81] Ye Qiao, Zhiheng Chen, Yifan Zhang, Yian Wang, and Sitao Huang. Tellme: An energy-efficient ternary llm accelerator for prefill and decode on edge fpgas, 2025. URL <https://arxiv.org/abs/2504.16266>. arXiv:2504.16266.
- [82] Ye Qiao, Zhiheng Chen, Yifan Zhang, Yian Wang, and Sitao Huang. Tellme v2: An efficient end-to-end ternary llm prefill and decode accelerator with table-lookup matmul on edge fpgas, 2025. URL <https://arxiv.org/abs/2510.15926>. arXiv:2510.15926.
- [83] Chenyang Yin, Zhenyu Bai, Pranav Venkatram, Shivam Aggarwal, Zhaoying Li, and Tulika Mitra. Tereffci: Highly efficient ternary llm inference on fpga, 2025. URL <https://arxiv.org/abs/2502.16473>. arXiv:2502.16473.
- [84] Yifan Zhang, Zhiheng Chen, Ye Qiao, and Sitao Huang. Pd-swap: Prefill-decode logic swapping for end-to-end llm inference on edge fpgas via dynamic partial reconfiguration, 2025. URL <https://arxiv.org/abs/2512.11550>. arXiv:2512.11550.

- [85] Zi-Wei Lin and Tian-Sheuan Chang. Vitallm: A versatile, ultra-compact ternary llm accelerator with dependency-aware scheduling, 2026. URL <https://arxiv.org/abs/2604.27396>. arXiv:2604.27396.
- [86] Zhen Li, Xin Huang, Liang Wang, Nan Yang, Ting Song, Yan Xia, Xun Wu, Shaohan Huang, Huishuai Zhang, Furu Wei, and Dongyan Zhao. Bitnet text embeddings, 2026. URL <https://arxiv.org/abs/2606.25674>. arXiv:2606.25674.
- [87] Jiayi Chen, Chen Wu, Shaoqun Zhang, Nan Li, Liangjie Zhang, and Qi Zhang. Efficient ternary weight embedding model: Bridging scalability and performance, 2024. URL <https://arxiv.org/abs/2411.15438>. arXiv:2411.15438.
- [88] Richard Connor, Alan Dearle, and Ben Claydon. Ultra-quantisation: Efficient embedding search via 1.58-bit encodings, 2025. URL <https://arxiv.org/abs/2506.00528>. arXiv:2506.00528.
- [89] Masaya Kawamura, Takuya Hasumi, Yuma Shirahata, and Ryuichi Yamamoto. Bittts: Highly compact text-to-speech using 1.58-bit quantization and weight indexing. In *Interspeech*, 2025. doi: 10.48550/arXiv.2506.03515. URL <https://arxiv.org/abs/2506.03515>. Also arXiv:2506.03515.
- [90] Zhaoqing Li, Haoning Xu, Zengrui Jin, Lingwei Meng, Tianzi Wang, Huimeng Wang, Youjun Chen, Mingyu Cui, Shujie Hu, and Xunying Liu. Towards one-bit asr: Extremely low-bit conformer quantization using co-training and stochastic precision. In *Interspeech*, 2025. doi: 10.48550/arXiv.2505.21245. URL <https://arxiv.org/abs/2505.21245>. Also arXiv:2505.21245.
- [91] Hongyu Wang, Chuyan Xiong, Ruiping Wang, and Xilin Chen. Bitvla: 1-bit vision-language-action models for robotics manipulation, 2025. URL <https://arxiv.org/abs/2506.07530>. arXiv:2506.07530.
- [92] Jainaveen Sundaram and Ravi Iyer. Llavaolmobitnet1b: Ternary llm goes multimodal!, 2024. URL <https://arxiv.org/abs/2408.13402>. arXiv:2408.13402.
- [93] Shu-Hao Zhang, Wei-Cheng Tang, Chen Wu, Peng Hu, Nan Li, Liang-Jie Zhang, Qi Zhang, and Shao-Qun Zhang. Ternaryclip: Efficiently compressing vision-language models with ternary weights and distilled knowledge, 2025. URL <https://arxiv.org/abs/2510.21879>. arXiv:2510.21879.
- [94] Hongyu Wang, Jiayu Xu, Ruiping Wang, Yan Feng, Yitao Zhai, Peng Pei, Xunliang Cai, and Xilin Chen. Mote: Mixture of ternary experts for memory-efficient large multimodal models, 2025. URL <https://arxiv.org/abs/2506.14435>. arXiv:2506.14435.
- [95] Ramprasath Ganesaraja, Sahil Dilip Panse, and Swathika N. Ternary mamba: Grouped quantization-aware training of w1.58a16 state space models, 2026. URL <https://arxiv.org/abs/2606.18114>. arXiv:2606.18114.
- [96] Md. Ashiq Ul Islam Sajid, Mohammad Sakib Mahmood, Md. Tareq Hasan, Md Abdur Rahim, Rafat Ara, and Md. Arafat Hossain. Bitrl: Reinforcement learning with 1-bit quantized language models for resource-constrained edge deployment, 2026. URL <https://arxiv.org/abs/2604.24273>. arXiv:2604.24273.
- [97] Mincheol Kang, Hyunjin Lim, Bomin Kang, and Daehee Park. Bittp: The lightweight trajectory prediction model with bitllm for edge-devices, 2026. URL <https://arxiv.org/abs/2605.29705>. arXiv:2605.29705.
- [98] Minsoo Kim, Sihwa Lee, Janghwan Lee, Sukjin Hong, Du-Seong Chang, Wonyong Sung, and Jungwook Choi. Token-scaled logit distillation for ternary weight generative language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. doi: 10.48550/arXiv.2308.06744. URL <https://arxiv.org/abs/2308.06744>. Also arXiv:2308.06744.

- [99] Wazib Ansar, Saptarsi Goswami, and Amlan Chakrabarti. Beexformer: A fast inferencing binarized transformer with early exits. *IEEE Transactions on Sustainable Computing*, 2024. doi: 10.48550/arXiv.2412.05225. URL <https://arxiv.org/abs/2412.05225>. Also arXiv:2412.05225.
- [100] Hong Chen, Chengtao Lv, Liang Ding, Haotong Qin, Xiabin Zhou, Yifu Ding, Xuebo Liu, Min Zhang, Jinyang Guo, Xianglong Liu, and Dacheng Tao. Db-llm: Accurate dual-binarization for efficient llms. In *International Conference on Learning Representations (ICLR)*, 2024. doi: 10.48550/arXiv.2402.11960. URL <https://arxiv.org/abs/2402.11960>. Also arXiv:2402.11960.
- [101] Mengzhao Chen, Wenqi Shao, Peng Xu, Jiahao Wang, Peng Gao, Kaipeng Zhang, and Ping Luo. Efficientqat: Efficient quantization-aware training for large language models, 2024. URL <https://arxiv.org/abs/2407.11062>. arXiv:2407.11062.
- [102] Sanjay Das, Swastik Bhattacharya, Souvik Kundu, Shamik Kundu, Anand Menon, Arnab Raha, and Kanad Basu. Genbfa: An evolutionary optimization approach to bit-flip attacks on llms, 2024. URL <https://arxiv.org/abs/2411.13757>. arXiv:2411.13757.
- [103] Dongwon Jo, Taesu Kim, Yulhwa Kim, and Jae-Joon Kim. Mixture of scales: Memory-efficient token-adaptive binarization for large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. doi: 10.48550/arXiv.2406.12311. URL <https://arxiv.org/abs/2406.12311>. Also arXiv:2406.12311.
- [104] Jacob Nielsen, Lukas Galke, and Peter Schneider-Kamp. When are 1.58 bits enough? a bottom-up exploration of bitnet quantization, 2024. URL <https://arxiv.org/abs/2411.05882>. arXiv:2411.05882.
- [105] Mohsen Dehghankar, Mahdi Erfanian, and Abolfazl Asudeh. An efficient matrix multiplication algorithm for accelerating inference in binary and ternary neural networks. In *International Conference on Machine Learning (ICML)*, 2024. doi: 10.48550/arXiv.2411.06360. URL <https://arxiv.org/abs/2411.06360>. Also arXiv:2411.06360.
- [106] Xiaojing Fan and Chunliang Tao. Towards resilient and efficient llms: A comparative study of efficiency, performance, and adversarial robustness, 2024. URL <https://arxiv.org/abs/2408.04585>. arXiv:2408.04585.
- [107] Zijian Ye, Wei Huang, Yifei Yu, Tianhe Ren, Zhongrui Wang, and Xiaojuan Qi. Dbellquant: Breaking the bell with double-bell transformation for llms post training binarization, 2025. URL <https://arxiv.org/abs/2507.01027>. arXiv:2507.01027.
- [108] Hoang Anh Dung, Cuong Pham, Cuong Nguyen, Trung Le, Thanh-Toan Do, and Jianfei Cai. Rethinking output alignment for 1-bit post-training quantization of large language models, 2025. URL <https://arxiv.org/abs/2512.21651>. arXiv:2512.21651.
- [109] Hong Huang, Decheng Wu, Rui Cen, Guanghua Yu, Zonghang Li, Kai Liu, Jianchen Zhu, Peng Chen, Xue Liu, and Dapeng Wu. Tequila: Trapping-free ternary quantization for large language models, 2025. URL <https://arxiv.org/abs/2509.23809>. arXiv:2509.23809.
- [110] Mahsa Ardakani, Jinendra Malekar, and Ramtin Zand. Llmpi: Optimizing llms for high-throughput on raspberry pi. In *IEEE/CVF CVPR Workshops*, 2025. doi: 10.48550/arXiv.2504.02118. URL <https://arxiv.org/abs/2504.02118>. Also arXiv:2504.02118.
- [111] Siqing Song, Chuang Wang, Ruiqi Wang, Yi Yang, and Xu-Yao Zhang. Achieving binary weight and activation for llms using post-training quantization, 2025. URL <https://arxiv.org/abs/2504.05352>. arXiv:2504.05352.

- [112] Vladimír Boža and Vladimír Macko. Addition is almost all you need: Compressing large language models with double binary factorization. *Transactions on Machine Learning Research*, 2025. doi: 10.48550/arXiv.2505.11076. URL <https://arxiv.org/abs/2505.11076>. Also arXiv:2505.11076.
- [113] Hao Gu, Lujun Li, Hao Wang, Lei Wang, Zheyu Wang, Bei Liu, Jiacheng Liu, Qiyuan Zhu, Sirui Han, and Yike Guo. Btc-llm: Efficient sub-1-bit llm quantization via learnable transformation and binary codebook, 2025. URL <https://arxiv.org/abs/2506.12040>. arXiv:2506.12040.
- [114] Haolin Zhang and Jeff Huang. Challenging gpu dominance: When cpus outperform for on-device llm inference, 2025. URL <https://arxiv.org/abs/2505.06461>. arXiv:2505.06461.
- [115] Elias Frantar, Utku Evci, Wonpyo Park, Neil Houlsby, and Dan Alistarh. Compression scaling laws: Unifying sparsity and quantization, 2025. URL <https://arxiv.org/abs/2502.16440>. arXiv:2502.16440.
- [116] Ningning Chen, Weicai Ye, and Ying Jiang. Hbllm: Wavelet-enhanced high-fidelity 1-bit quantization for llms, 2025. URL <https://arxiv.org/abs/2512.00862>. arXiv:2512.00862.
- [117] Junyu Chen, Junzhuo Li, Zhen Peng, Wenjie Wang, Yuxiang Ren, Long Shi, and Xuming Hu. Lota-qaf: Lossless ternary adaptation for quantization-aware fine-tuning, 2025. URL <https://arxiv.org/abs/2505.18724>. arXiv:2505.18724.
- [118] Ba-Hien Tran and Van Minh Nguyen. Highly efficient and effective llms with multi-boolean architectures. In *International Conference on Learning Representations (ICLR)*, 2025. doi: 10.48550/arXiv.2505.22811. URL <https://arxiv.org/abs/2505.22811>. Also arXiv:2505.22811.
- [119] Xianglong Yan, Tianao Zhang, Zhiteng Li, Haotong Qin, and Yulun Zhang. Progressive binarization with semi-structured pruning for llms, 2025. URL <https://arxiv.org/abs/2502.01705>. arXiv:2502.01705.
- [120] Jiaqi Zhao, Miao Zhang, Ming Wang, Yuzhang Shang, Kaihao Zhang, Weili Guan, Yaowei Wang, and Min Zhang. Ptl1.61: Push the real limit of extremely low-bit post-training quantization methods for large language models. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025. doi: 10.48550/arXiv.2502.13179. URL <https://arxiv.org/abs/2502.13179>. Also arXiv:2502.13179.
- [121] Junhao Xia, Ming Zhao, Limin Xiao, and Xiujun Zhang. Sdq-llm: Sigma-delta quantization for 1-bit llms of any size. In *European Conference on Artificial Intelligence (ECAI)*, 2025. doi: 10.48550/arXiv.2510.03275. URL <https://arxiv.org/abs/2510.03275>. Also arXiv:2510.03275.
- [122] Ben Crulis, Cyril De Runz, Barthélemy Serres, and Gilles Venturini. Ternarization of vision language models for use on edge devices, 2025. URL <https://arxiv.org/abs/2504.06298>. arXiv:2504.06298.
- [123] Nii Osae Osae Dade, Tony Morri, Sayandip Pal, Moinul Hossain Rahat, and Rickston Pinto. Litespark inference for cpus: Ultra-fast simd framework for ternary (1.58-bit) language models, 2026. URL <https://arxiv.org/abs/2605.06485>. arXiv:2605.06485.
- [124] Siqing Song, Chuang Wang, Yong Lang, Yi Yang, and Xu-Yao Zhang. Lbllm: Lightweight binarization of large language models via three-stage distillation, 2026. URL <https://arxiv.org/abs/2604.19167>. arXiv:2604.19167.
- [125] Nisharg Nargund and Priyesh Shukla. Ternarylm: Memory-efficient language modeling via native 1.5-bit quantization with adaptive layer-wise scaling, 2026. URL <https://arxiv.org/abs/2602.07374>. arXiv:2602.07374.

- [126] Shigeng Wang, Chao Li, Yangyuxuan Kang, Jiawei Fan, and Anbang Yao. Cat-q: Cost-efficient and accurate ternary quantization for llms, 2026. URL <https://arxiv.org/abs/2606.26650>. arXiv:2606.26650.
- [127] Guoan Wang, Feiyu Wang, Zongwei Lv, Yikun Zong, and Tong Yang. Hestia: A hessian-guided differentiable quantization-aware training framework for extremely low-bit llms, 2026. URL <https://arxiv.org/abs/2601.20745>. arXiv:2601.20745.
- [128] David Alejandro Trejo Pizzo. Hybrid gated flow (hgf): Stabilizing 1.58-bit llms via selective low-rank correction, 2026. URL <https://arxiv.org/abs/2602.05269>. arXiv:2602.05269.
- [129] Hyochan Chong, Dongkyu Kim, Changdong Kim, and Minseop Choi. Nanoquant: Efficient sub-1-bit quantization of large language models, 2026. URL <https://arxiv.org/abs/2602.06694>. arXiv:2602.06694.
- [130] Youngcheon You, Banseok Lee, Minseop Choi, Seonyoung Kim, Hyochan Chong, Changdong Kim, Youngmin Kim, and Dongkyu Kim. Rabbit: Residual-aware binarization training for accuracy and efficient llms, 2026. URL <https://arxiv.org/abs/2602.05367>. arXiv:2602.05367.
- [131] Ekaterina Alimaskina, Darya Rudas, Denis Shveykin, Gleb Molodtsov, Pavel Vasiliev, and Aleksandr Beznosikov. Extreme low-bit inference in reasoning models: Failure modes and targeted recovery, 2026. URL <https://arxiv.org/abs/2606.02011>. arXiv:2606.02011.
- [132] Yuzhuang Xu, Xu Han, Yuxuan Li, Pengzhan Li, and Wanxiang Che. Fitting is not enough: Smoothness in extremely quantized llms, 2026. URL <https://arxiv.org/abs/2605.08894>. arXiv:2605.08894.
- [133] Sohir Maskey, Constantin Eichenberg, Johannes Messner, and Douglas Orr. 1-bit wonder: Improving qat performance in the low-bit regime through k-means quantization, 2026. URL <https://arxiv.org/abs/2602.15563>. arXiv:2602.15563.
- [134] Xunjie Wang, Jiacheng Shi, Zihan Zhao, Yang Yu, Zhichao Hua, and Jinyu Gu. Tz-llm: Protecting on-device large language models with arm trustzone. In *European Conference on Computer Systems (EuroSys)*, 2025. doi: 10.48550/arXiv.2511.13717. URL <https://arxiv.org/abs/2511.13717>. Also arXiv:2511.13717.
- [135] Xiangyu Li, Chengyu Yin, Weijun Wang, Jianyu Wei, Ting Cao, and Yunxin Liu. Vec-lut: Vector table lookup for parallel ultra-low-bit llm inference on edge devices. In *ACM MobiSys*, 2025. doi: 10.48550/arXiv.2512.06443. URL <https://arxiv.org/abs/2512.06443>. Also arXiv:2512.06443.
- [136] Tollef Emil Jørgensen. Resource-efficient language models: Quantization for fast and accessible inference, 2025. URL <https://arxiv.org/abs/2505.08620>. arXiv:2505.08620.
- [137] Euhid Aman, Esteban Carlin, Hsing-Kuo Pao, Giovanni Beltrame, Ghaluh Indah Permata Sari, and Yie-Tarng Chen. Bitmar: Low-bit multimodal fusion with episodic memory for edge devices, 2025. URL <https://arxiv.org/abs/2510.10560>. arXiv:2510.10560.
- [138] Jiayi Chen, Jieqi Shi, Jing Huo, and Chen Wu. R2q: Towards robust 2-bit large language models via residual refinement quantization, 2025. URL <https://arxiv.org/abs/2511.21736>. arXiv:2511.21736.
- [139] Shu-Hao Zhang, Le-Tong Huang, Xiang-Sheng Deng, Xin-Yi Zou, Chen Wu, Nan Li, Shao-Qun Zhang, and Zhi-Hua Zhou. Edgerazor: A lightweight framework for llms via mixed-precision quantization-aware distillation, 2026. URL <https://arxiv.org/abs/2605.04062>. arXiv:2605.04062.

- [140] Oliver Grainge, Michael Milford, Indu Bodala, Sarvapali D. Ramchurn, and Shoaib Ehsan. Tetra-vpr: A ternary transformer approach for compact visual place recognition. *IEEE Robotics and Automation Letters*, 2025. doi: 10.48550/arXiv.2503.02511. URL <https://arxiv.org/abs/2503.02511>. Also arXiv:2503.02511.
- [141] Microsoft. bitnet.cpp: Official Inference Framework for 1-bit LLMs. <https://github.com/microsoft/BitNet>, 2026. accessed 2026-09-03.
- [142] Falcon-LLM Team, TII. Falcon-Edge: Powerful, Universal, Fine-tunable 1.58bit Language Models. <https://huggingface.co/blog/tiiuae/falcon-edge>, 2025. accessed 2026-09-03.
- [143] Microsoft. microsoft/bitnet-b1.58-2B-4T Model Card. <https://huggingface.co/microsoft/bitnet-b1.58-2B-4T>, 2025. accessed 2026-09-03.
- [144] grave0x. Current bitnet-b1.58-2B-4T checkpoints contain all-zero MLP weight tensors (model generates garbage), GitHub issue #608. <https://github.com/microsoft/BitNet/issues/608>, 2026. opened 2026-08-11, accessed 2026-09-14.